



US012404535B2

(12) **United States Patent**
Ouyang et al.

(10) **Patent No.:** **US 12,404,535 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **NAD(P) DEPOT FOR NAD(P)-DEPENDENT
ENZYME-BASED SENSORS**

(71) Applicant: **ABBOTT DIABETES CARE INC.**,
Alameda, CA (US)

(72) Inventors: **Tianmei Ouyang**, Fremont, CA (US);
Cade Fox, Burlingame, CA (US);
Zenghe Liu, Alameda, CA (US);
Stephen Oja, Oakland, CA (US);
Benjamin J. Feldman, Berkeley, CA
(US)

(73) Assignee: **ABBOTT DIABETES CARE INC.**,
Alameda, CA (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 260 days.

(21) Appl. No.: **17/552,103**

(22) Filed: **Dec. 15, 2021**

(65) **Prior Publication Data**

US 2022/0186277 A1 Jun. 16, 2022

Related U.S. Application Data

(60) Provisional application No. 63/125,846, filed on Dec.
15, 2020.

(51) **Int. Cl.**
C12Q 1/00 (2006.01)

C12Q 1/32 (2006.01)

(52) **U.S. Cl.**
CPC **C12Q 1/005** (2013.01); **C12Q 1/32**
(2013.01); **C12Y 101/0103** (2013.01); **C12Y**
112/01002 (2013.01)

(58) **Field of Classification Search**
CPC C12Q 1/005; C12Q 1/32; C12Y 101/0103;
C12Y 112/01002
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,134,461 A 10/2000 Say et al.
6,605,200 B1 8/2003 Mao et al.
6,605,201 B1 8/2003 Mao et al.

(Continued)

FOREIGN PATENT DOCUMENTS

DE 10246303 A1 * 5/2003 G01N 27/301
EP 2089531 A1 8/2009

(Continued)

OTHER PUBLICATIONS

Ning Yang, Xianping Chen, Tianling Ren, Ping Zhang, Daoguo
Yang, Carbon nanotube based biosensors, *Sensors and Actuators B:*
Chemical, vol. 207, Part A, 2015, <https://doi.org/10.1016/j.snb.2014.10.040>. (Year: 2015).
International Search Report and Written Opinion for International
Application No. PCT/US2021/063600, mailed on May 5, 2022, 9
pages.

(Continued)

Primary Examiner — Louise W Humphrey

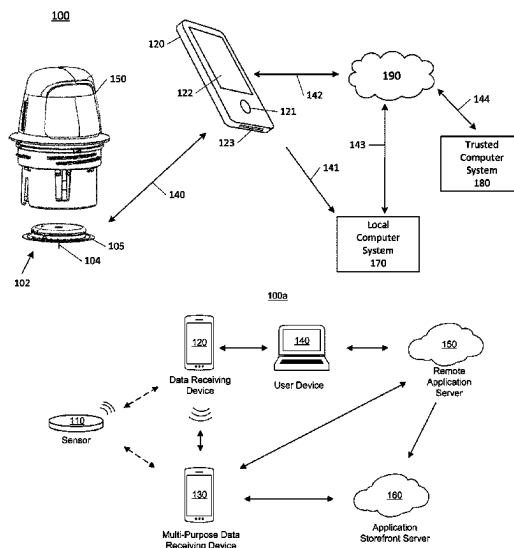
Assistant Examiner — Rachel Emily Martin

(74) *Attorney, Agent, or Firm* — STERNE, KESSLER,
GOLDSTEIN & FOX P.L.L.C.

(57) **ABSTRACT**

The present disclosure provides analyte sensors including
one or more NAD(P)-dependent enzymes and an internal
supply of NAD(P) for the detection of an analyte. The
present disclosure further provides methods of using such
analyte sensors for detecting one or more analytes present in
a biological sample of a subject, and methods of manufac-
turing said analyte sensors.

18 Claims, 40 Drawing Sheets
(6 of 40 Drawing Sheet(s) Filed in Color)



(56)

References Cited**U.S. PATENT DOCUMENTS**

6,736,957	B1	5/2004	Forrow et al.
7,501,053	B2	3/2009	Karinka et al.
7,754,093	B2	7/2010	Forrow et al.
8,268,143	B2	9/2012	Liu et al.
8,444,834	B2	5/2013	Liu et al.
8,761,857	B2	6/2014	Feldman et al.
2003/0042137	A1	3/2003	Mao et al.
2005/0205422	A1	9/2005	Moser et al.
2010/0230285	A1	9/2010	Hoss et al.
2013/0150691	A1	6/2013	Pace et al.
2014/0171771	A1	6/2014	Feldman et al.
2016/0331283	A1	11/2016	Rao et al.
2018/0235520	A1	8/2018	Rao et al.
2019/0024130	A1	1/2019	Ouyang et al.
2019/0274598	A1	9/2019	Scott et al.
2020/0196919	A1	6/2020	Rao et al.
2020/0237276	A1*	7/2020	Oja A61B 5/14865
2021/0204841	A1	7/2021	Thomas et al.
2022/0304599	A1*	9/2022	Liang C12Q 1/004

FOREIGN PATENT DOCUMENTS

EP	2608715	B1 *	6/2016	 A61B 5/14532
JP	2012502689	A	2/2012	
WO	WO 98/33936	A1	8/1998	
WO	WO 2007/012494	A1	2/2007	
WO	WO-2011091363	A2	7/2011	
WO	WO-2017151952	A1	9/2017	
WO	WO 2018/136898	A1	7/2018	
WO	WO-2019222615	A1	11/2019	
WO	WO 2019/236850	A1	12/2019	
WO	WO 2019/236859	A1	12/2019	
WO	WO 2019/236876	A1	12/2019	
WO	WO-2020159956	A1	8/2020	

OTHER PUBLICATIONS

Vidal et al., "Review of NAD(P)H-dependent oxidoreductases: Properties, engineering and application," BBA—Proteins and Proteomics 1866(2):327-347 (2018).

* cited by examiner

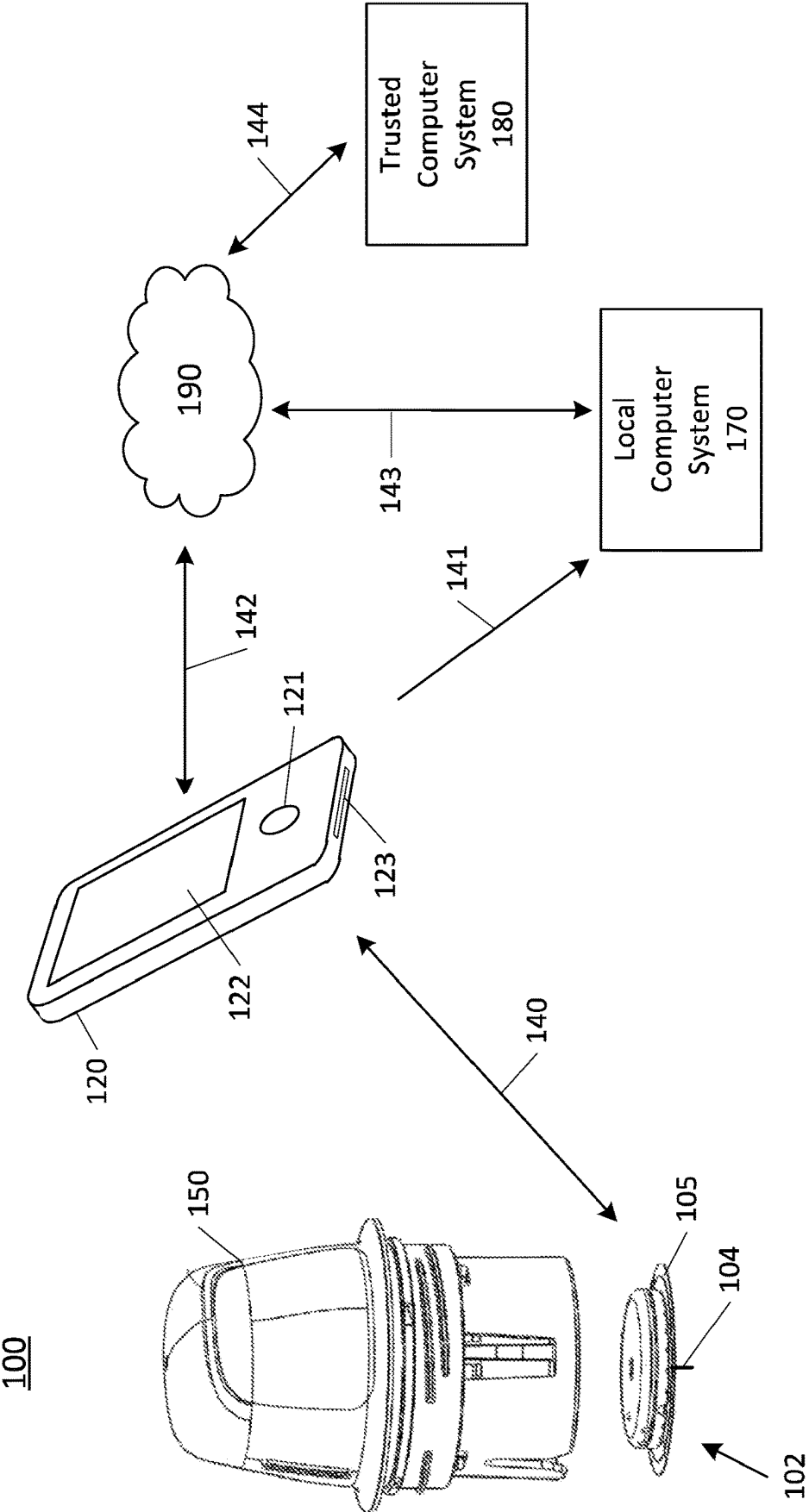


FIG. 1A

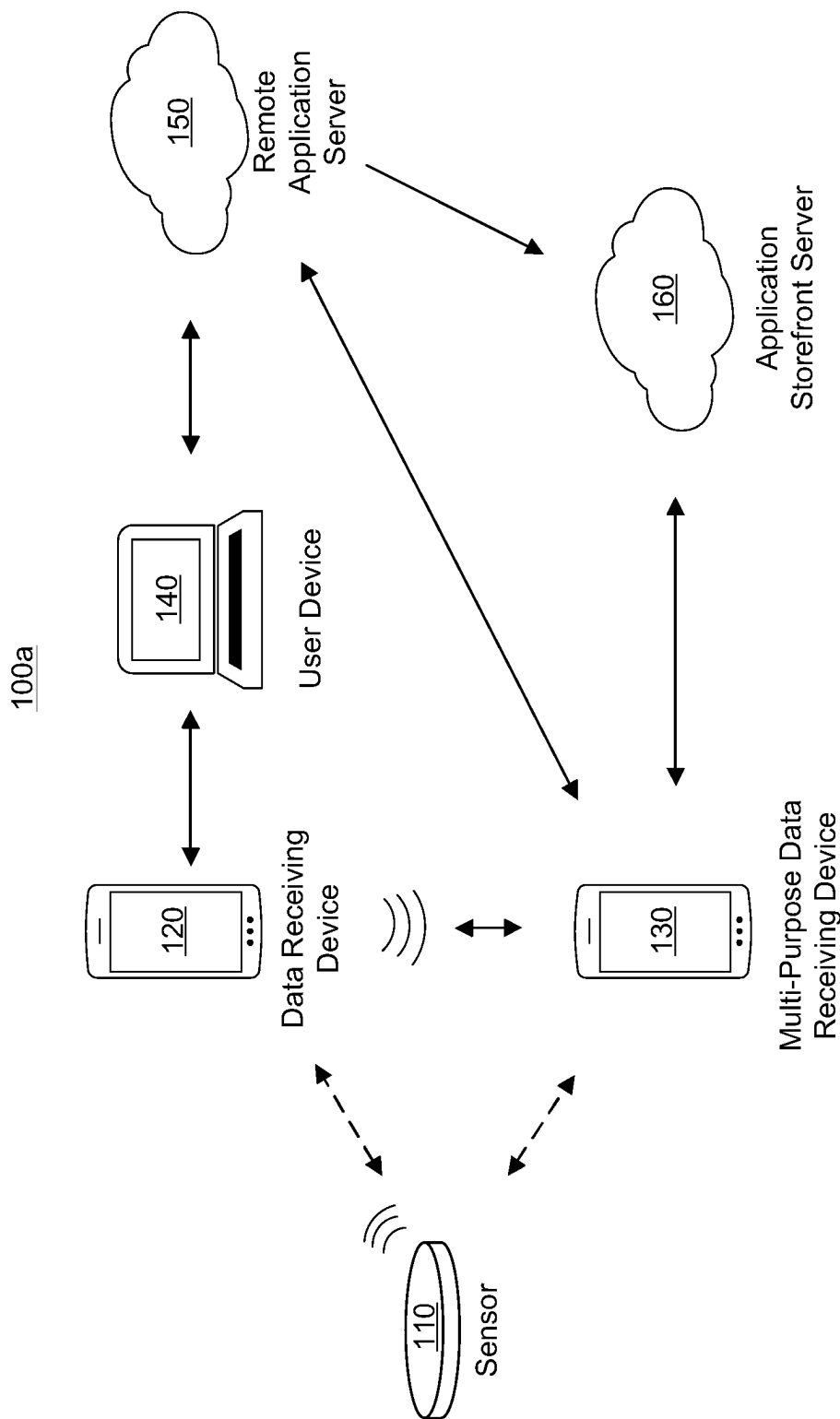


FIG. 1B

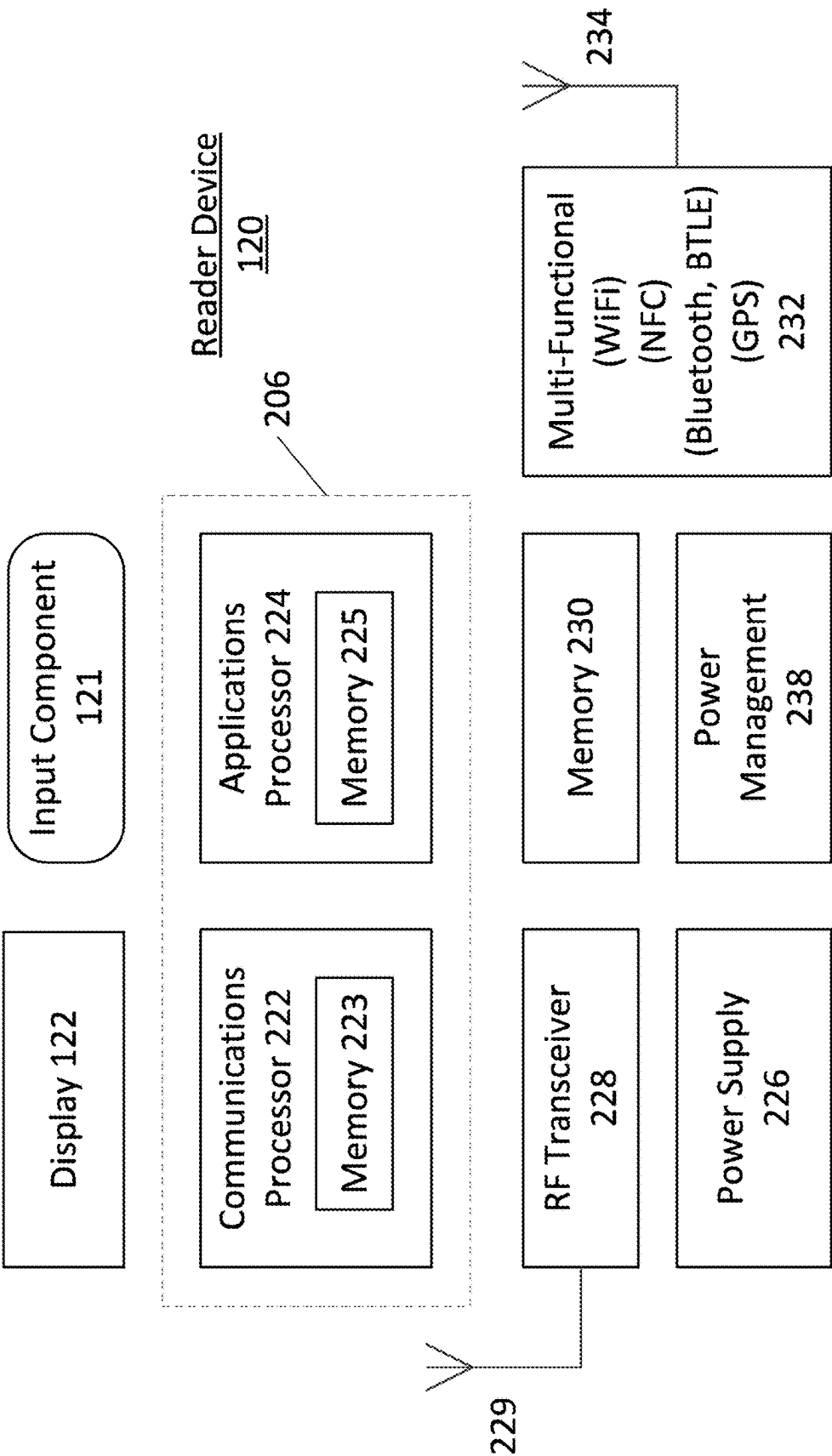


FIG. 2A

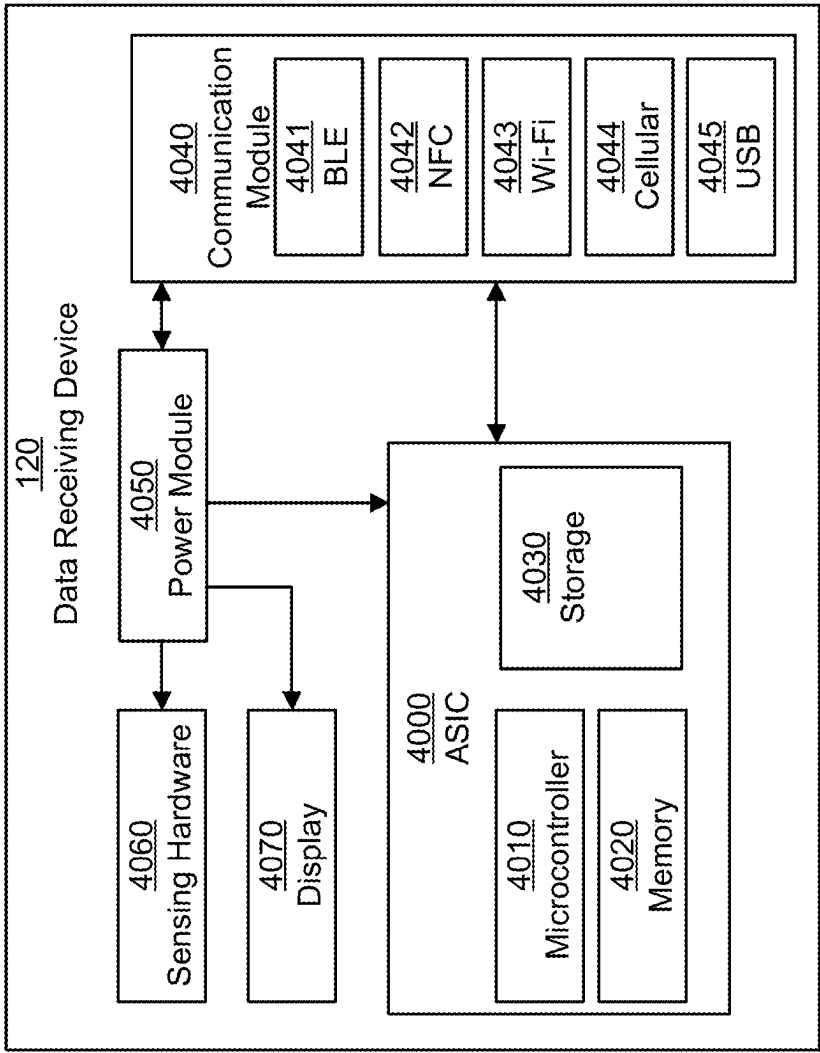
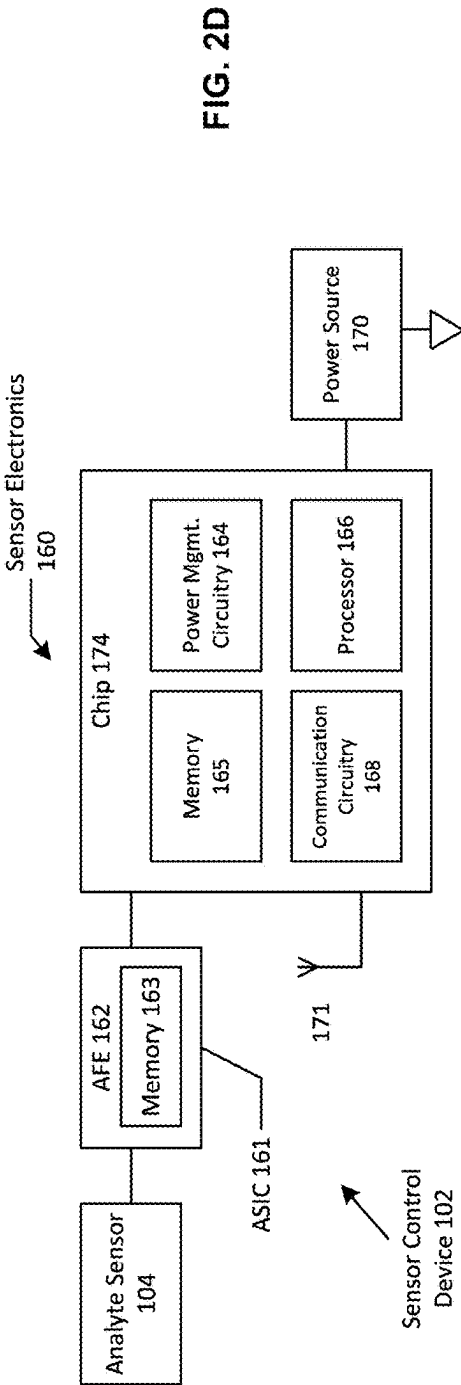
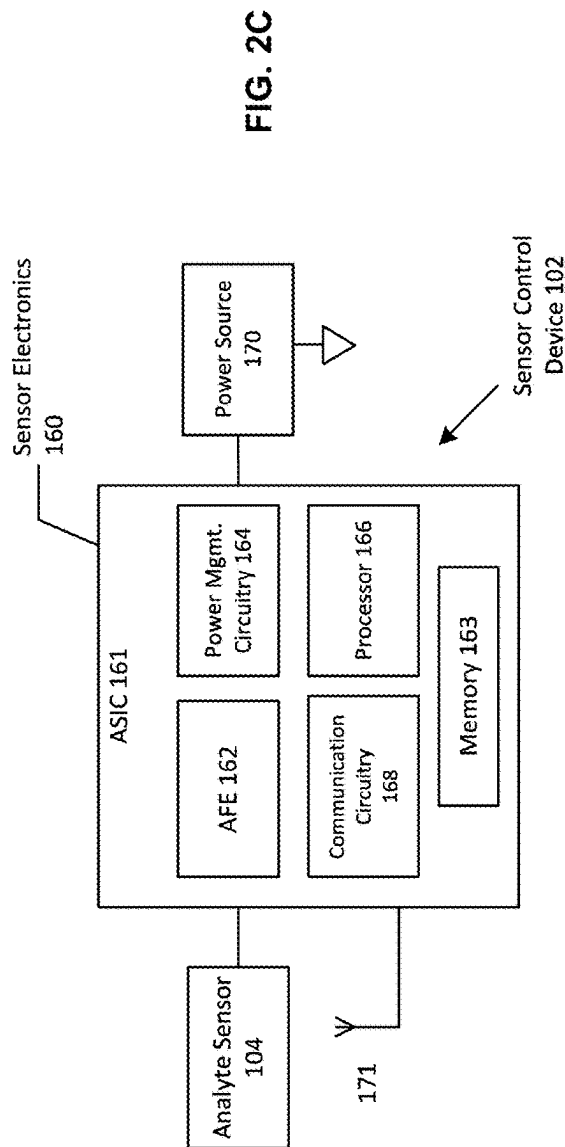


FIG. 2B



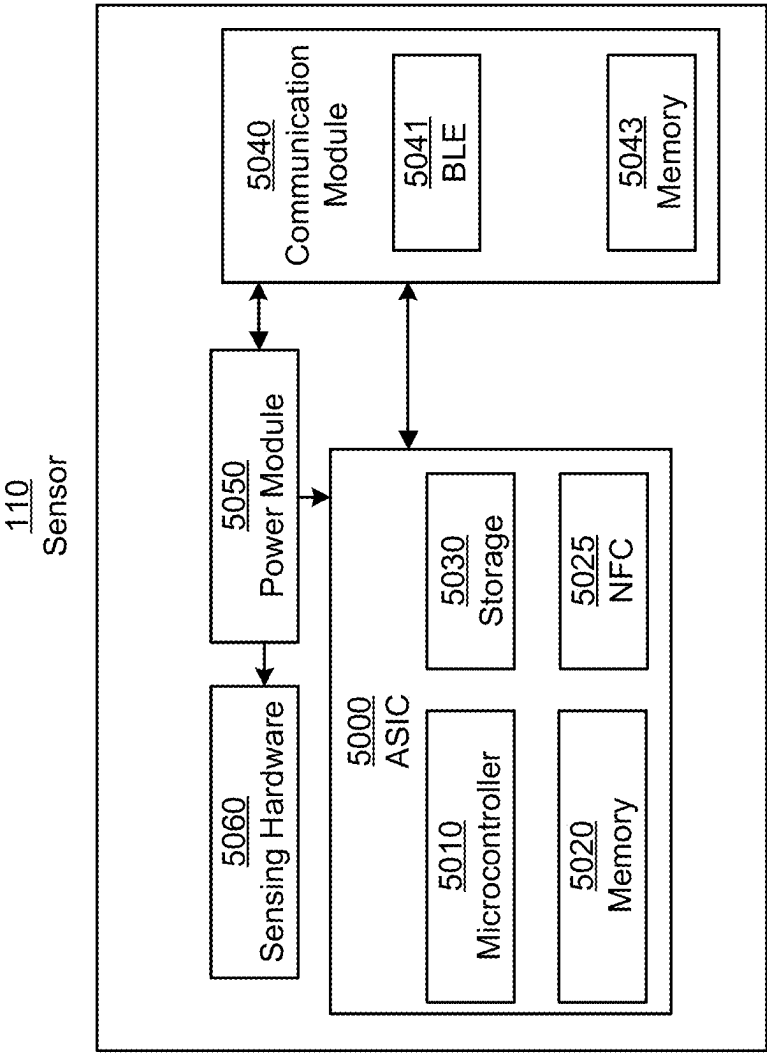


FIG. 2E

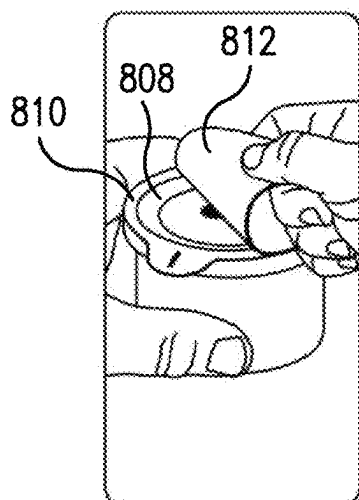


FIG. 3A

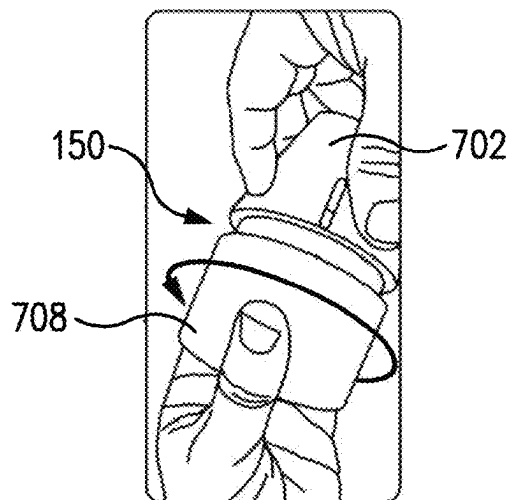


FIG. 3B

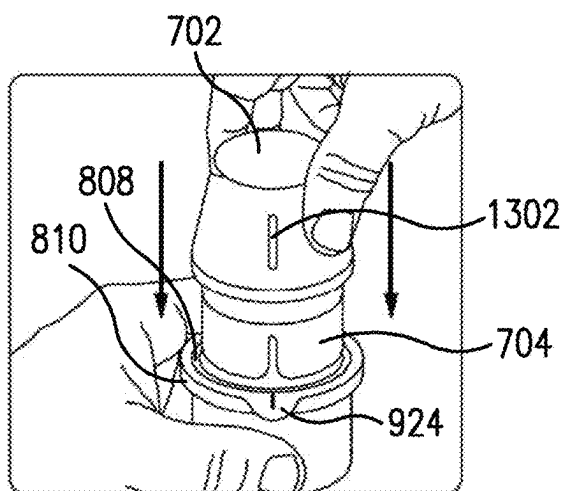


FIG. 3C

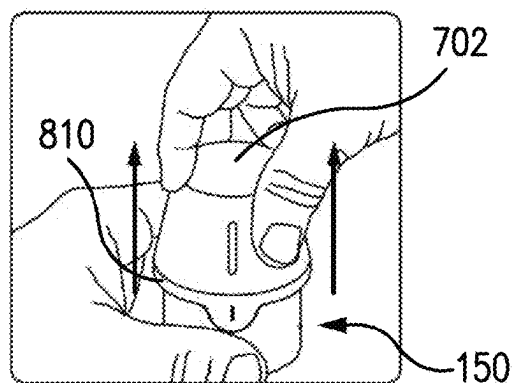


FIG. 3D

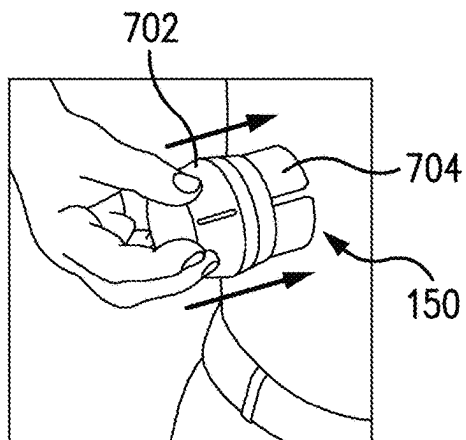


FIG. 3E

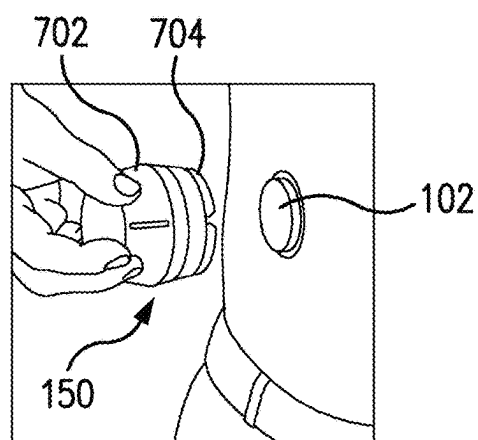
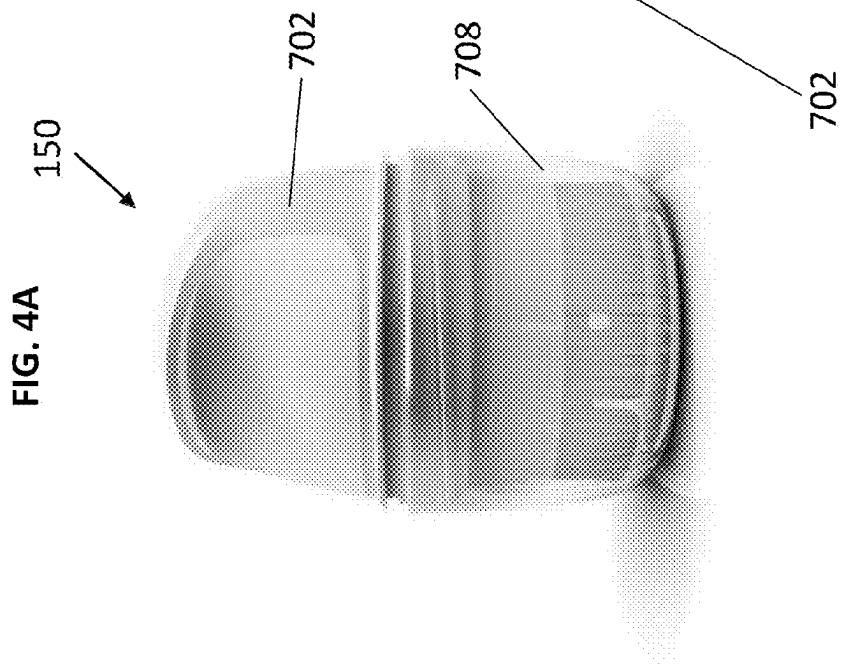
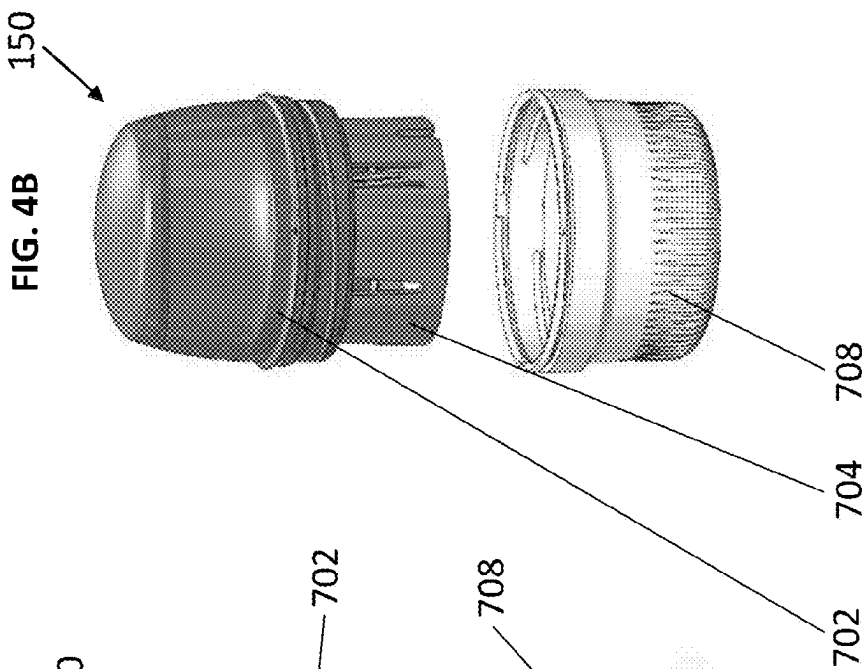
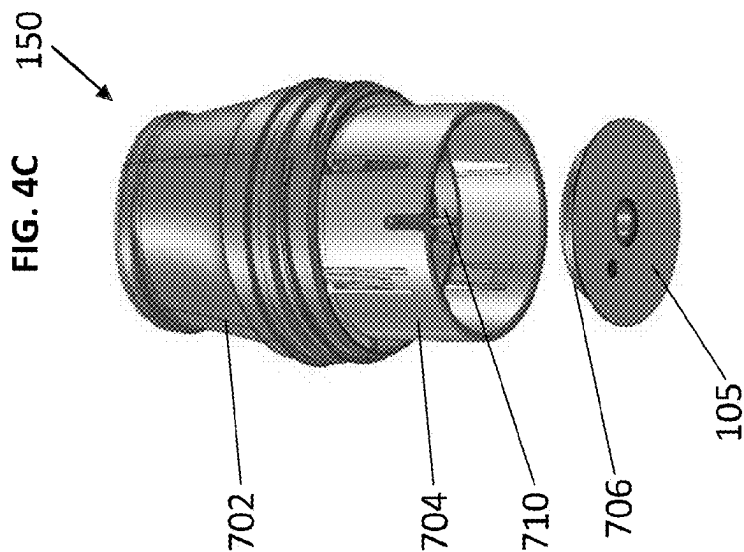
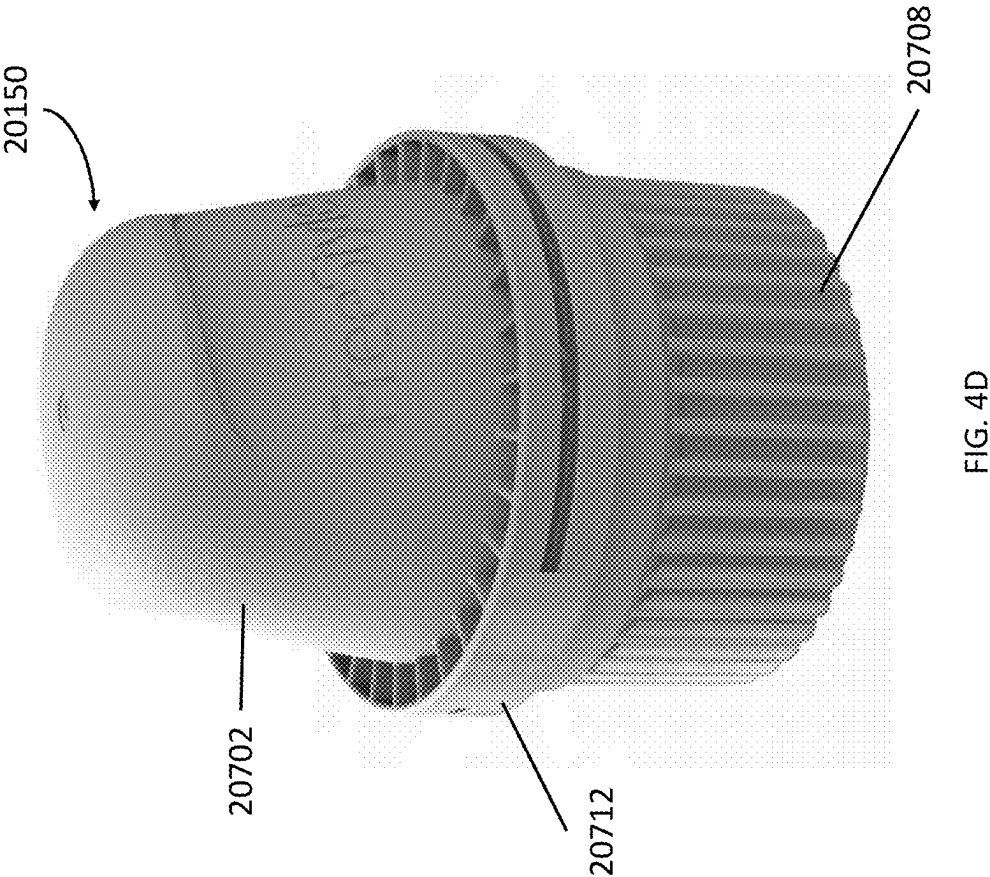
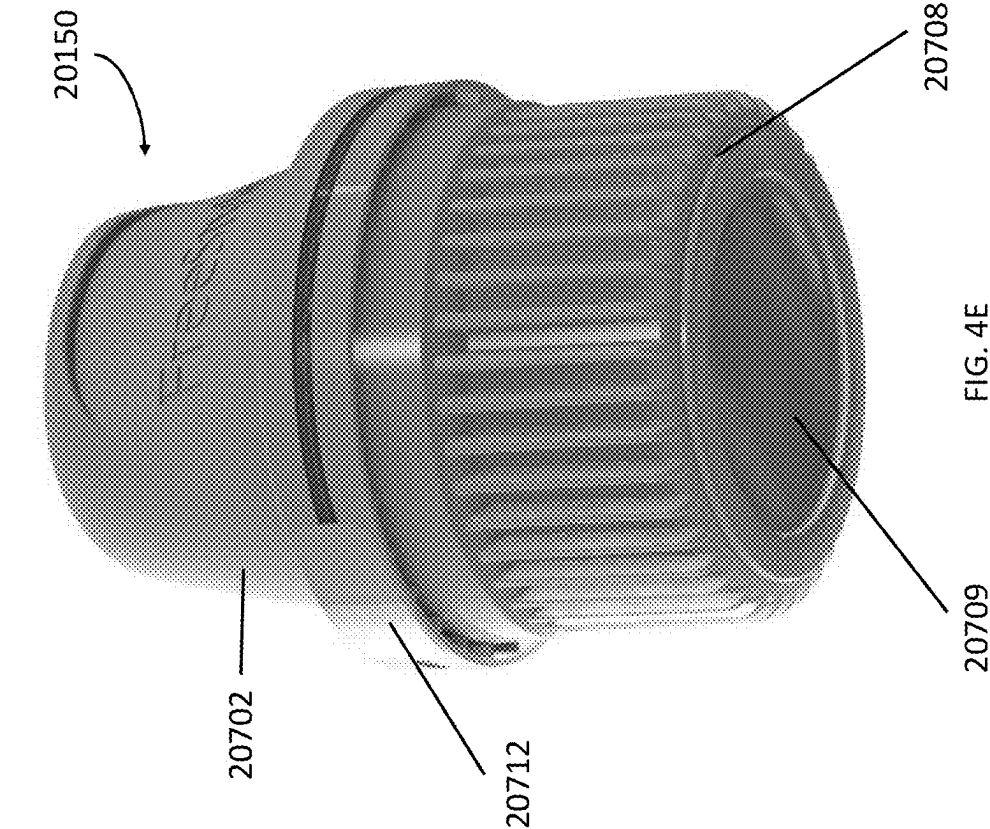


FIG. 3F





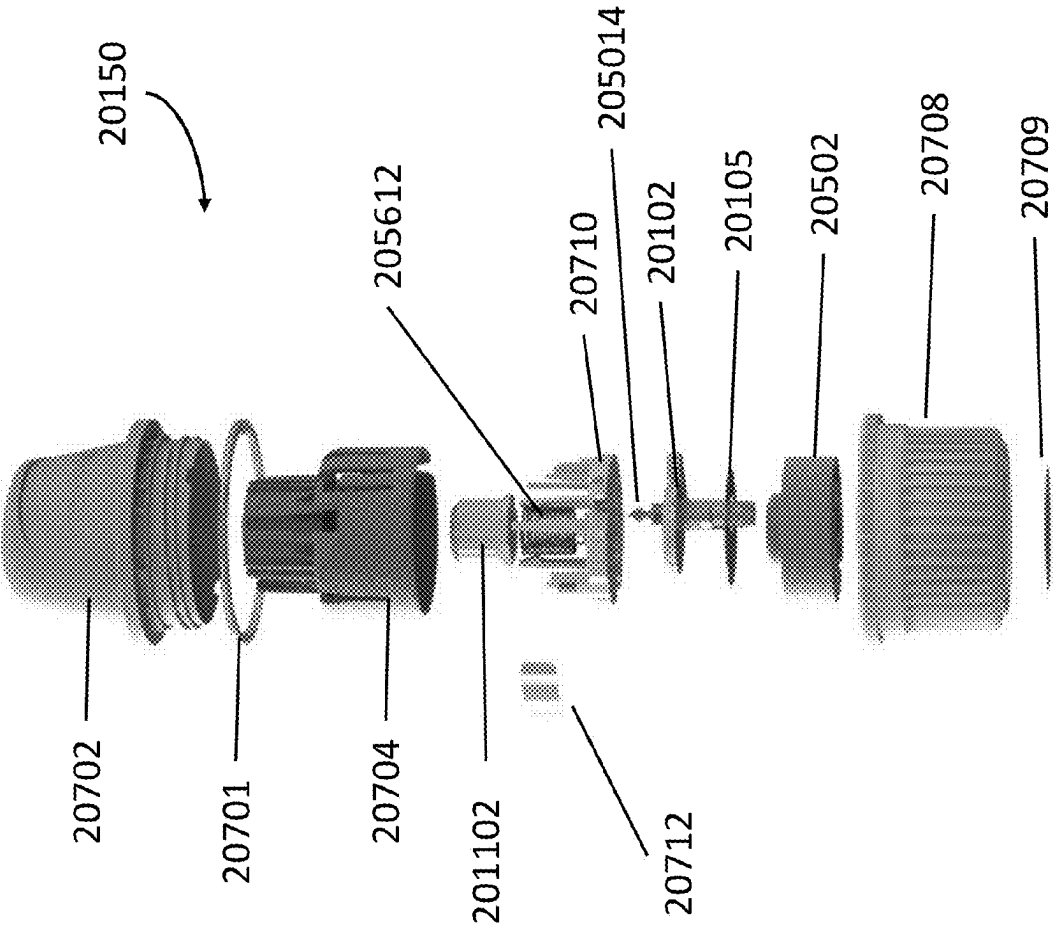


FIG. 4F

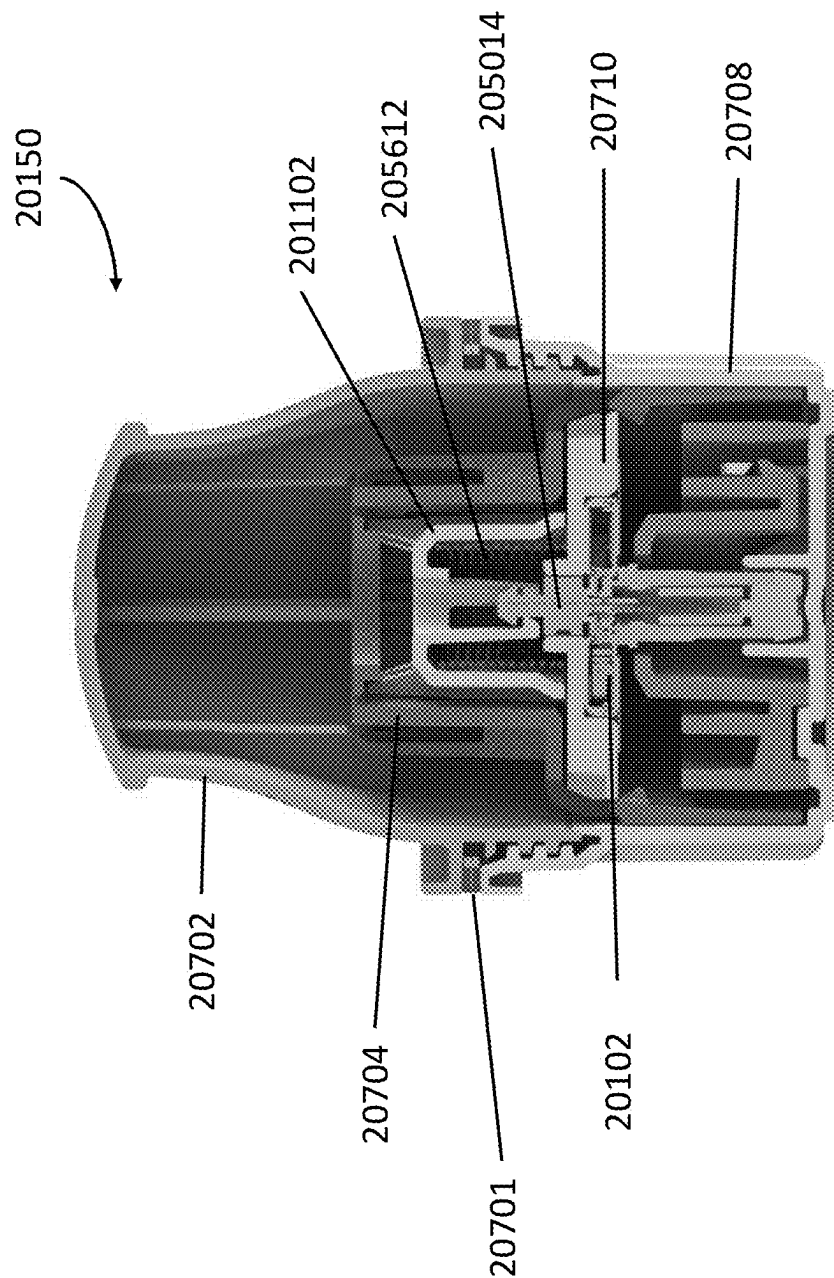


FIG. 4G

FIG. 6A

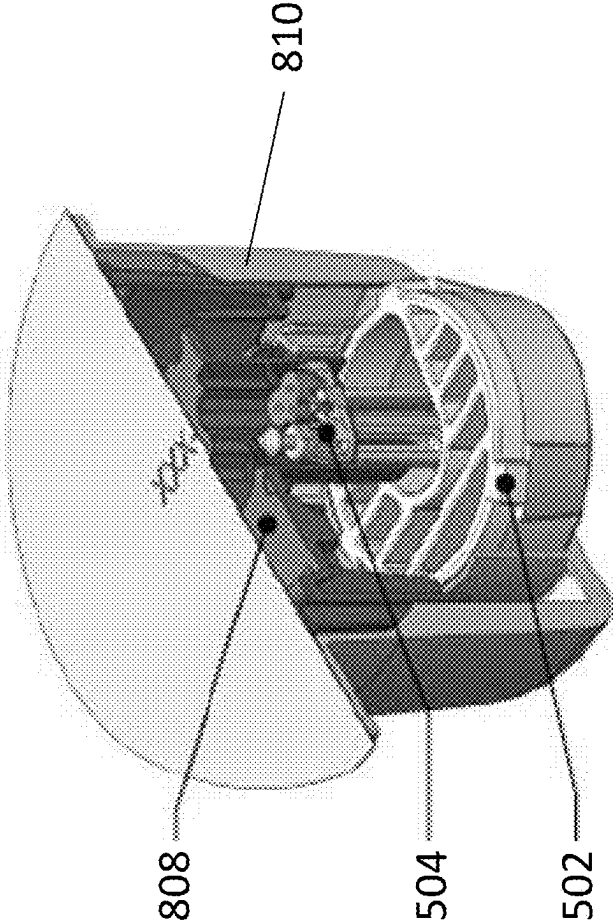
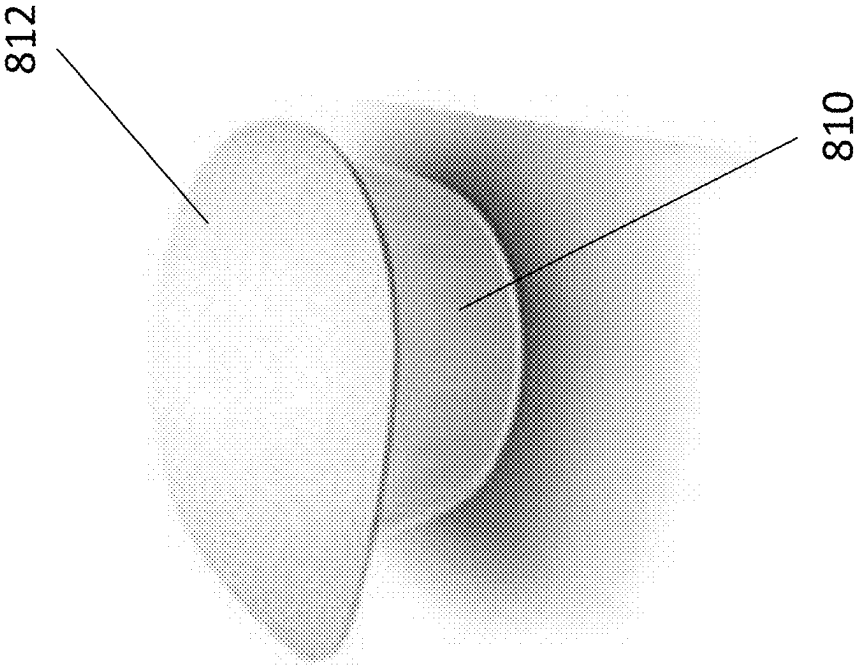


FIG. 5



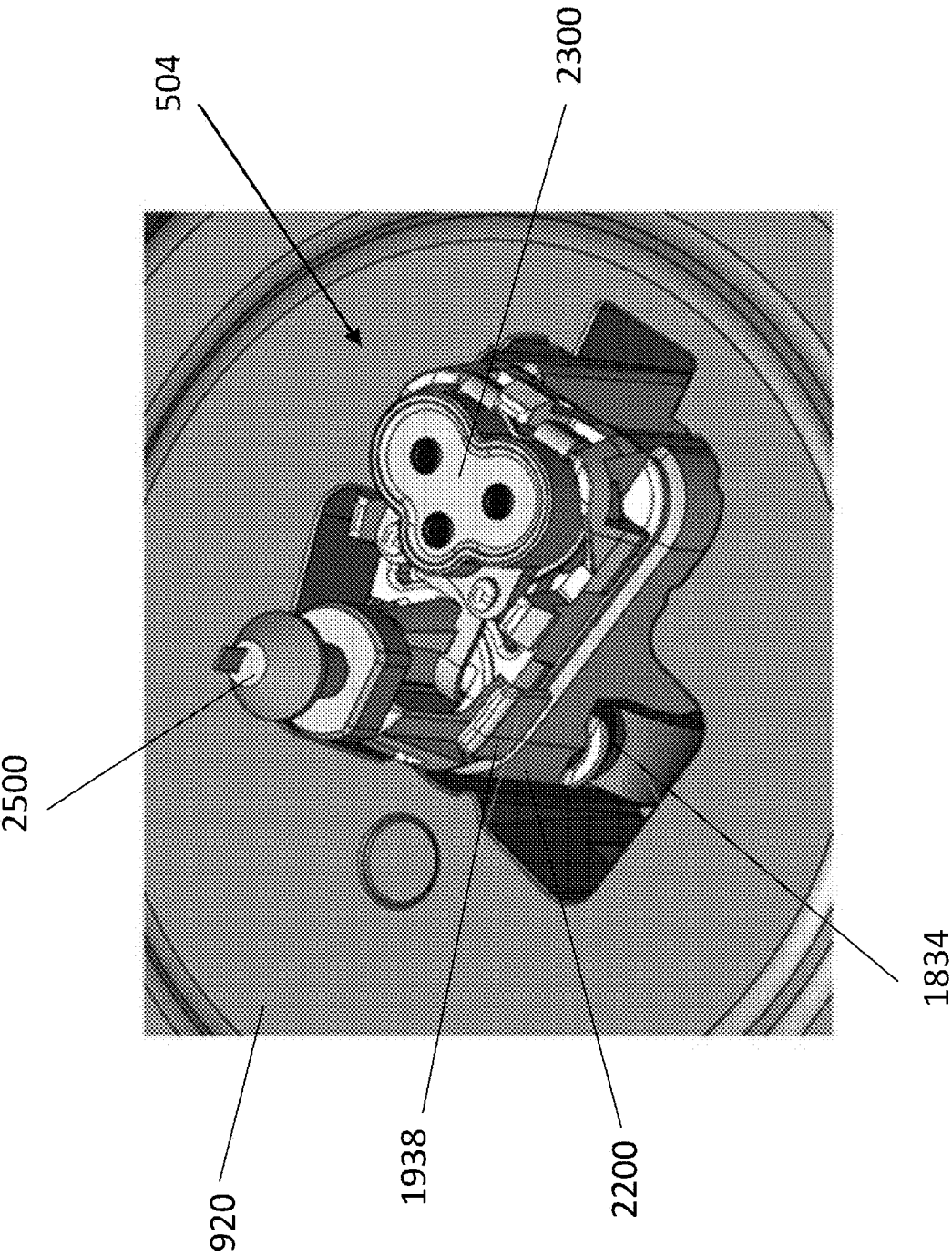


FIG. 6B

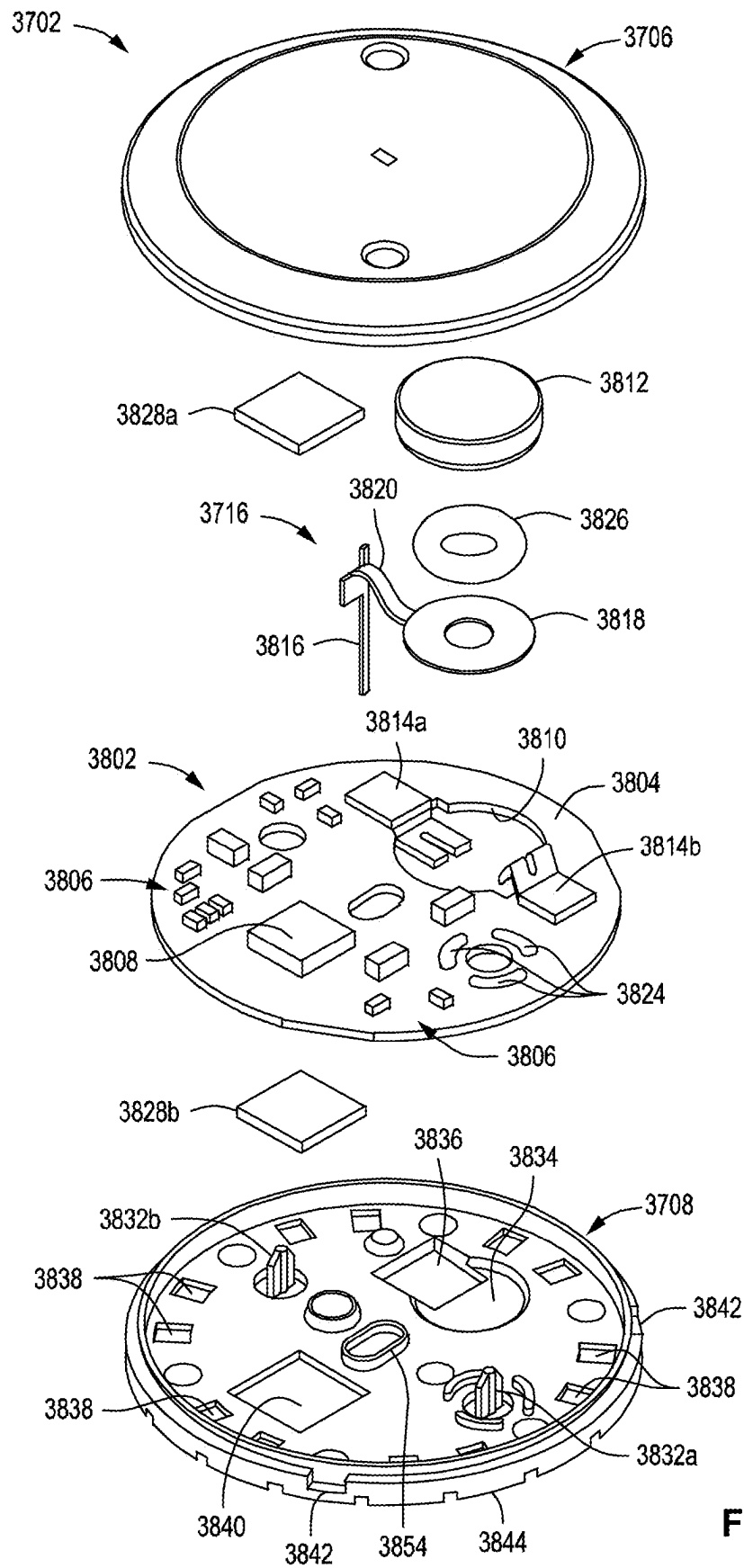


FIG. 7A

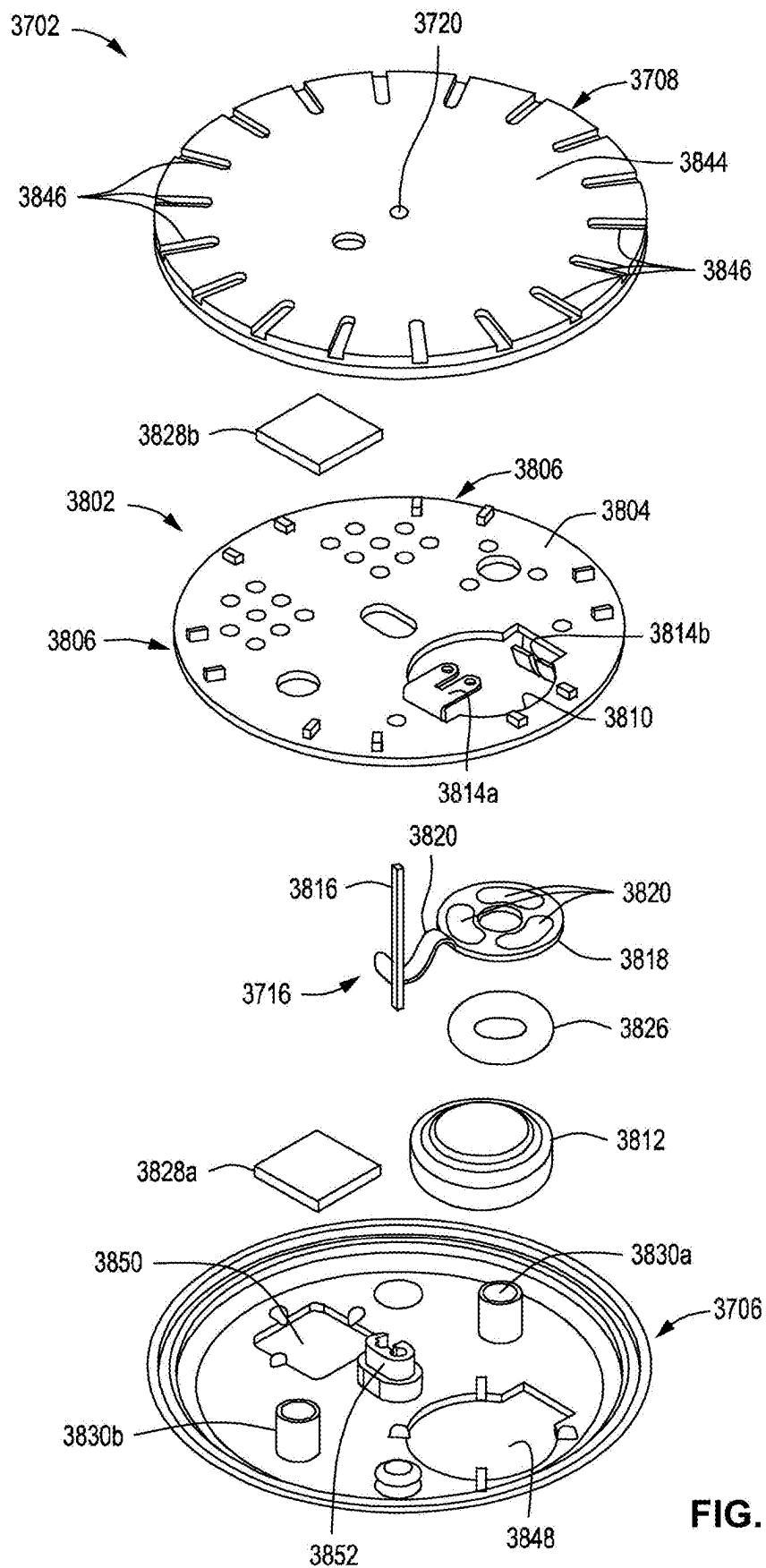


FIG. 7B

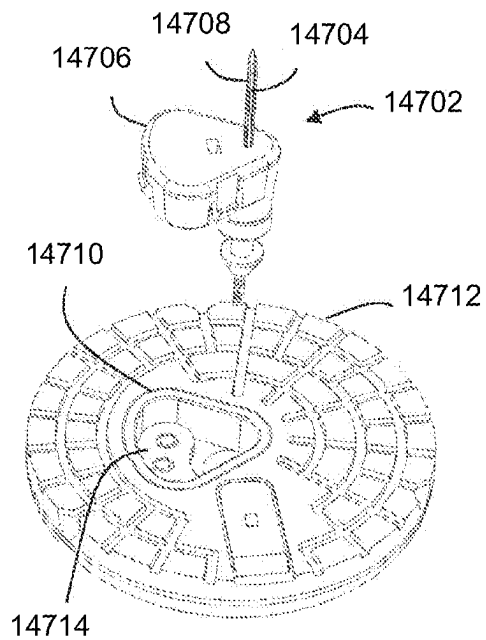


FIG. 8A

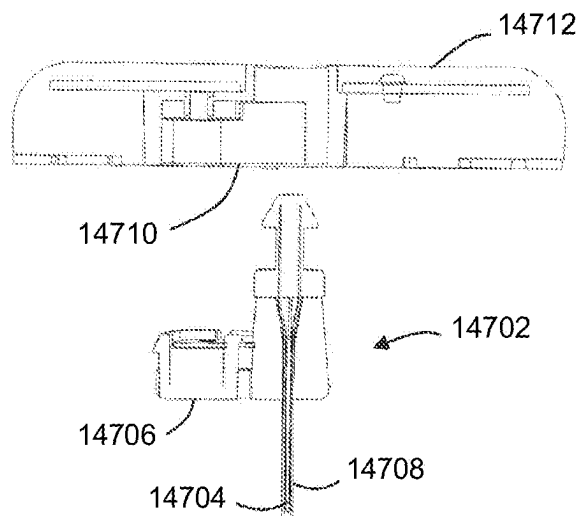


FIG. 8B

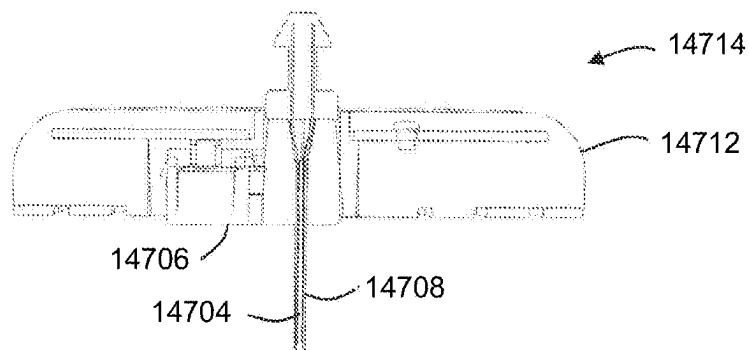


FIG. 8C

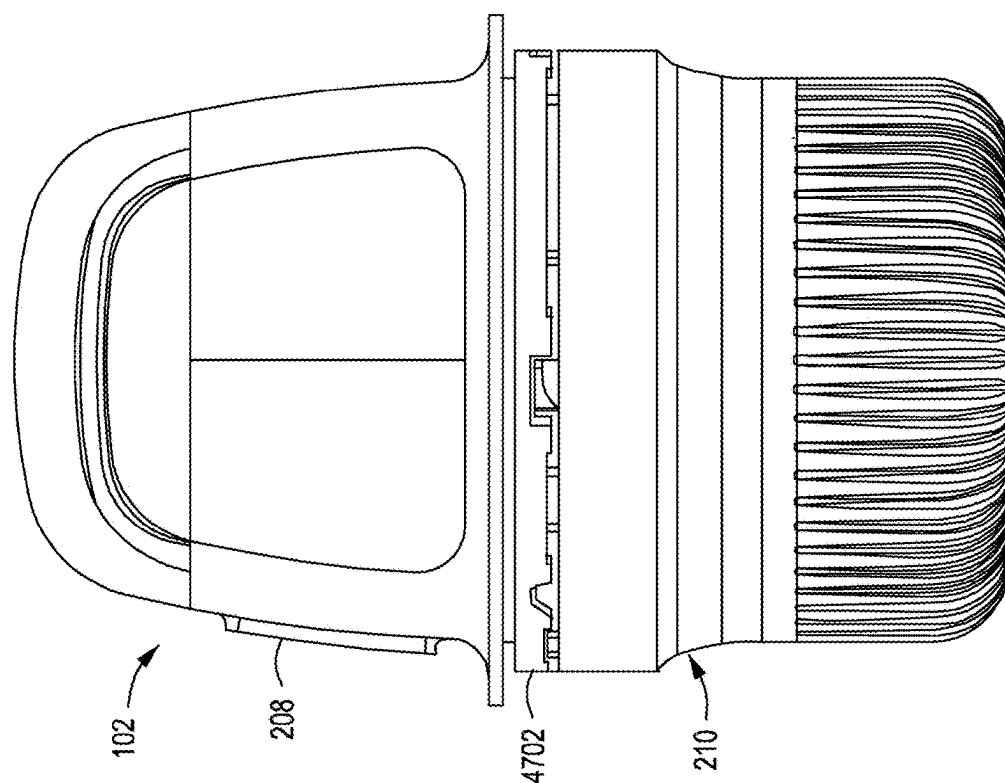


FIG. 9A

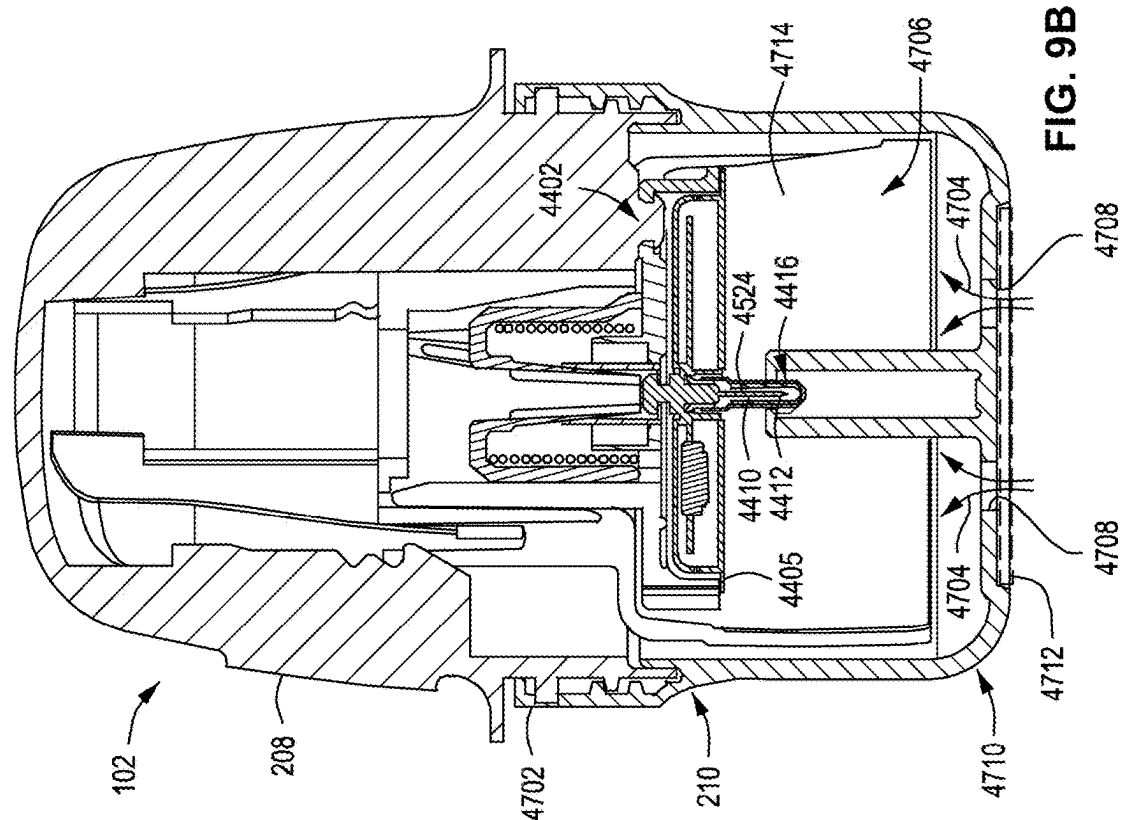


FIG. 9B

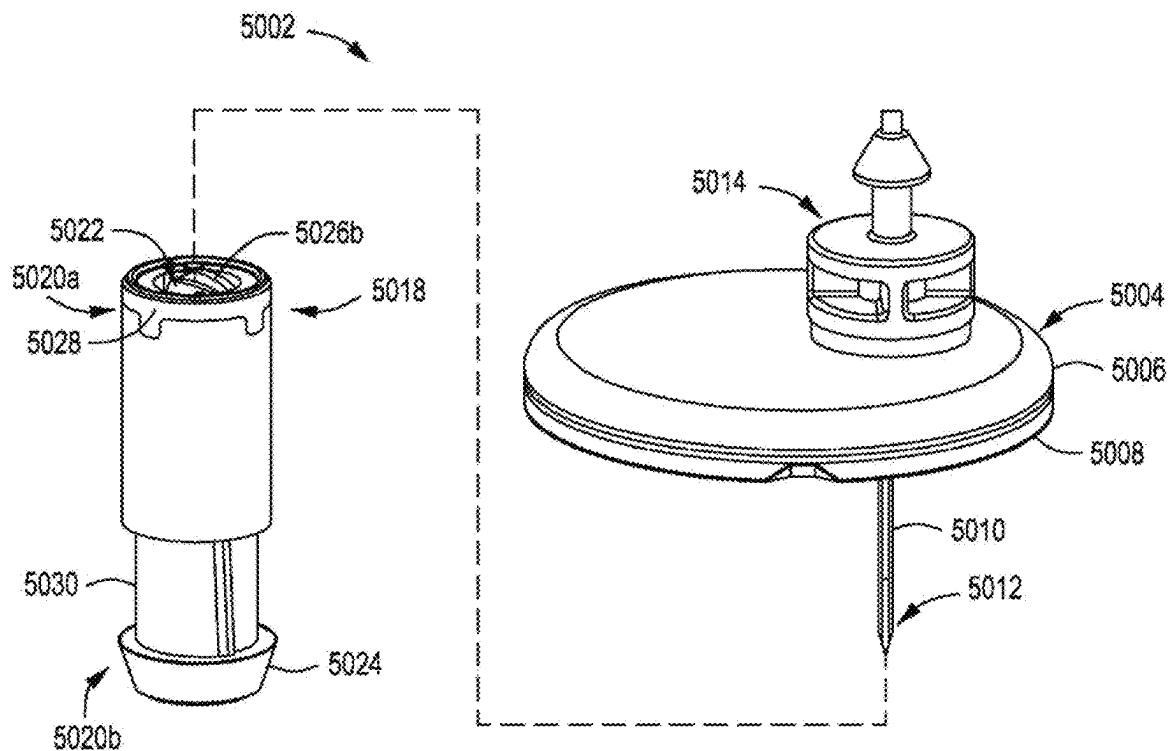


FIG. 10A

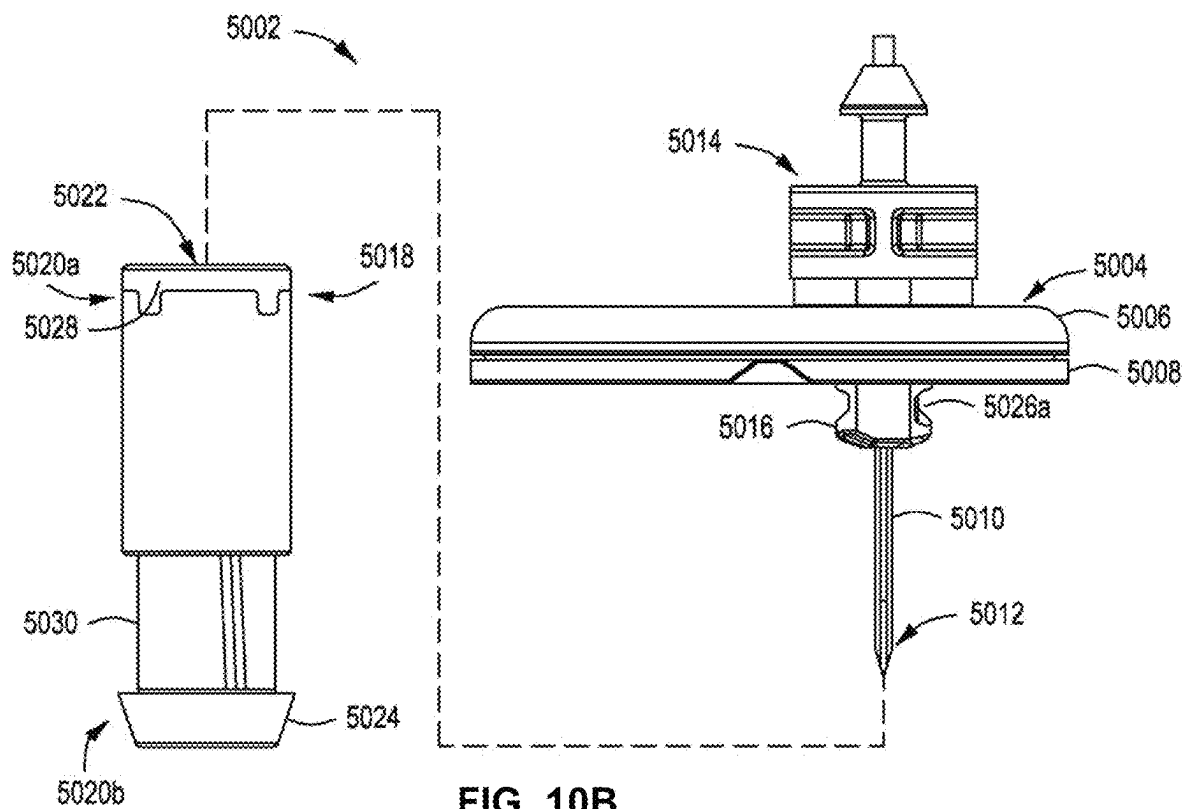


FIG. 10B

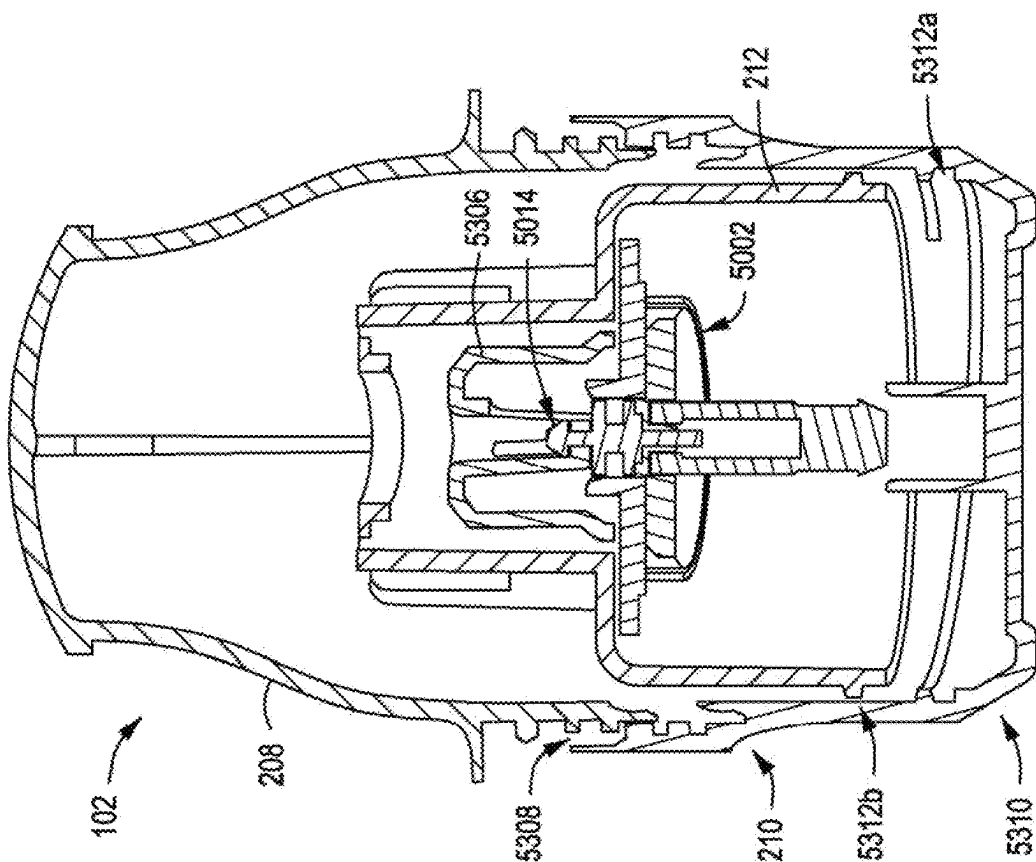


FIG. 11B

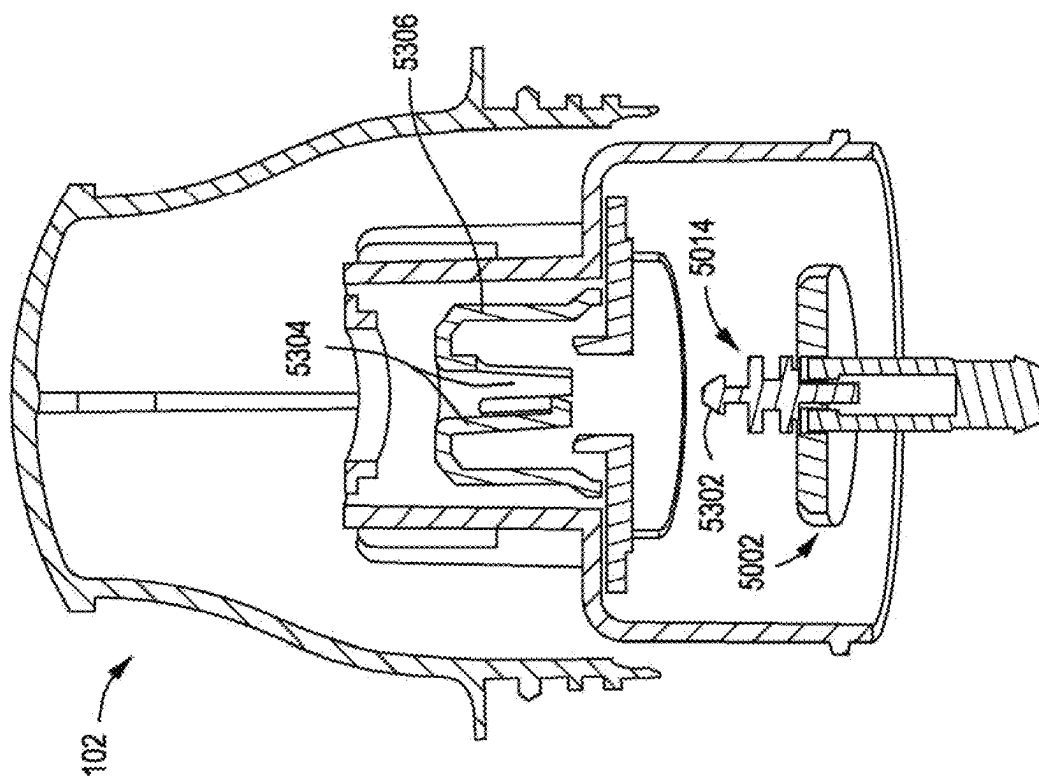


FIG. 11A

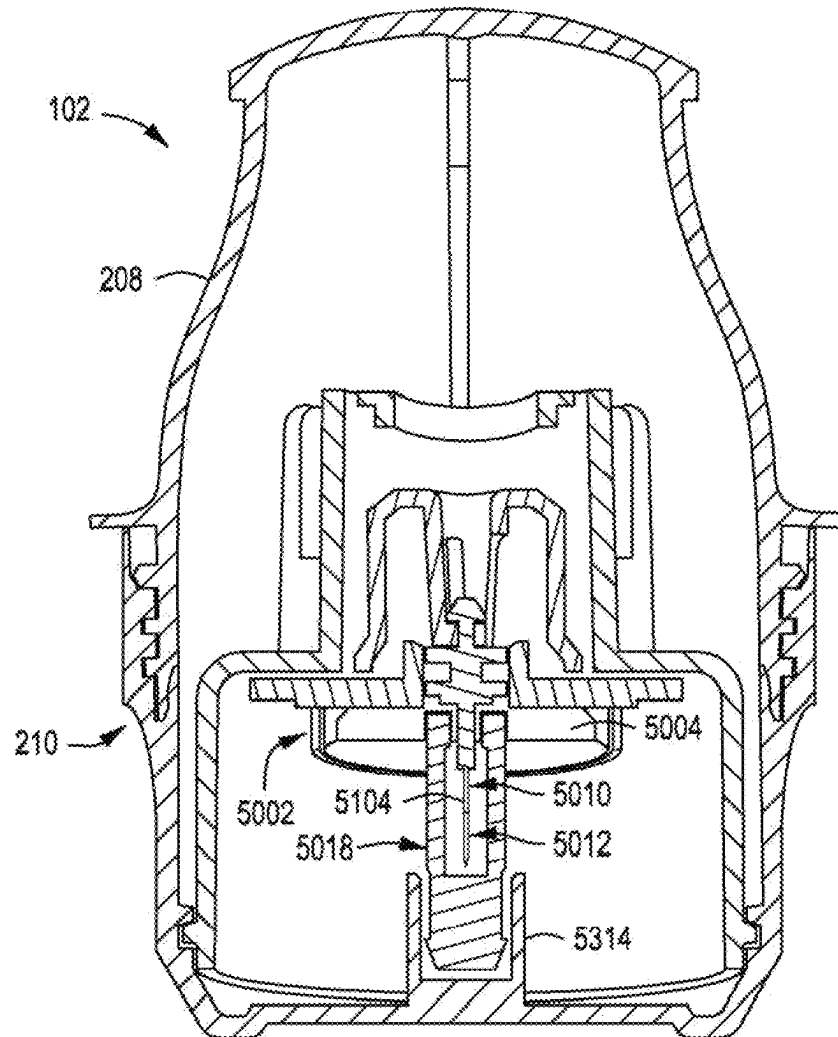


FIG. 11C

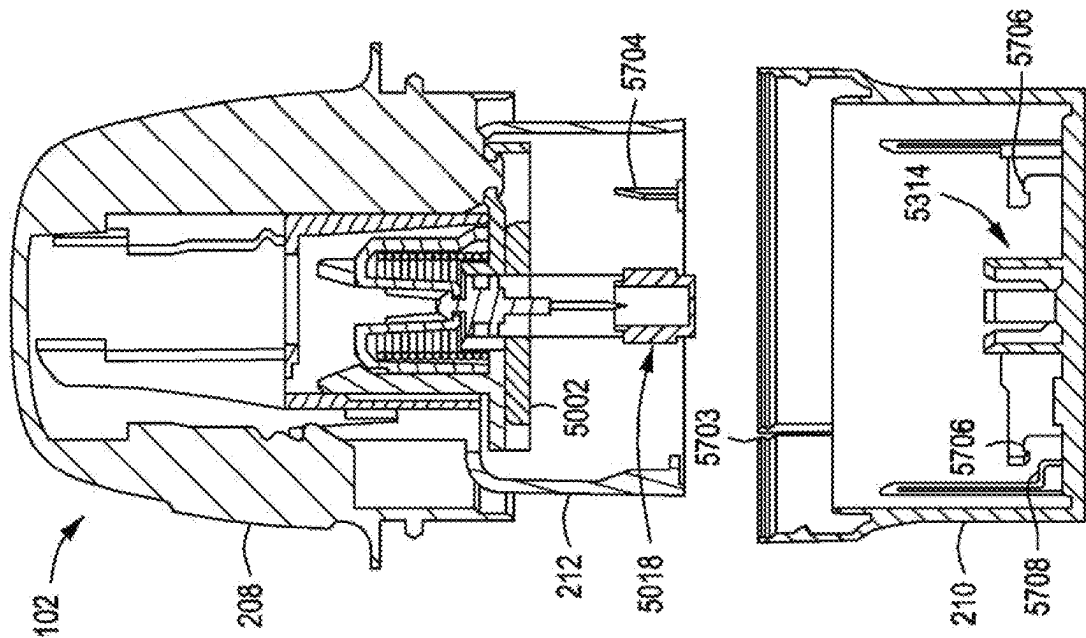


FIG. 12A

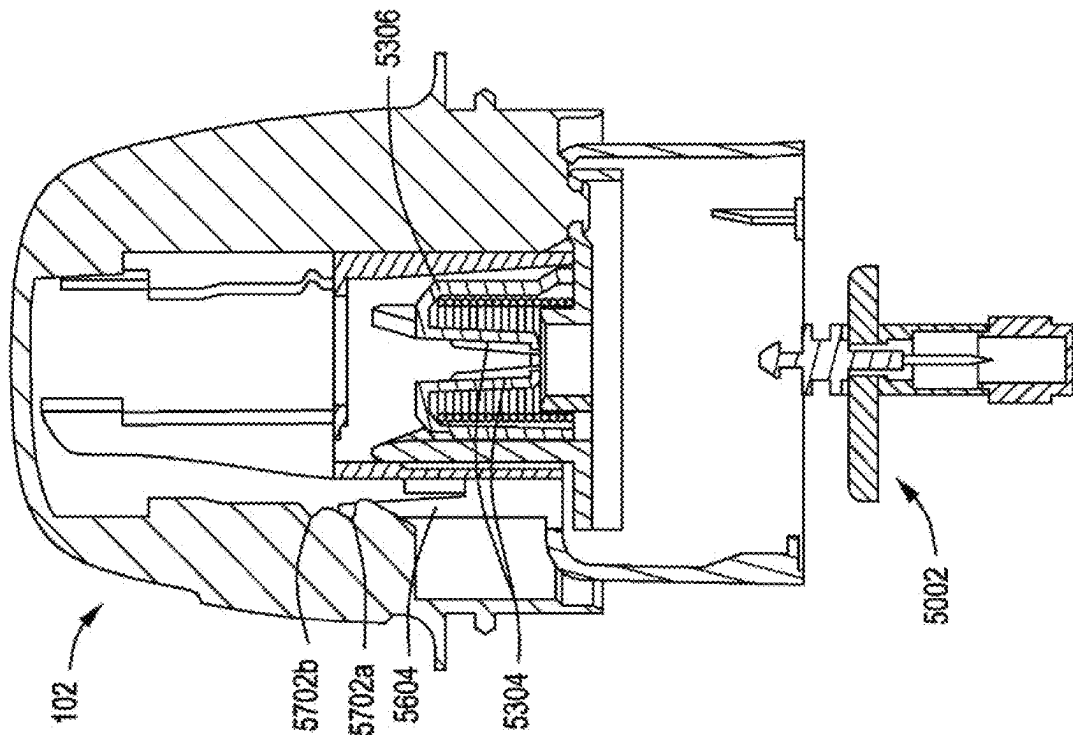


FIG. 12B

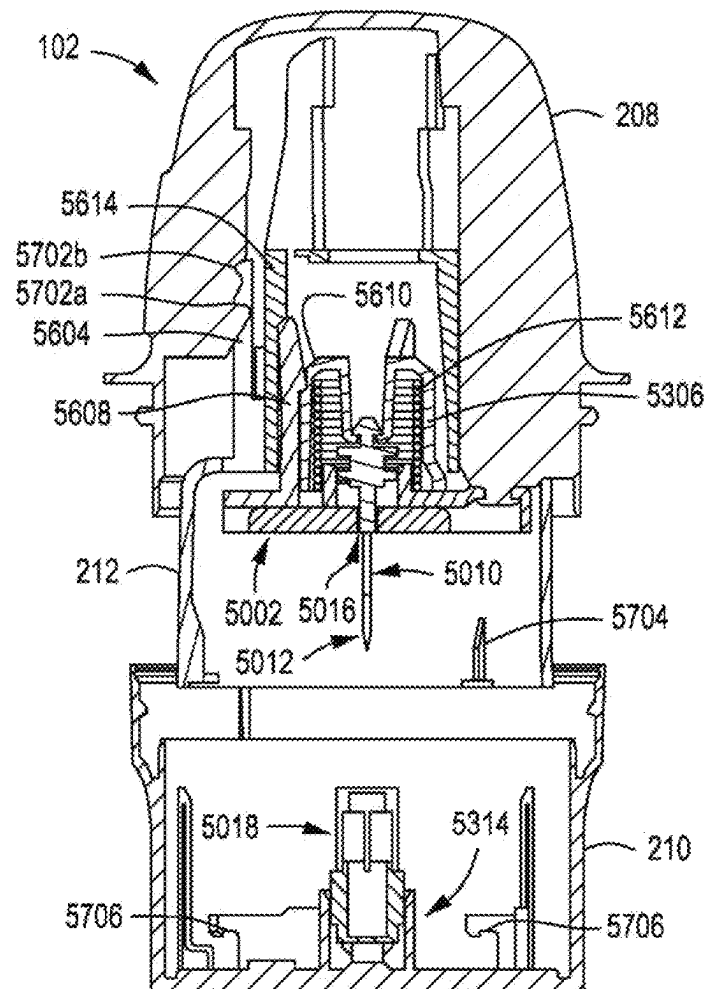


FIG. 12C

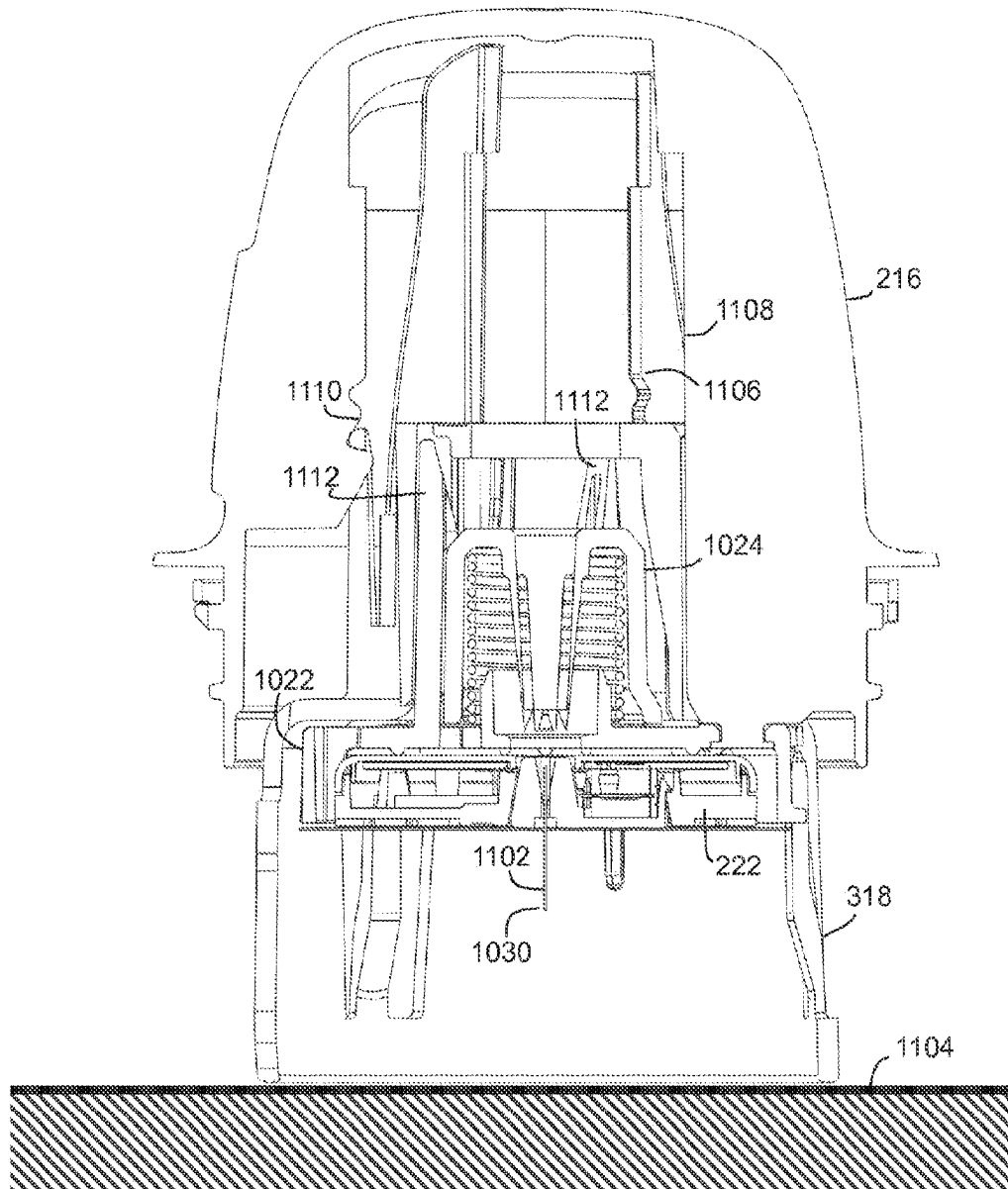


FIG. 13A

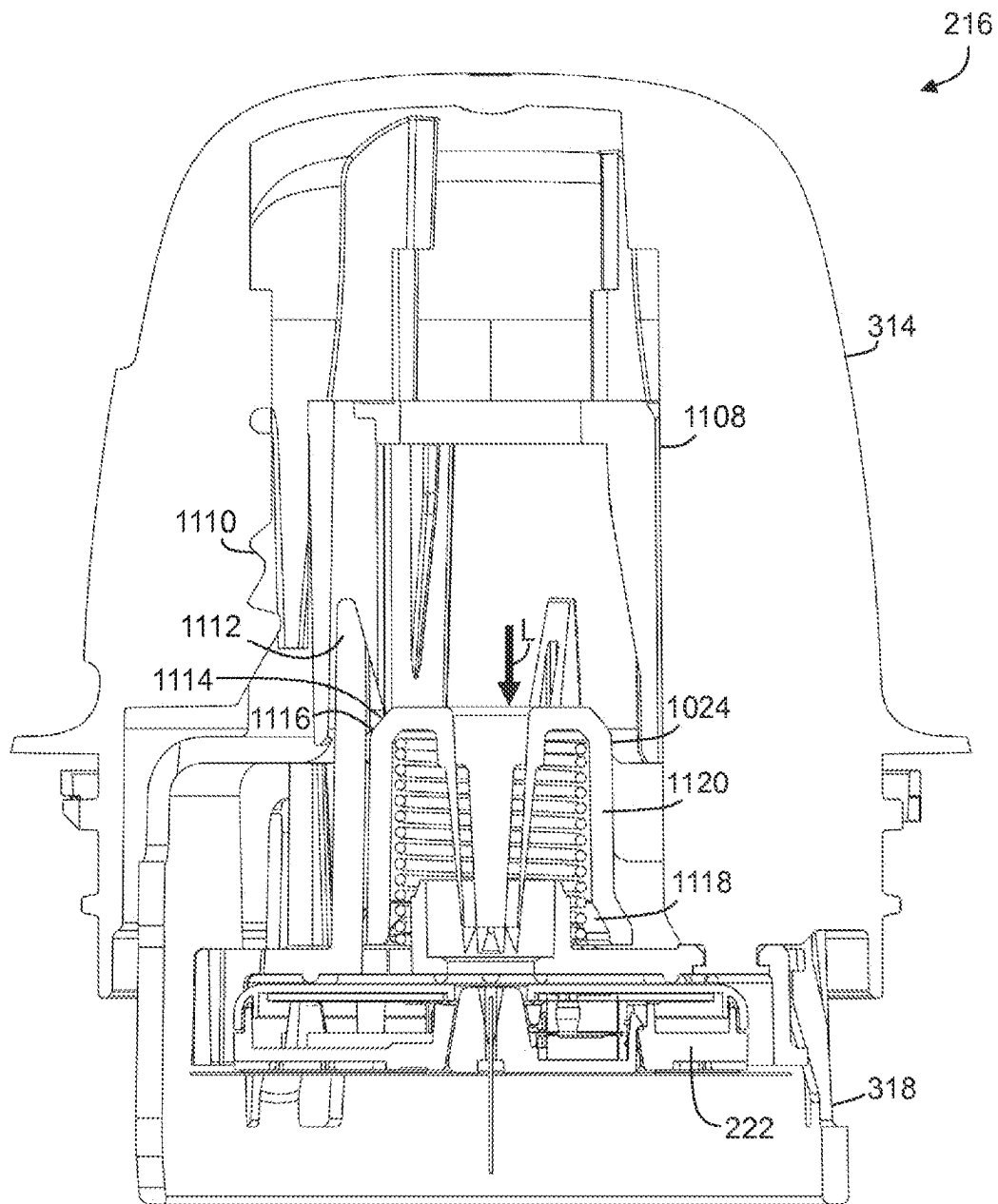


FIG. 13B

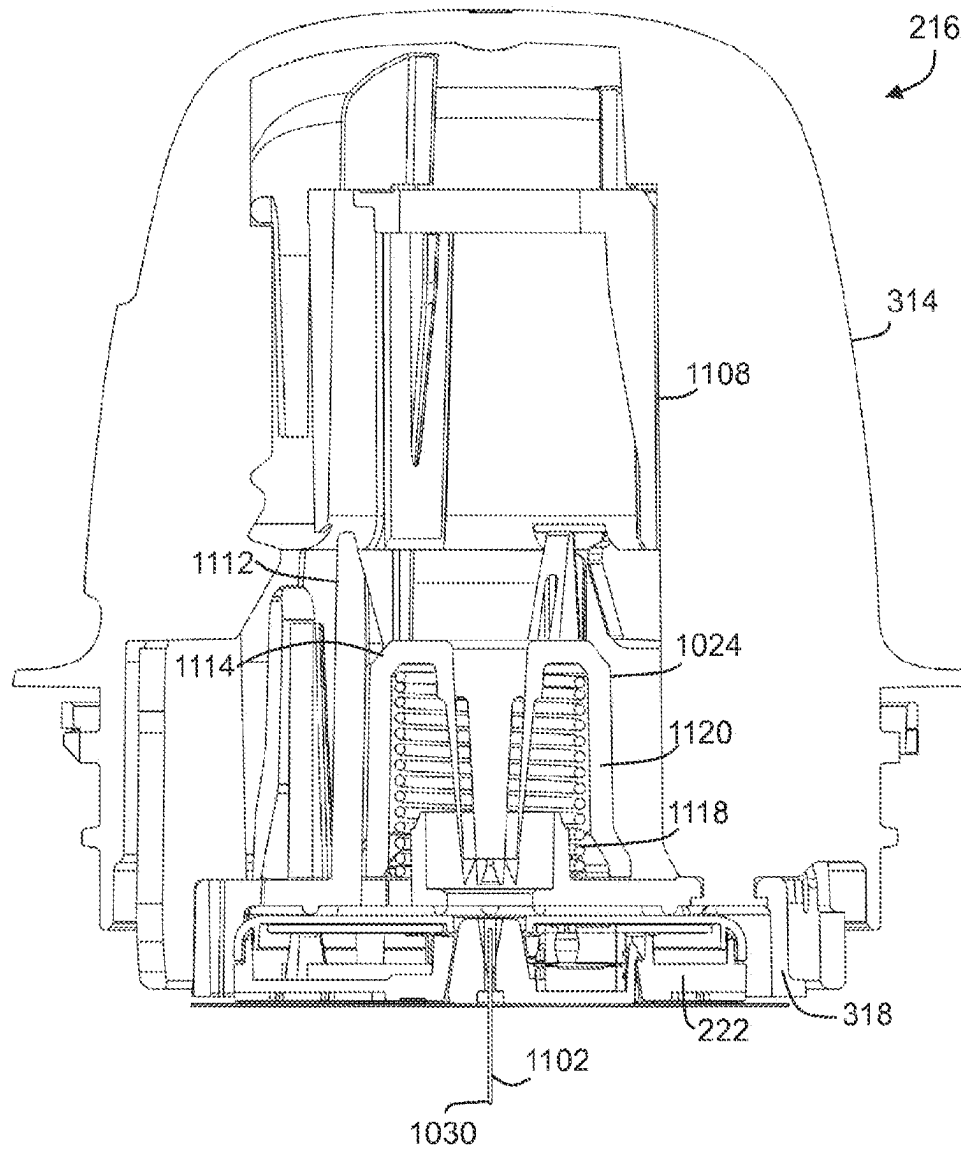


FIG. 13C

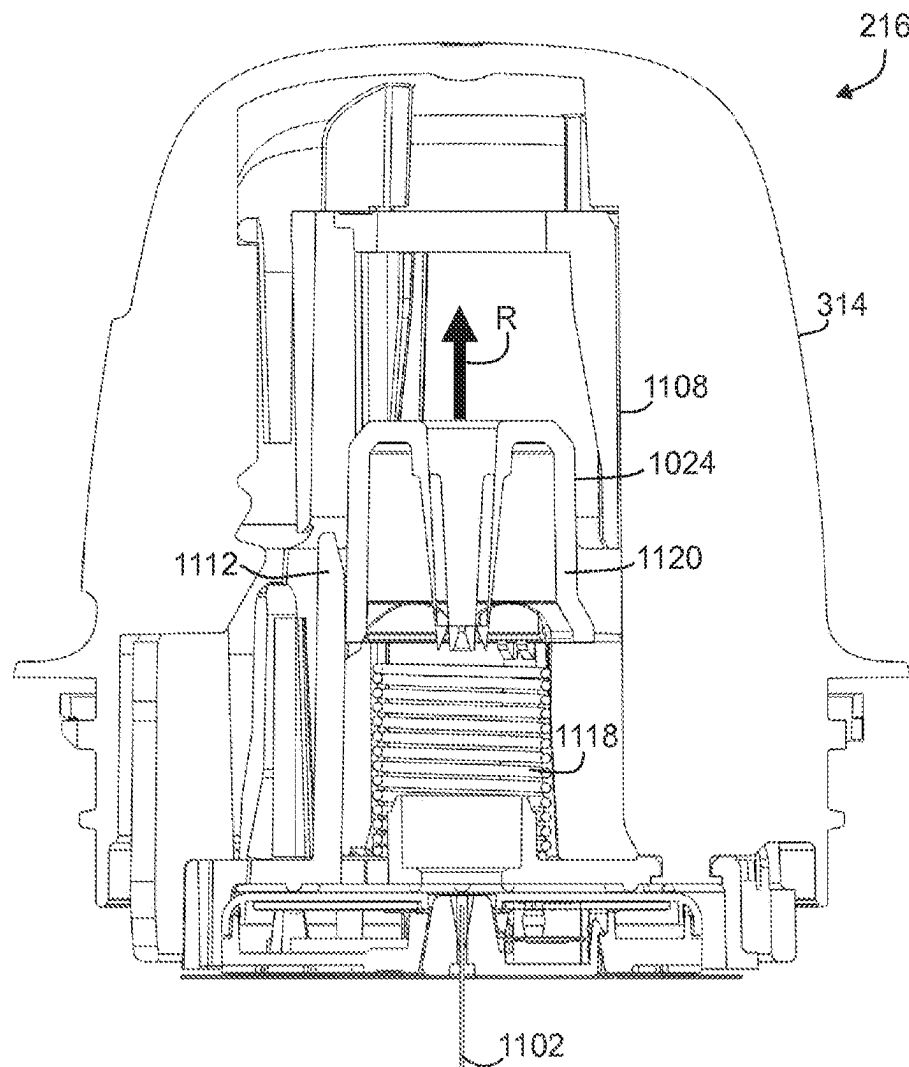


FIG. 13D

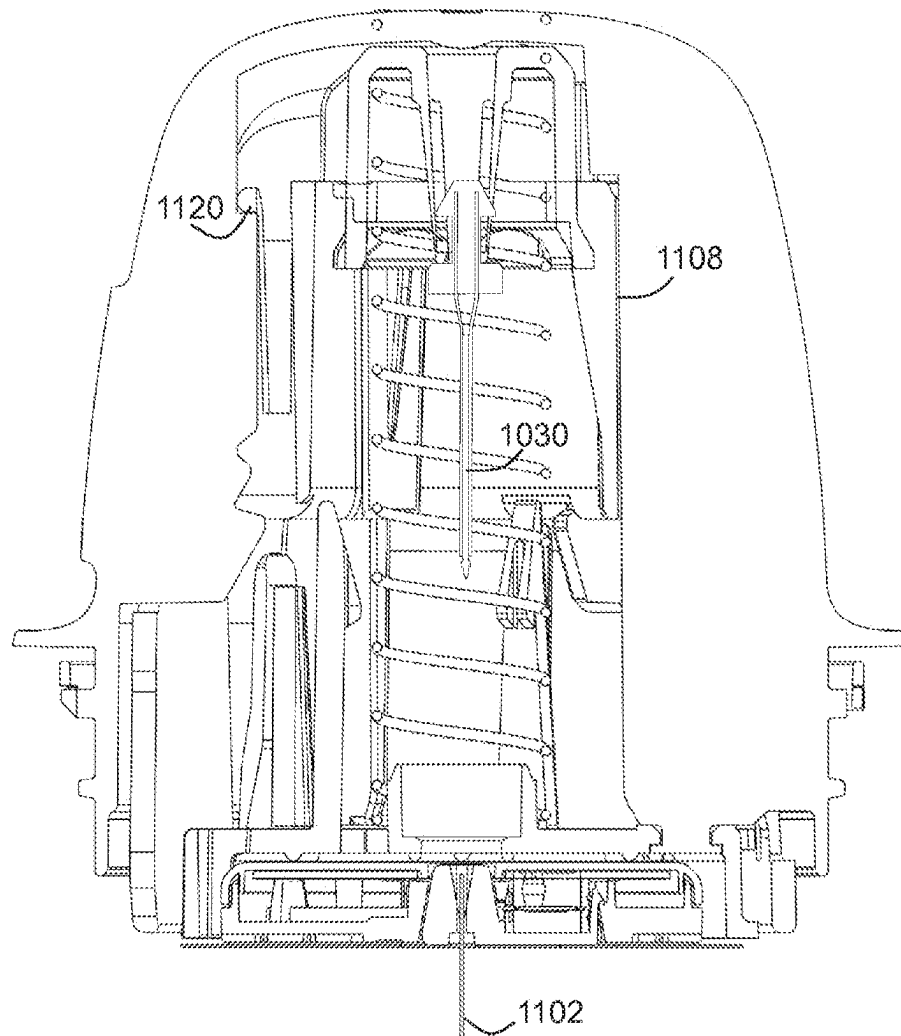


FIG. 13E

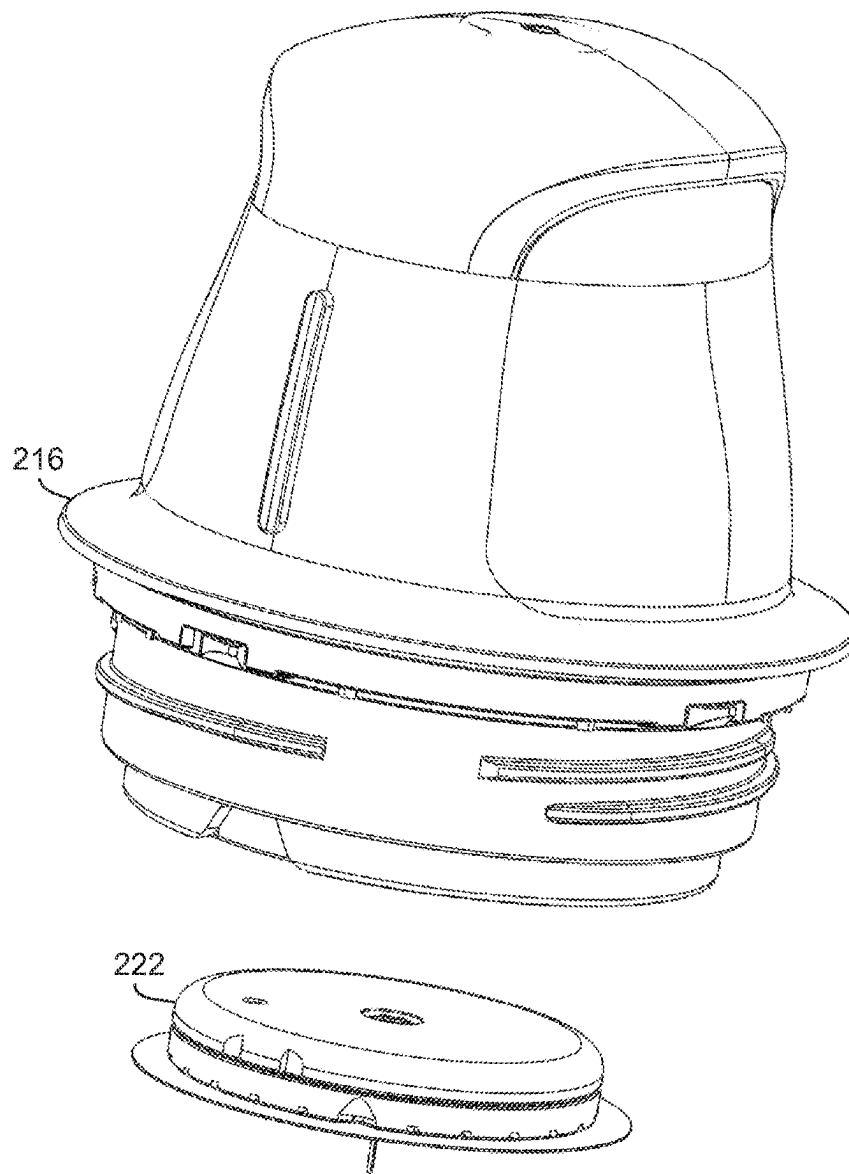
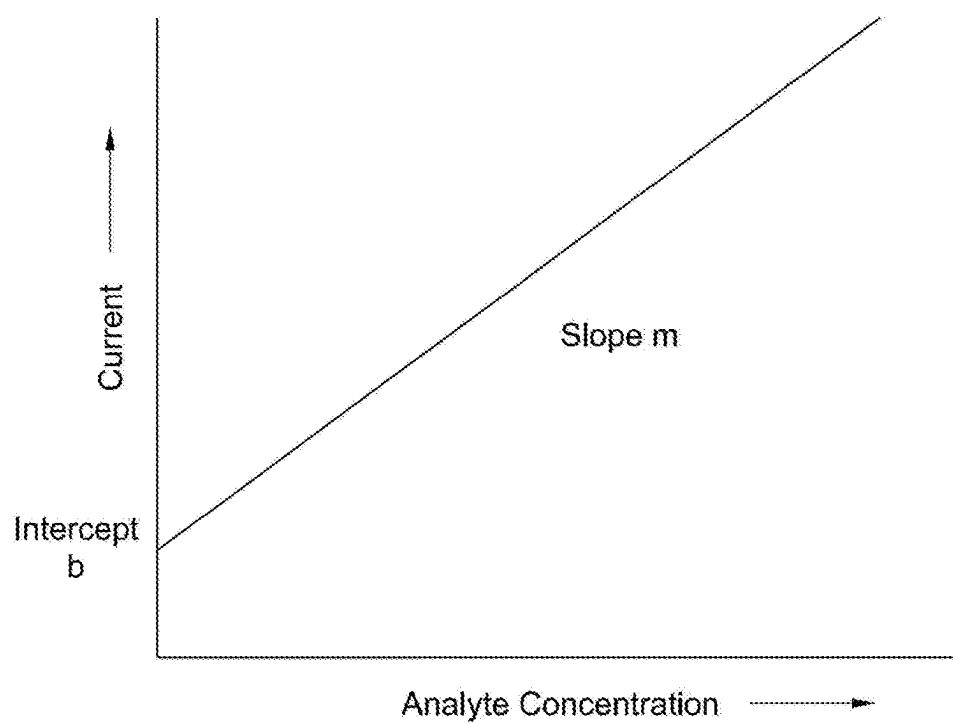
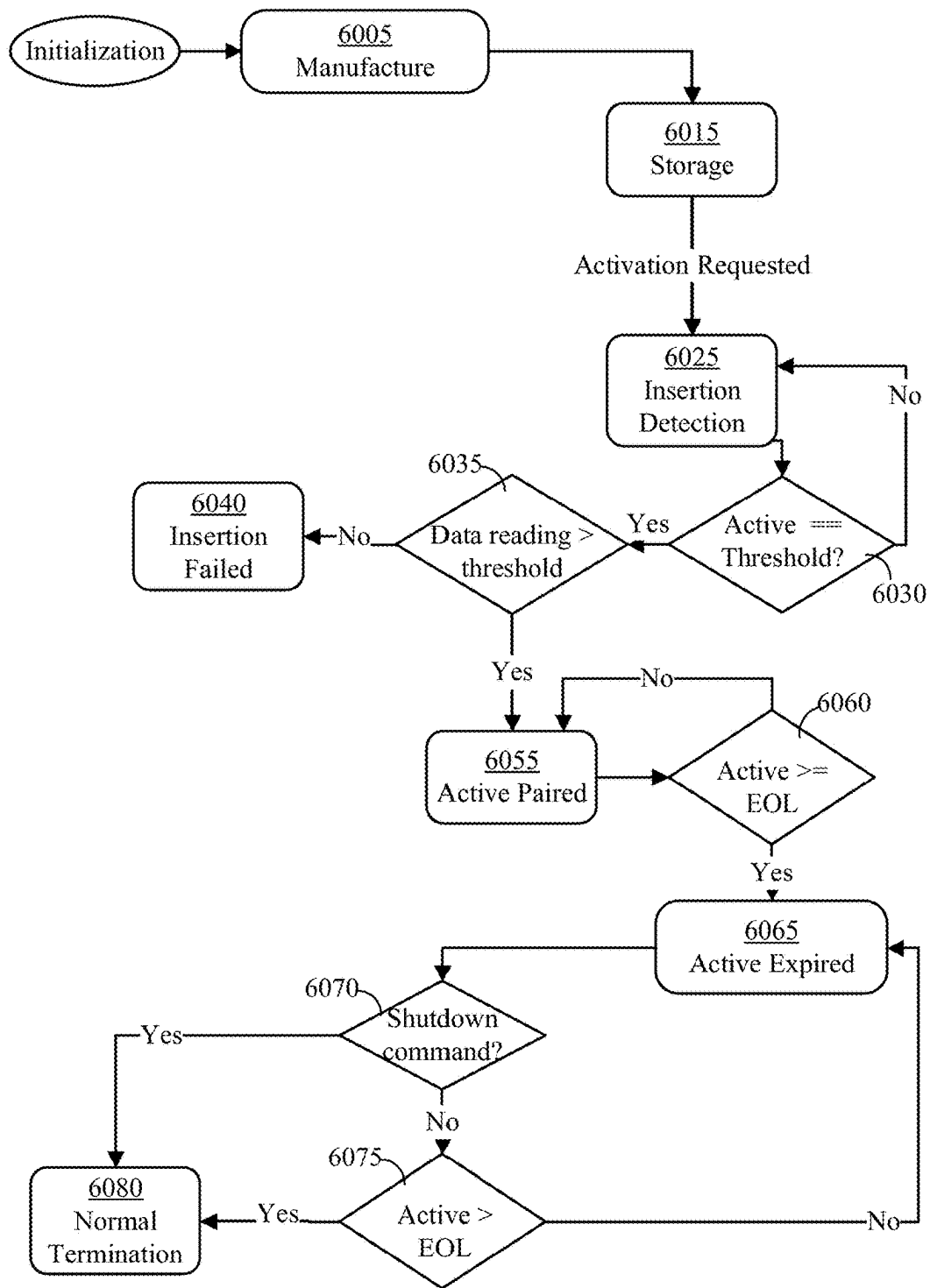


FIG. 13F

**FIG. 14**

6000**FIG. 15**

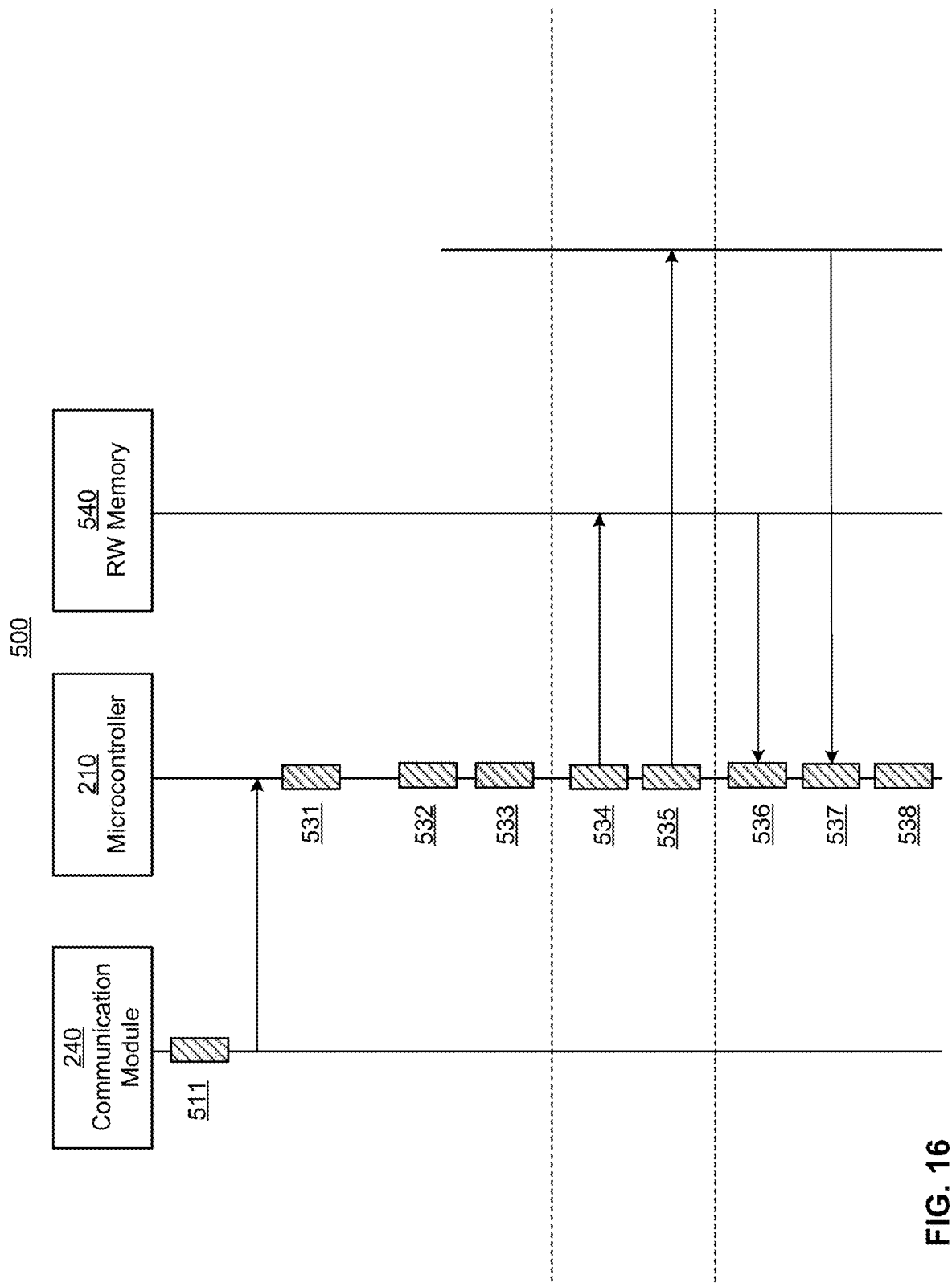


FIG. 16

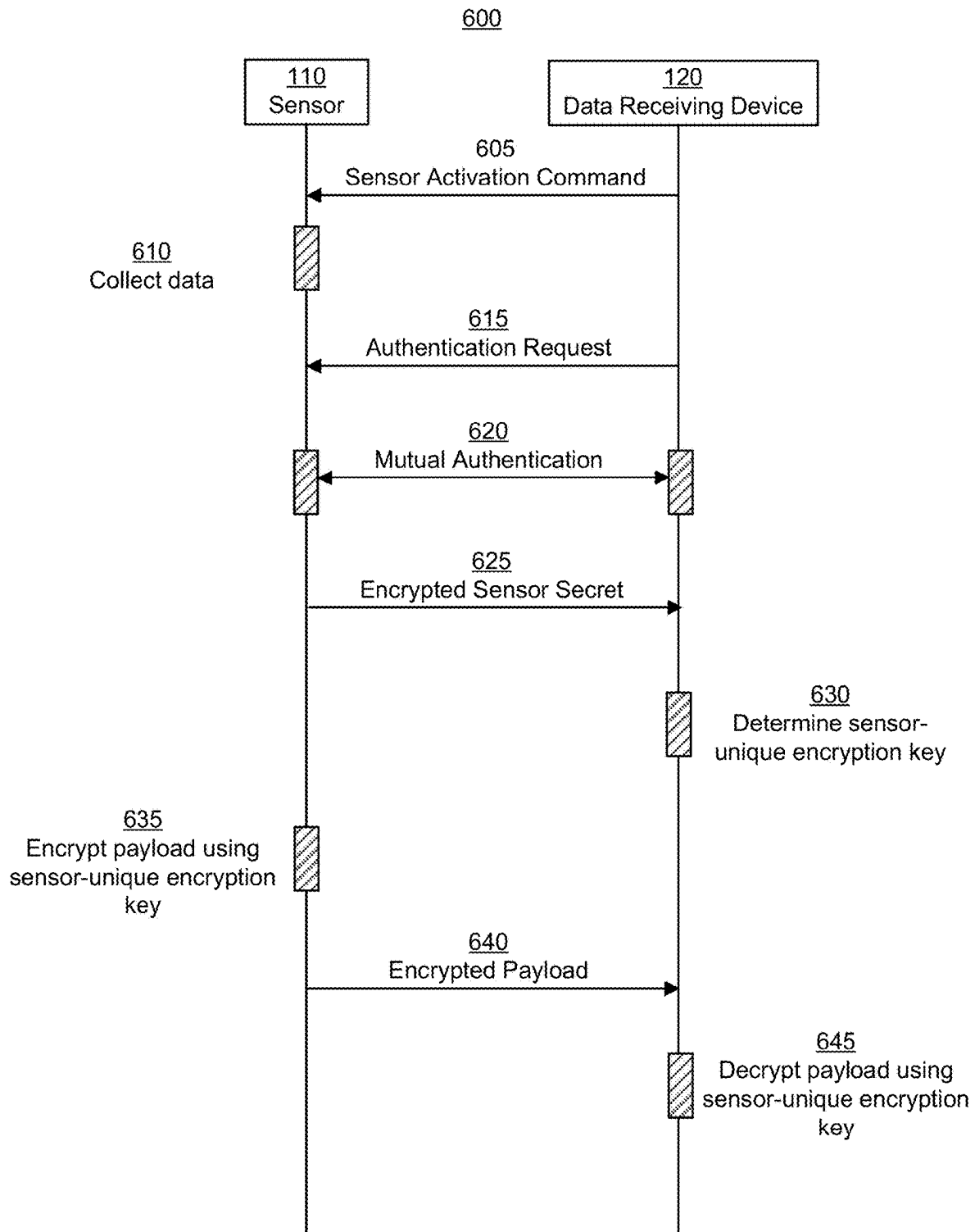


FIG. 17

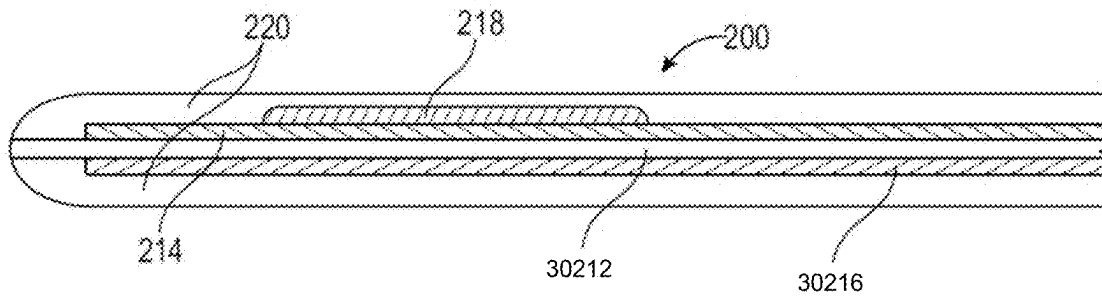


FIG. 18A

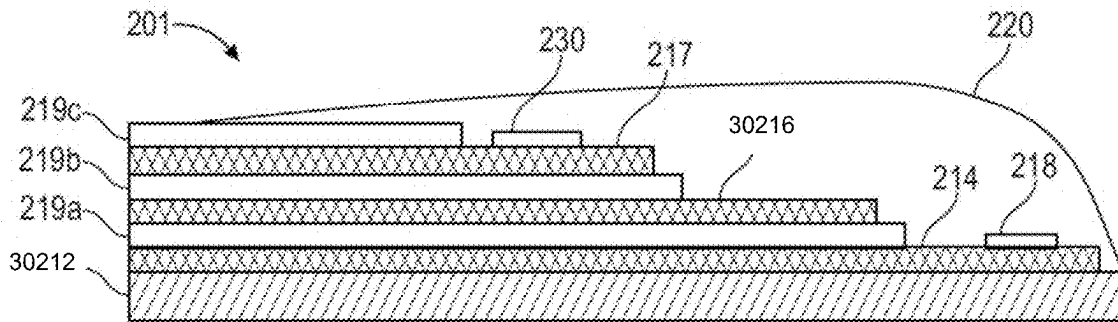


FIG. 18B

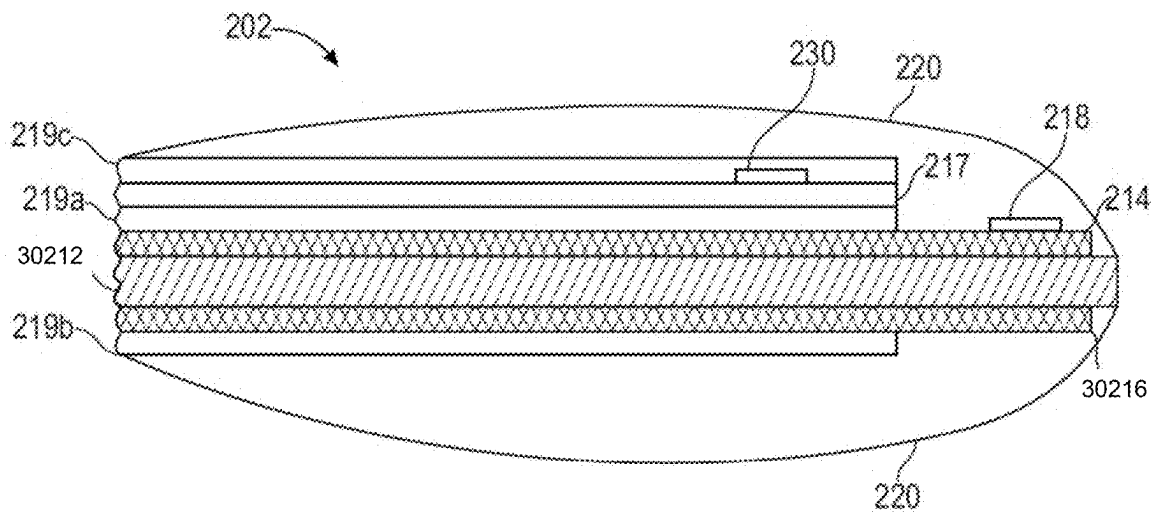


FIG. 18C

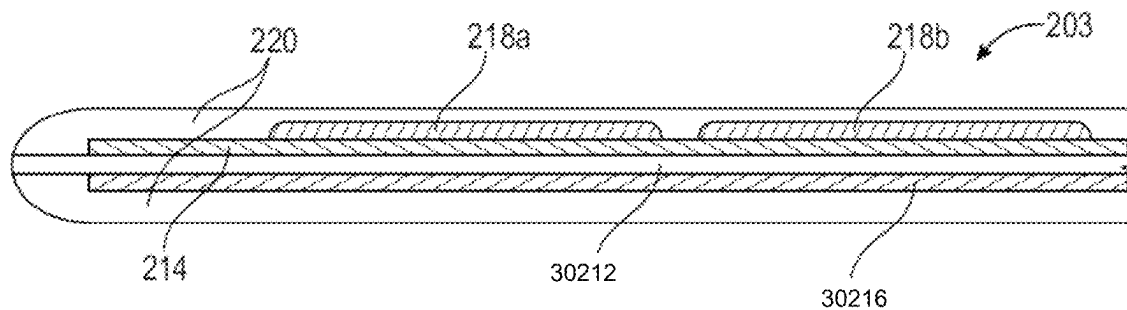


FIG. 19A

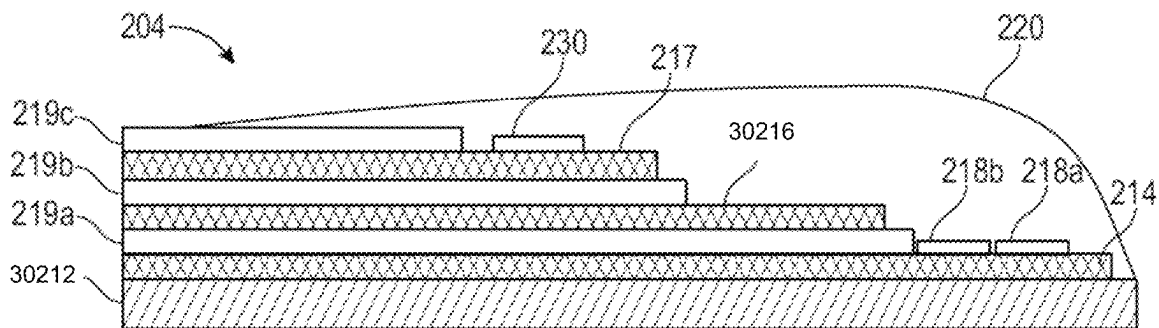


FIG. 19B

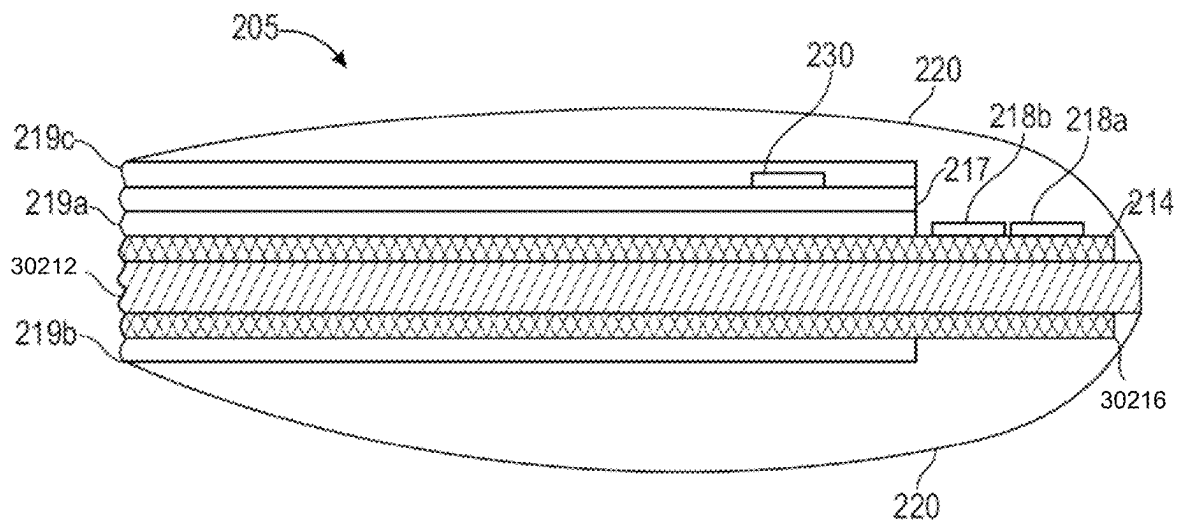


FIG. 19C

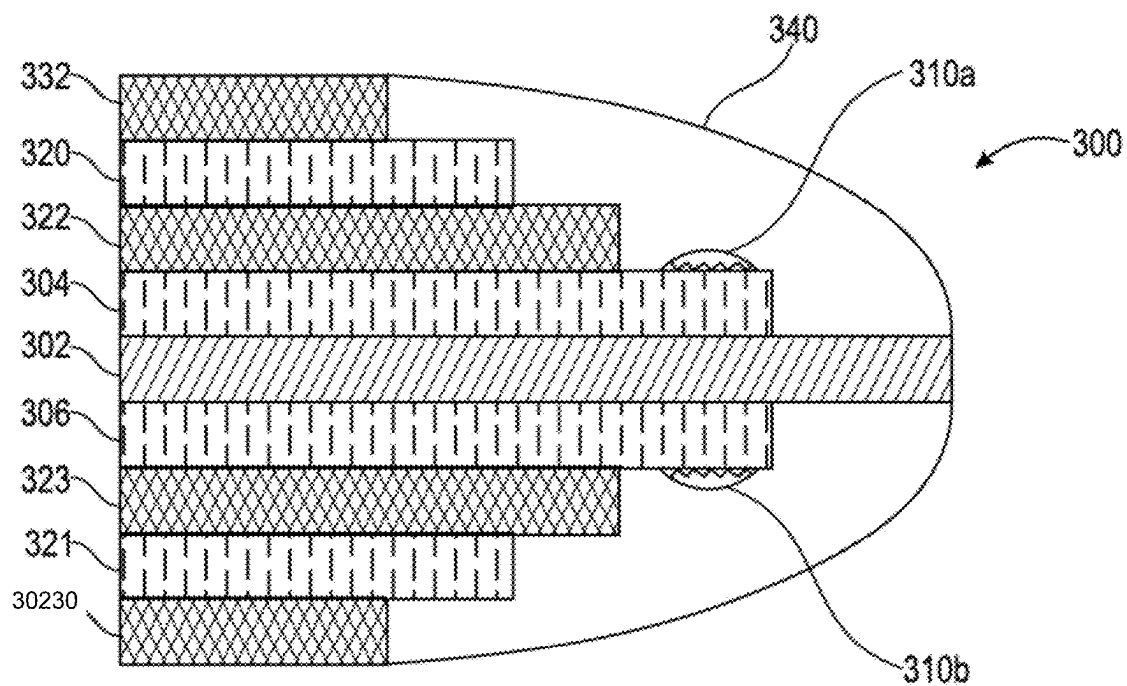


FIG. 20

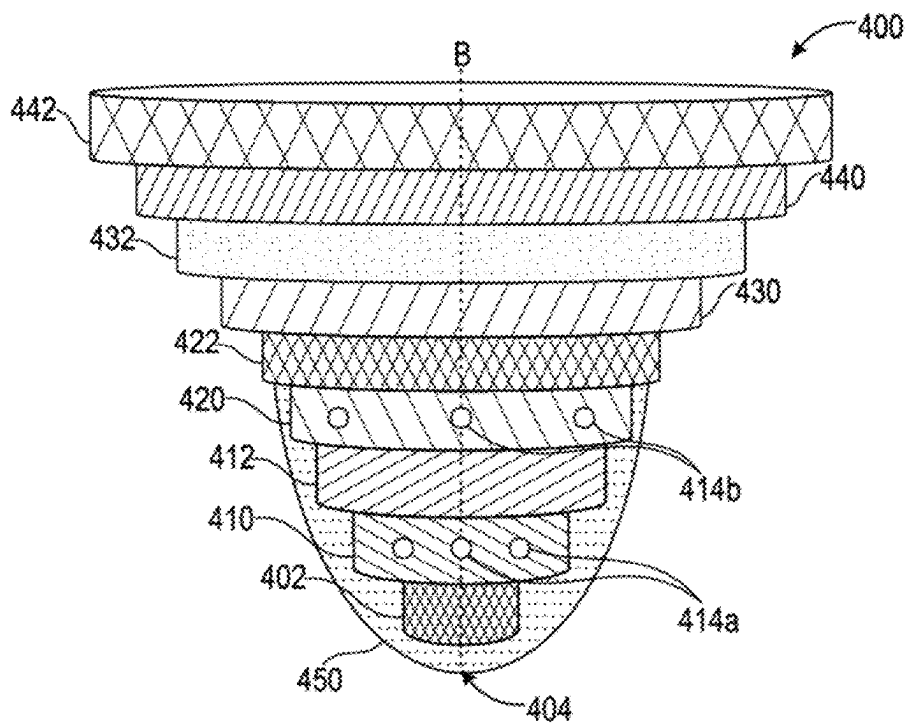


FIG. 21A

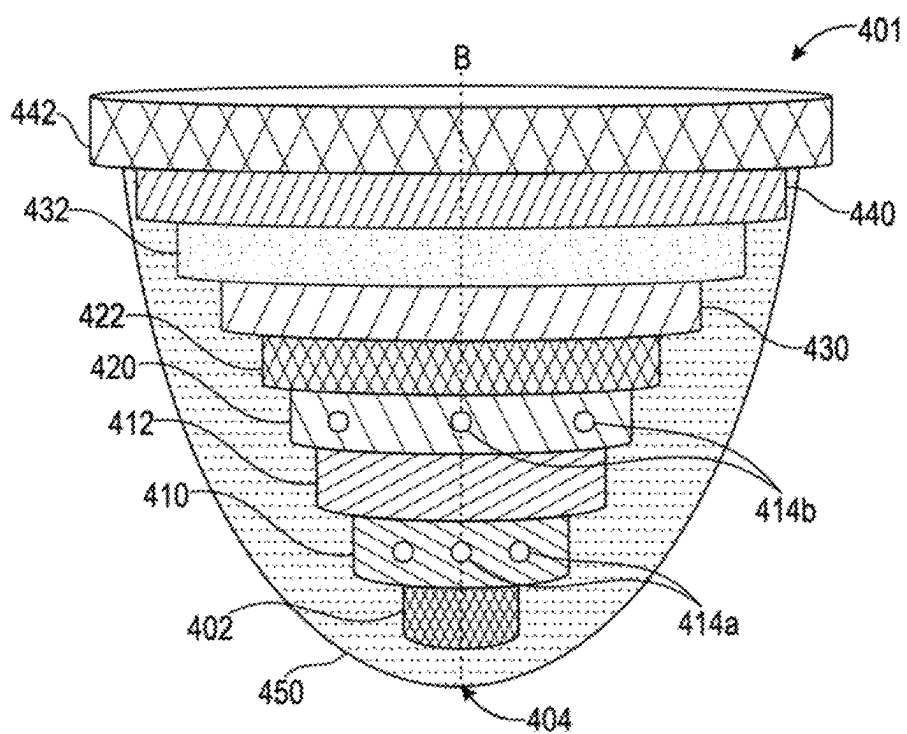


FIG. 21B

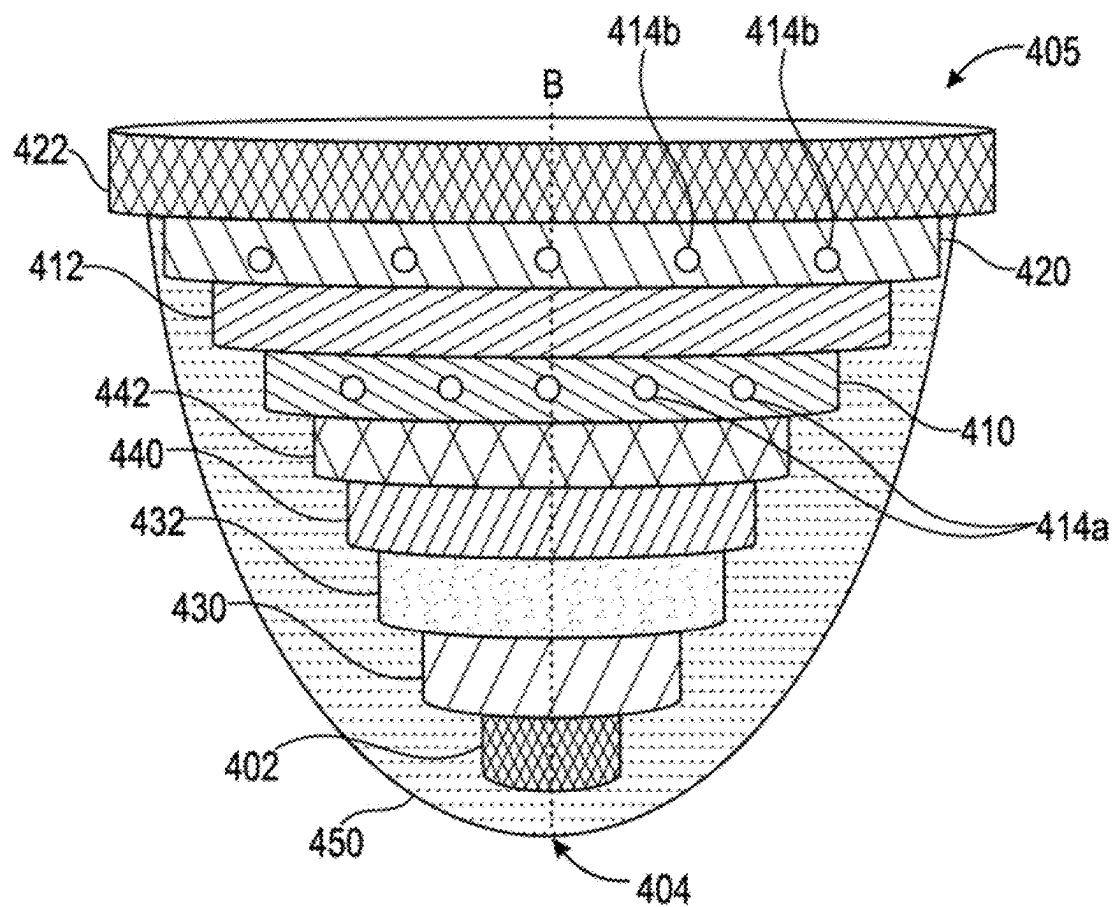


FIG. 21C

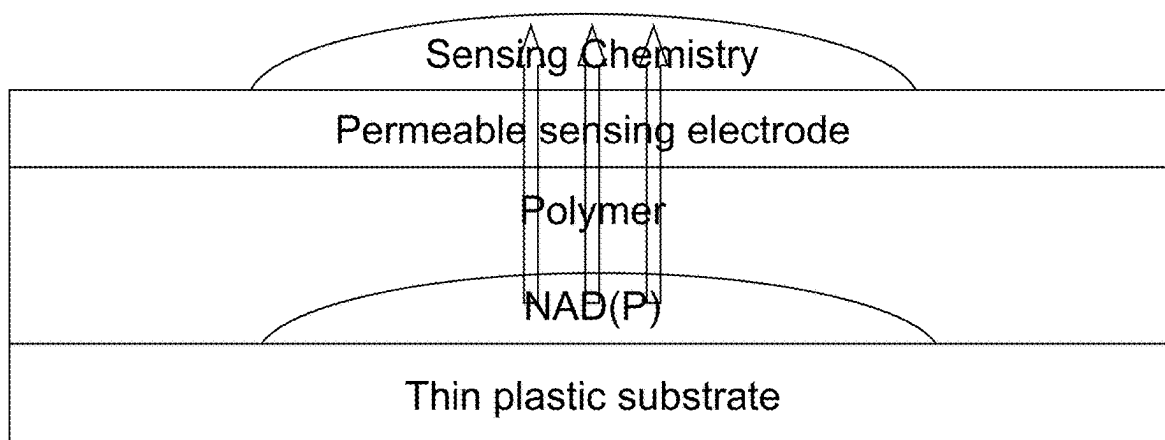


FIG. 22

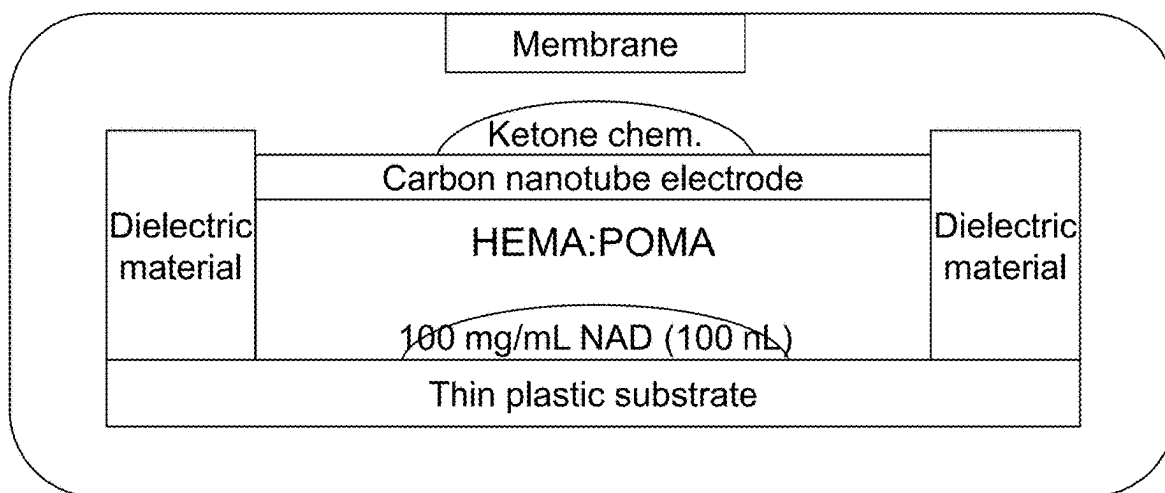
NAD depot

FIG. 23A

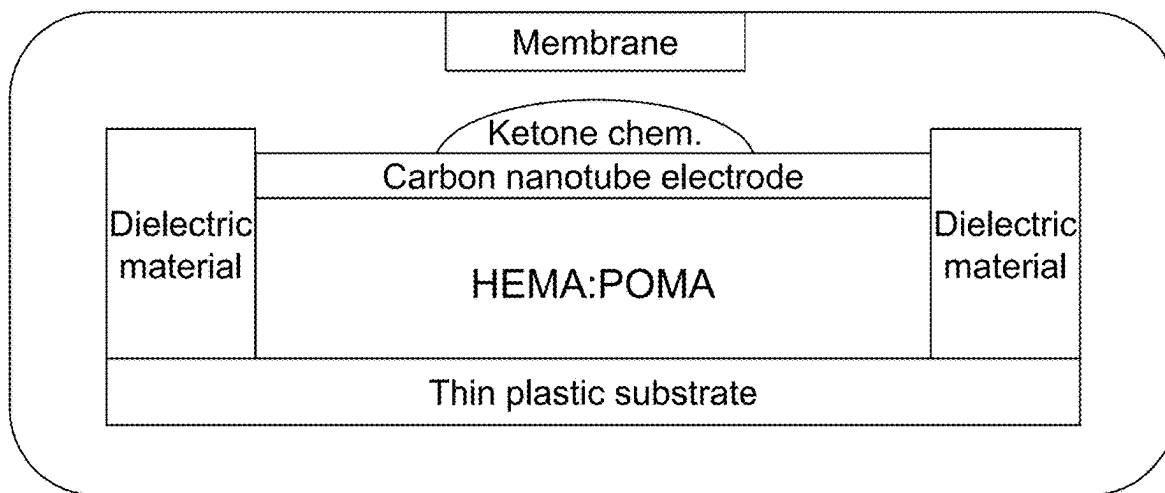
No NAD depot (control)

FIG. 23B

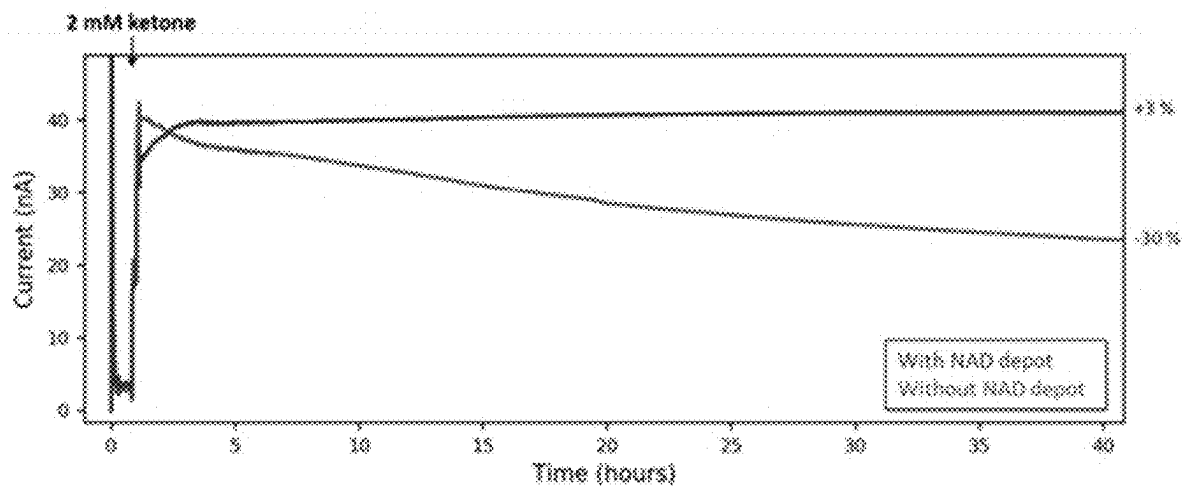


FIG. 24

1

NAD(P) DEPOT FOR NAD(P)-DEPENDENT ENZYME-BASED SENSORS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. Provisional Application No. 63/125,846, filed Dec. 15, 2020, the contents of which is incorporated by reference in its entirety.

FIELD

The subject matter described herein relates to analyte sensors comprising an NAD(P) depot and methods of using the same.

BACKGROUND

The detection of various analytes within an individual can sometimes be vital for monitoring the condition of their health as deviations from normal analyte levels can be indicative of a physiological condition. For example, monitoring glucose levels can enable people suffering from diabetes to take appropriate corrective action including administration of medicine or consumption of particular food or beverage products to avoid significant physiological harm. Other analytes can be desirable to monitor for other physiological conditions. In certain instances, it can be desirable to monitor more than one analyte to monitor single or multiple physiological conditions, particularly if a person is suffering from comorbid conditions that result in simultaneous dysregulation of two or more analytes in combination with one another.

Analyte monitoring in an individual can take place periodically or continuously over a period of time. Periodic analyte monitoring can take place by withdrawing a sample of bodily fluid, such as blood or urine, at set time intervals and analyzing *ex vivo*. Periodic, *ex vivo* analyte monitoring can be sufficient to determine the physiological condition of many individuals. However, *ex vivo* analyte monitoring can be inconvenient or painful in some instances. Moreover, there is no way to recover lost data if an analyte measurement is not obtained at an appropriate time.

Continuous analyte monitoring can be conducted using one or more sensors that remain at least partially implanted within a tissue of an individual, such as dermally, subcutaneously or intravenously, so that analyses can be conducted *in vivo*. Implanted sensors can collect analyte data on-demand, at a set schedule, or continuously, depending on an individual's particular health needs and/or previously measured analyte levels. Analyte monitoring with an *in vivo* implanted sensor can be a more desirable approach for individuals having severe analyte dysregulation and/or rapidly fluctuating analyte levels, although it can also be beneficial for other individuals as well. Since implanted analyte sensors often remain within a tissue of an individual for an extended period of time, it can be highly desirable for such analyte sensors to be made from stable materials exhibiting a high degree of biocompatibility.

However, implantable sensors can be plagued by short life spans or reduced sensitivity. For example, many implantable sensors use enzymes for continuous monitoring of analyte levels *in vivo* and many of these enzymes rely on coenzymes for activity. For example, nicotinamide adenine dinucleotide (NAD) and nicotinamide adenine dinucleotide phosphate (NADP) are two of the most important coenzymes found in living cells, and are frequently required for the activity of

2

enzymes such as dehydrogenases that are found in implantable sensors. The amount of NAD or NADP available for use by the enzymes present in implantable sensors can affect the sensitivity of the sensor to accurately monitor analyte levels *in vivo*. Under certain circumstances, exogenous NAD or NADP may not be present in sufficient quantities to support sensor operation or, even if sufficient exogenous quantities exists, such molecules are too large to readily diffuse to the area of the sensor that retains the enzyme dependent on NAD or NADP, which can lead to reduced sensitivity. Accordingly, there is a need in the art for sensors that retain sensitivity over longer periods of time.

SUMMARY

The purpose and advantages of the disclosed subject matter will be set forth in and are apparent from the description that follows, as well as will be learned by practice of the disclosed subject matter. Additional advantages of the disclosed subject matter will be realized and attained by the devices particularly pointed out in the written description and claims hereof, as well as from the appended drawings.

To achieve these and other advantages and in accordance with the purpose of the disclosed subject matter, as embodied and broadly described, the disclosed subject matter provides analyte sensors that include an internal supply of NAD(P). For example, and not by way of limitation, an analyte sensor of the present disclosure includes an internal supply of NAD(P), a permeable polymer that overcoats the internal supply of NAD(P), at least a first working electrode that is disposed upon a surface of the permeable polymer, an analyte-responsive active area disposed upon a surface of the first working electrode and, optionally, a mass transport limiting membrane permeable to the analyte that overcoats at least the analyte-responsive area.

In certain embodiments, the analyte is selected from the group consisting of glucose, a ketone, an alcohol, lactate and a combination thereof. In certain embodiments, the analyte is glucose. In certain embodiments, the analyte is a ketone. In certain embodiments, the analyte is lactate. In certain embodiments, the analyte is an alcohol, e.g., ethanol.

In certain embodiments, the first working electrode is a permeable working electrode. In certain embodiments, the permeable working electrode comprises a carbon nanotube.

In certain embodiments, the analyte-responsive active area comprises an NAD(P)-dependent enzyme. In certain embodiments, the NAD(P)-dependent enzyme is an NAD(P)-dependent dehydrogenase. In certain embodiments, the NAD(P)-dependent enzyme present within the analyte-responsive active area is β -hydroxybutyrate dehydrogenase. In certain embodiments, the NAD(P)-dependent enzyme present within the analyte-responsive active area is glucose dehydrogenase. In certain embodiments, the NAD(P)-dependent enzyme present within the analyte-responsive active area is a lactate dehydrogenase. In certain embodiments, the NAD(P)-dependent enzyme present within the analyte-responsive active area is an alcohol dehydrogenase. In certain embodiments, the NAD(P)-dependent enzyme present within the analyte-responsive active area is diaphorase.

In certain embodiments, the permeable polymer comprises a poly(propylene glycol)-based polymer. In certain embodiments, the permeable polymer comprises poly(propylene glycol) methacrylate and/or 2-hydroxyethyl methacrylate.

3

In certain embodiments, the analyte-responsive active area further includes a diaphorase. In certain embodiments, the analyte-responsive active area further includes a redox mediator.

In certain embodiments, an analyte sensor of the present disclosure further includes a second working electrode and a second active area disposed upon a surface of the second working electrode and responsive to a second analyte differing from the first analyte. In certain embodiments, the second active area includes at least one enzyme responsive to the second analyte. In certain embodiments, a second portion of the mass transport limiting membrane overcoats the second active area.

The present disclosure further provides methods for monitoring an analyte in vivo. In certain embodiments, the method can include providing an analyte sensor comprising (a) an internal supply of NAD(P), (b) a permeable polymer that overcoats the internal supply of NAD(P), (c) at least a first working electrode that is disposed upon a surface of the permeable polymer, wherein the first working electrode is a permeable working electrode, (d) an analyte-responsive active area disposed upon a surface of the first working electrode, wherein the analyte-responsive active area comprises an NAD(P)-dependent enzyme and (e) a mass transport limiting membrane permeable to the analyte that overcoats at least the analyte-responsive area. In certain embodiments, the method further includes applying a potential to the first working electrode, obtaining a first signal at or above an oxidation-reduction potential of the first active area, where the first signal is proportional to a concentration of a first analyte in a fluid contacting the first active area, and correlating the first signal to the concentration of the first analyte in the fluid.

In certain embodiments, the analyte sensor for use in the disclosed methods can further include a second working electrode and a second active area disposed upon a surface of the second working electrode and responsive to a second analyte differing from the first analyte. In certain embodiments, the second active area includes at least one enzyme responsive to the second analyte and a second portion of the mass transport limiting membrane overcoats the second active area.

BRIEF DESCRIPTION OF THE DRAWINGS

The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

The following figures are included to illustrate certain aspects of the present disclosure and should not be viewed as exclusive embodiments. The subject matter disclosed is capable of considerable modifications, alterations, combinations, and equivalents in form and function, without departing from the scope of this disclosure.

FIG. 1A is a system overview of a sensor applicator, reader device, monitoring system, network and remote system.

FIG. 1B is a diagram illustrating an operating environment of an example analyte monitoring system for use with the techniques described herein.

FIG. 2A is a block diagram depicting an example embodiment of a reader device.

FIG. 2B is a block diagram illustrating an example data receiving device for communicating with the sensor according to exemplary embodiments of the disclosed subject matter.

4

FIGS. 2C and 2D are block diagrams depicting example embodiments of sensor control devices.

FIG. 2E is a block diagram illustrating an example analyte sensor according to exemplary embodiments of the disclosed subject matter.

FIG. 3A is a proximal perspective view depicting an example embodiment of a user preparing a tray for an assembly.

FIG. 3B is a side view depicting an example embodiment of a user preparing an applicator device for an assembly.

FIG. 3C is a proximal perspective view depicting an example embodiment of a user inserting an applicator device into a tray during an assembly.

FIG. 3D is a proximal perspective view depicting an example embodiment of a user removing an applicator device from a tray during an assembly.

FIG. 3E is a proximal perspective view depicting an example embodiment of a patient applying a sensor using an applicator device.

FIG. 3F is a proximal perspective view depicting an example embodiment of a patient with an applied sensor and a used applicator device.

FIG. 4A is a side view depicting an example embodiment of an applicator device coupled with a cap.

FIG. 4B is a side perspective view depicting an example embodiment of an applicator device and cap decoupled.

FIG. 4C is a perspective view depicting an example embodiment of a distal end of an applicator device and electronics housing.

FIG. 4D is a top perspective view of an exemplary applicator device in accordance with the disclosed subject matter.

FIG. 4E is a bottom perspective view of the applicator device of FIG. 4D.

FIG. 4F is an exploded view of the applicator device of FIG. 4D.

FIG. 4G is a side cutaway view of the applicator device of FIG. 4D.

FIG. 5 is a proximal perspective view depicting an example embodiment of a tray with sterilization lid coupled.

FIG. 6A is a proximal perspective cutaway view depicting an example embodiment of a tray with sensor delivery components.

FIG. 6B is a proximal perspective view depicting sensor delivery components.

FIGS. 7A and 7B are isometric exploded top and bottom views, respectively, of an exemplary sensor control device.

FIG. 8A-8C are assembly and cross-sectional views of an on-body device including an integrated connector for the sensor assembly.

FIGS. 9A and 9B are side and cross-sectional side views, respectively, of an example embodiment of the sensor applicator of FIG. 1A with the cap of FIG. 2C coupled thereto.

FIGS. 10A and 10B are isometric and side views, respectively, of another example sensor control device.

FIGS. 11A-11C are progressive cross-sectional side views showing assembly of the sensor applicator with the sensor control device of FIGS. 10A-10B.

FIGS. 12A-12C are progressive cross-sectional side views showing assembly and disassembly of an example embodiment of the sensor applicator with the sensor control device of FIGS. 10A-10B.

FIGS. 13A-13F illustrate cross-sectional views depicting an example embodiment of an applicator during a stage of deployment.

FIG. 14 is a graph depicting an example of an in vitro sensitivity of an analyte sensor.

5

FIG. 15 is a diagram illustrating example operational states of the sensor according to exemplary embodiments of the disclosed subject matter.

FIG. 16 is a diagram illustrating an example operational and data flow for over-the-air programming of a sensor according to the disclosed subject matter.

FIG. 17 is a diagram illustrating an example data flow for secure exchange of data between two devices according to the disclosed subject matter.

FIGS. 18A-18C show cross-sectional diagrams of analyte sensors including a single active area.

FIGS. 19A-19C show cross-sectional diagrams of analyte sensors including two active areas.

FIG. 20 shows a cross-sectional diagram of analyte sensors including two active areas.

FIGS. 21A-21C show perspective views of analyte sensors including two active areas disposed upon separate working electrodes.

FIG. 22 provides a cross-sectional diagram of an exemplary sensor that includes an NAD(P) depot for controlled NAD(P) release.

FIG. 23A provides a schematic of an exemplary analyte sensor that includes an NAD depot.

FIG. 23B provides a cross-sectional diagram of an exemplary analyte sensor that excludes an NAD depot for use as a control.

FIG. 24 provides stability profiles of ketone detection over time with an analyte sensor that includes an NAD depot (as shown in FIG. 23A) compared to an analyte sensor that does not include an NAD depot (as shown in FIG. 23B).

DETAILED DESCRIPTION

The present disclosure is directed to analyte sensors comprising one or more active areas that include a nicotinamide adenine dinucleotide (NAD) or nicotinamide adenine dinucleotide phosphate (NADP)-dependent enzyme (referred to herein collectively as an “NAD(P)-dependent enzyme”). In particular, analyte sensors of the present disclosure include an internal reservoir of the cofactor NAD and/or NADP (referred to herein collectively as “NAD(P)”) for the NAD(P)-dependent enzyme.

The use of an internal reservoir of NAD(P) within an analyte sensor can overcome some of the limitations associated with analyte sensors that include an NAD(P)-dependent enzyme. For example, the amount of exogenous NAD(P) present in environment surrounding the sensor might not be in sufficient quantities to support analyte sensor operation, which can result in reduced sensitivity of the sensor. In addition, even if sufficient exogenous NAD(P) exists in the environment surrounding the analyte sensor, the molecular size of NAD(P) can prevent the molecule from diffusing through the surrounding sensor membrane to reach the one or more NAD(P)-dependent enzymes present in the sensing chemistry layer, e.g., active areas, of the analyte sensor.

The present disclosure provides analyte sensors that include an internal supply of NAD(P) that can release NAD(P) over an extended period of time to allow monitoring of an analyte in vivo. In certain embodiments, the NAD(P) internal supply (also referred to herein as “an NAD(P) depot”) can be coated with or distributed within a permeable layer (e.g., a polymeric permeable layer) that controls diffusion of NAD(P) from the NAD(P) depot to maintain a sufficient concentration of NAD(P) for the sensing chemistry during use of the analyte sensor.

6

The present disclosure further provides methods of detecting an analyte using the disclosed sensors and methods of manufacturing the disclosed analyte sensors.

For clarity, but not by way of limitation, the detailed description of the presently disclosed subject matter is divided into the following subsections:

- I. Definitions;
- II. Analyte Sensors;
 1. General Structure of Analyte Sensor Systems;
 2. NAD(P)-depot;
 3. Enzymes;
 4. Redox Mediators;
 5. Polymeric Backbone;
 6. Mass Transport Limiting Membrane;
 7. Interference Domain; and
 8. Manufacturing;
- III. Analyte Monitoring.

I. Definitions

The terms used in this specification generally have their ordinary meanings in the art, within the context of this disclosure and in the specific context where each term is used. Certain terms are discussed below, or elsewhere in the specification, to provide additional guidance to the practitioner in describing the compositions and methods of the present disclosure and how to make and use them.

As used herein, the use of the word “a” or “an” when used in conjunction with the term “comprising” in the claims and/or the specification can mean “one,” but it is also consistent with the meaning of “one or more,” “at least one,” and “one or more than one.”

The terms “comprise(s),” “include(s),” “having,” “has,” “can,” “contain(s),” and variants thereof, as used herein, are intended to be open-ended transitional phrases, terms or words that do not preclude additional acts or structures. The present disclosure also contemplates other embodiments “comprising,” “consisting of” and “consisting essentially of,” the embodiments or elements presented herein, whether explicitly set forth or not.

The term “about” or “approximately” means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, which depends in part on how the value is measured or determined, i.e., the limitations of the measurement system. For example, “about” can mean within 3 or more than 3 standard deviations, per the practice in the art. Alternatively, “about” can mean a range of up to 20%, preferably up to 10%, more preferably up to 5%, and more preferably still up to 1% of a given value. Alternatively, particularly with respect to biological systems or processes, the term can mean within an order of magnitude, preferably within 5-fold, and more preferably within 2-fold, of a value.

As used herein, “analyte sensor” or “sensor” can refer to any device capable of receiving sensor information from a user, including for purpose of illustration but not limited to, body temperature sensors, blood pressure sensors, pulse or heart-rate sensors, glucose level sensors, analyte sensors, physical activity sensors, body movement sensors, or any other sensors for collecting physical or biological information. Analytes measured by the analyte sensors can include, by way of example and not limitation, glucose, ketones, lactate, oxygen, hemoglobin A1C, albumin, alcohol, alkaline phosphatase, alanine transaminase, aspartate aminotransferase, bilirubin, blood urea nitrogen, calcium, carbon dioxide,

chloride, creatinine, hematocrit, lactate, magnesium, oxygen, pH, phosphorus, potassium, sodium, total protein, uric acid, etc.

The term “biological fluid,” as used herein, refers to any bodily fluid or bodily fluid derivative in which the analyte can be measured. Non-limiting examples of a biological fluid include dermal fluid, interstitial fluid, plasma, blood, lymph, synovial fluid, cerebrospinal fluid, saliva, broncho-alveolar lavage, amniotic fluid, sweat, tears or the like. In certain embodiments, the biological fluid is dermal fluid or interstitial fluid. In certain embodiments, the biological fluid is interstitial fluid.

The term “electrolysis,” as used herein, refers to electrooxidation or electroreduction of a compound either directly at an electrode or via one or more electron transfer agents (e.g., redox mediators or enzymes).

The terms “enzyme composition” and “sensing chemistry,” as used interchangeably herein, refer to a composition that includes one or more enzymes for detecting and/or measuring an analyte. In certain non-limiting embodiments, the enzyme compositions can include one or more enzymes, polymers, redox mediators and/or crosslinkers.

As used herein, the term “homogenous membrane” refers to a membrane comprising a single type of membrane polymer.

As used herein, the term “multi-component membrane” refers to a membrane comprising two or more types of membrane polymers.

As used herein, the term “NAD(P)” refers to the cofactor NAD (and its reduced form NADH) or NADP (and its reduced form NADPH) or a derivative thereof.

As used herein, the term “NAD(P)-dependent enzyme” refers to an enzyme that uses NAD (and its reduced form NADH) or NADP (and its reduced form NADPH) as a coenzyme in a redox reaction.

As used herein, the term “permeable electrode” refers to an electrode that is composed of a material that allows the passing of molecules, e.g., NAD(P), through the material of the electrode.

As used herein, the term “polyvinylpyridine-based polymer” refers to a polymer or copolymer that comprises polyvinylpyridine (e.g., poly(2-vinylpyridine) or poly(4-vinylpyridine)) or a derivative thereof.

As used herein, the term “redox mediator” refers to an electron transfer agent for carrying electrons between an analyte or an analyte-reduced or analyte oxidized enzyme and an electrode, either directly, or via one or more additional electron transfer agents. In certain embodiments, redox mediators that include a polymeric backbone can also be referred to as “redox polymers.”

The term “reference electrode” as used herein, can refer to either reference electrodes or electrodes that function as both, a reference and a counter electrode. Similarly, the term “counter electrode,” as used herein, can refer to both, a counter electrode and a counter electrode that also functions as a reference electrode.

II. Analyte Sensors

Before the present subject matter is described in detail, it is to be understood that this disclosure is not limited to the particular embodiments described, as such may, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting, since the scope of the present disclosure will be limited only by the appended claims.

The publications discussed herein are provided solely for their disclosure prior to the filing date of the present application. Nothing herein is to be construed as an admission that the present disclosure is not entitled to antedate such publication by virtue of prior disclosure. Further, the dates of publication provided may be different from the actual publication dates which may need to be independently confirmed.

Generally, embodiments of the present disclosure include systems, devices and methods for the use of analyte sensor insertion applicators for use with in vivo analyte monitoring systems. An applicator can be provided to the user in a sterile package with an electronics housing of the sensor control device contained therein. According to some embodiments, a structure separate from the applicator, such as a container, can also be provided to the user as a sterile package with a sensor module and a sharp module contained therein. The user can couple the sensor module to the electronics housing, and can couple the sharp to the applicator with an assembly process that involves the insertion of the applicator into the container in a specified manner. In other embodiments, the applicator, sensor control device, sensor module, and sharp module can be provided in a single package. The applicator can be used to position the sensor control device on a human body with a sensor in contact with the wearer’s bodily fluid. The embodiments provided herein are improvements to reduce the likelihood that a sensor is improperly inserted or damaged, or elicits an adverse physiological response. Other improvements and advantages are provided as well. The various configurations of these devices are described in detail by way of the embodiments which are only examples.

Furthermore, many embodiments include in vivo analyte sensors structurally configured so that at least a portion of the sensor is, or can be, positioned in the body of a user to obtain information about at least one analyte of the body. It should be noted, however, that the embodiments disclosed herein can be used with in vivo analyte monitoring systems that incorporate in vitro capability, as well as purely in vitro or ex vivo analyte monitoring systems, including systems that are entirely non-invasive.

Furthermore, for each and every embodiment of a method disclosed herein, systems and devices capable of performing each of those embodiments are covered within the scope of the present disclosure. For example, embodiments of sensor control devices are disclosed and these devices can have one or more sensors, analyte monitoring circuits (e.g., an analog circuit), memories (e.g., for storing instructions), power sources, communication circuits, transmitters, receivers, processors and/or controllers (e.g., for executing instructions) that can perform any and all method steps or facilitate the execution of any and all method steps. These sensor control device embodiments can be used and can be capable of use to implement those steps performed by a sensor control device from any and all of the methods described herein.

Furthermore, the systems and methods presented herein can be used for operations of a sensor used in an analyte monitoring system, such as but not limited to wellness, fitness, dietary, research, information or any purposes involving analyte sensing over time. As used herein, “analyte sensor” or “sensor” can refer to any device capable of receiving sensor information from a user, including for purpose of illustration but not limited to, body temperature sensors, blood pressure sensors, pulse or heart-rate sensors, glucose level sensors, analyte sensors, physical activity sensors, body movement sensors, or any other sensors for

collecting physical or biological information. In certain embodiments, an analyte sensor of the present disclosure can further measure analytes including, but not limited to, glucose, ketones, lactate, oxygen, hemoglobin A1C, albumin, alcohol, alkaline phosphatase, alanine transaminase, aspartate aminotransferase, bilirubin, blood urea nitrogen, calcium, carbon dioxide, chloride, creatinine, hematocrit, lactate, magnesium, oxygen, pH, phosphorus, potassium, sodium, total protein, uric acid, etc.

As mentioned, a number of embodiments of systems, devices, and methods are described herein that provide for the improved assembly and use of dermal sensor insertion devices for use with in vivo analyte monitoring systems. In particular, several embodiments of the present disclosure are designed to improve the method of sensor insertion with respect to in vivo analyte monitoring systems and, in particular, to prevent the premature retraction of an insertion sharp during a sensor insertion process. Some embodiments, for example, include a dermal sensor insertion mechanism with an increased firing velocity and a delayed sharp retraction. In other embodiments, the sharp retraction mechanism can be motion-actuated such that the sharp is not retracted until the user pulls the applicator away from the skin. Consequently, these embodiments can reduce the likelihood of prematurely withdrawing an insertion sharp during a sensor insertion process; decrease the likelihood of improper sensor insertion; and decrease the likelihood of damaging a sensor during the sensor insertion process, to name a few advantages. Several embodiments of the present disclosure also provide for improved insertion sharp modules to account for the small scale of dermal sensors and the relatively shallow insertion path present in a subject's dermal layer. In addition, several embodiments of the present disclosure are designed to prevent undesirable axial and/or rotational movement of applicator components during sensor insertion. Accordingly, these embodiments can reduce the likelihood of instability of a positioned dermal sensor, irritation at the insertion site, damage to surrounding tissue, and breakage of capillary blood vessels resulting in fouling of the dermal fluid with blood, to name a few advantages. In addition, to mitigate inaccurate sensor readings which can be caused by trauma at the insertion site, several embodiments of the present disclosure can reduce the end-depth penetration of the needle relative to the sensor tip during insertion.

Before describing these aspects of the embodiments in detail, however, it is first desirable to describe examples of devices that can be present within, for example, an in vivo analyte monitoring system, as well as examples of their operation, all of which can be used with the embodiments described herein.

There are various types of in vivo analyte monitoring systems. "Continuous Analyte Monitoring" systems (or "Continuous Glucose Monitoring" systems), for example, can transmit data from a sensor control device to a reader device continuously without prompting, e.g., automatically according to a schedule. "Flash Analyte Monitoring" systems (or "Flash Glucose Monitoring" systems or simply "Flash" systems), as another example, can transfer data from a sensor control device in response to a scan or request for data by a reader device, such as with a Near Field Communication (NFC) or Radio Frequency Identification (RFID) protocol. In vivo analyte monitoring systems can also operate without the need for finger stick calibration.

In vivo analyte monitoring systems can be differentiated from "in vitro" systems that contact a biological sample outside of the body (or "ex vivo") and that typically include

a meter device that has a port for receiving an analyte test strip carrying bodily fluid of the user, which can be analyzed to determine the user's blood sugar level.

In vivo monitoring systems can include a sensor that, while positioned in vivo, makes contact with the bodily fluid of the user and senses the analyte levels contained therein. The sensor can be part of the sensor control device that resides on the body of the user and contains the electronics and power supply that enable and control the analyte sensing. The sensor control device, and variations thereof, can also be referred to as a "sensor control unit," an "on-body electronics" device or unit, an "on-body" device or unit, or a "sensor data communication" device or unit, to name a few.

In vivo monitoring systems can also include a device that receives sensed analyte data from the sensor control device and processes and/or displays that sensed analyte data, in any number of forms, to the user. This device, and variations thereof, can be referred to as a "handheld reader device," "reader device" (or simply a "reader"), "handheld electronics" (or simply a "handheld"), a "portable data processing" device or unit, a "data receiver," a "receiver" device or unit (or simply a "receiver"), or a "remote" device or unit, to name a few. Other devices such as personal computers have also been utilized with or incorporated into in vivo and in vitro monitoring systems.

1. General Structure of Analyte Sensor Systems

A. Exemplary In Vivo Analyte Monitoring System

FIG. 1A is a conceptual diagram depicting an example embodiment of an analyte monitoring system **100** that includes a sensor applicator **150**, a sensor control device **102**, and a reader device **120**. Here, sensor applicator **150** can be used to deliver sensor control device **102** to a monitoring location on a user's skin where a sensor **104** is maintained in position for a period of time by an adhesive patch **105**. Sensor control device **102** is further described in FIGS. 2B and 2C, and can communicate with reader device **120** via a communication path or link **140** using a wired or wireless, uni- or bi-directional, and encrypted or non-encrypted technique. Example wireless protocols include Bluetooth, Bluetooth Low Energy (BLE, BTLE, Bluetooth SMART, etc.), Near Field Communication (NFC) and others. Users can monitor applications installed in memory on reader device **120** using screen **122** and input **121** and the device battery can be recharged using power port **123**. More detail about reader device **120** is set forth with respect to FIG. 2A below. Reader device **120** can constitute an output medium for viewing analyte concentrations and alerts or notifications determined by sensor **104** or a processor associated therewith, as well as allowing for one or more user inputs, according to certain embodiments. Reader device **120** can be a multi-purpose smartphone or a dedicated electronic reader instrument. While only one reader device **120** is shown, multiple reader devices **120** can be present in certain instances.

Reader device **120** can communicate with local computer system **170** via a communication path **141**, which also can be wired or wireless, uni- or bi-directional, and encrypted or non-encrypted. Local computer system **170** can include one or more of a laptop, desktop, tablet, phablet, smartphone, set-top box, video game console, remote terminal or other computing device and wireless communication can include any of a number of applicable wireless networking protocols including Bluetooth, Bluetooth Low Energy (BTLE), Wi-Fi or others. Local computer system **170** can communicate via communications path **143** with a network **190** similar to how reader device **120** can communicate via a communications

11

path **142** with network **190**, by wired or wireless technique as described previously. Network **190** can be any of a number of networks, such as private networks and public networks, local area or wide area networks, and so forth. A trusted computer system **180** can include a server and can provide authentication services and secured data storage and can communicate via communications path **144** with network **190** by wired or wireless technique. Local computer system **170** and/or trusted computer system **180** can be accessible, according to certain embodiments, by individuals other than a primary user who have an interest in the user's analyte levels. Reader device **120** can include display **122** and optional input component **121**. Display **122** can include a touch-screen interface, according to certain embodiments.

Sensor control device **102** includes sensor housing, which can house circuitry and a power source for operating sensor **104**. Optionally, the power source and/or active circuitry can be omitted. A processor (not shown) can be communicatively coupled to sensor **104**, with the processor being physically located within the sensor housing or reader device **120**. Sensor **104** protrudes from the underside of the sensor housing and extends through adhesive layer **105**, which is adapted for adhering the sensor housing to a tissue surface, such as skin, according to certain embodiments.

FIG. 1B illustrates an operating environment of an analyte monitoring system **100a** capable of embodying the techniques described herein. The analyte monitoring system **100a** can include a system of components designed to provide monitoring of parameters, such as analyte levels, of a human or animal body or can provide for other operations based on the configurations of the various components. As embodied herein, the system can include a low-power analyte sensor **110**, or simply "sensor" worn by the user or attached to the body for which information is being collected. As embodied herein, the analyte sensor **110** can be a sealed, disposable device with a predetermined active use lifetime (e.g., 1 day, 14 days, 30 days, etc.). Sensors **110** can be applied to the skin of the user body and remain adhered over the duration of the sensor lifetime or can be designed to be selectively removed and remain functional when reapplied. The low-power analyte monitoring system **100a** can further include a data reading device **120** or multi-purpose data receiving device **130** configured as described herein to facilitate retrieval and delivery of data, including analyte data, from the analyte sensor **110**.

As embodied herein, the analyte monitoring system **100a** can include a software or firmware library or application provided, for example via a remote application server **150** or application storefront server **160**, to a third-party and incorporated into a multi-purpose hardware device **130** such as a mobile phone, tablet, personal computing device, or other similar computing device capable of communicating with the analyte sensor **110** over a communication link. Multi-purpose hardware can further include embedded devices, including, but not limited to insulin pumps or insulin pens, having an embedded library configured to communicate with the analyte sensor **110**. Although the illustrated embodiments of the analyte monitoring system **100a** include only one of each of the illustrated devices, this disclosure contemplates the analyte monitoring system **100a** incorporate multiples of each components interacting throughout the system. For example and without limitation, as embodied herein, data reading device **120** and/or multi-purpose data receiving device **130** can include multiples of each. As embodied herein, multiple data receiving devices **130** can communicate directly with sensor **110** as described herein.

12

Additionally or alternatively, a data receiving device **130** can communicate with secondary data receiving devices **130** to provide analyte data, or visualization or analysis of the data, for secondary display to the user or other authorized parties.

Sensor **104** of FIG. 1A is adapted to be at least partially inserted into a tissue of interest, such as within the dermal or subcutaneous layer of the skin. Sensor **104** can include a sensor tail of sufficient length for insertion to a desired depth in a given tissue. The sensor tail can include at least one working electrode. In certain configurations, the sensor tail can include an active area, e.g., including one or more NAD(P)-dependent enzymes, for detecting an analyte. A counter electrode can be present in combination with the at least one working electrode. Particular electrode configurations upon the sensor tail are described in more detail below.

One or more mass transport limiting membranes can overcoat the active area, as also described in further detail below.

The active area can be configured for detecting a particular analyte described herein. For example, but not by way of the limitation, the analyte can include glucose, ketones, lactate, oxygen, hemoglobin A1C, albumin, alcohol, alkaline phosphatase, alanine transaminase, aspartate aminotransferase, bilirubin, blood urea nitrogen, calcium, carbon dioxide, chloride, creatinine, hematocrit, lactate, magnesium, oxygen, pH, phosphorus, potassium, sodium, total protein, uric acid, etc. In certain embodiments, the analytes for detection using the disclosed analyte sensors include alcohols, ketones, creatinine, glucose, and lactate. In certain embodiments, the active area can be configured for detecting two or more analytes described herein. In certain embodiments, an active area of a presently disclosed sensor is configured to detect ketones. In certain embodiments, an active area of a presently disclosed sensor is configured to detect glucose. In certain embodiments, an active area of a presently disclosed sensor is configured to detect lactate. In certain embodiments, an active area of a presently disclosed sensor is configured to detect creatinine. In certain embodiments, an active area of a presently disclosed sensor is configured to detect an alcohol, e.g., ethanol.

In certain embodiments of the present disclosure, one or more analytes can be monitored in any biological fluid of interest such as dermal fluid, interstitial fluid, plasma, blood, lymph, synovial fluid, cerebrospinal fluid, saliva, bronchoalveolar lavage, amniotic fluid or the like. In certain particular embodiments, analyte sensors of the present disclosure can be adapted for assaying dermal fluid or interstitial fluid to determine a concentration of one or more analytes in vivo. In certain embodiments, the biological fluid is interstitial fluid.

An introducer can be present transiently to promote introduction of sensor **104** into a tissue. In certain illustrative embodiments, the introducer can include a needle or similar sharp. As would be readily recognized by a person skilled in the art, other types of introducers, such as sheaths or blades, can be present in alternative embodiments. More specifically, the needle or other introducer can transiently reside in proximity to sensor **104** prior to tissue insertion and then be withdrawn afterward. While present, the needle or other introducer can facilitate insertion of sensor **104** into a tissue by opening an access pathway for sensor **104** to follow. For example, and not by the way of limitation, the needle can facilitate penetration of the epidermis as an access pathway to the dermis to allow implantation of sensor **104** to take place, according to one or more embodiments. After opening the access pathway, the needle or other introducer can be withdrawn so that it does not represent a sharps hazard. In

13

certain embodiments, suitable needles can be solid or hollow, beveled or non-beveled, and/or circular or non-circular in cross-section. In more particular non-limiting embodiments, suitable needles can be comparable in cross-sectional diameter and/or tip design to an acupuncture needle, which can have a cross-sectional diameter of about 250 microns. However, suitable needles can have a larger or smaller cross-sectional diameter if needed for certain particular applications.

In certain embodiments, a tip of the needle (while present) can be angled over the terminus of sensor 104, such that the needle penetrates a tissue first and opens an access pathway for sensor 104. In certain embodiments, sensor 104 can reside within a lumen or groove of the needle, with the needle similarly opening an access pathway for sensor 104. In either case, the needle is subsequently withdrawn after facilitating sensor insertion.

B. Exemplary Reader Device

FIG. 2A is a block diagram depicting an example embodiment of a reader device configured as a smartphone. Here, reader device 120 can include a display 122, input component 121, and a processing core 206 including a communications processor 222 coupled with memory 223 and an applications processor 224 coupled with memory 225. Also included can be separate memory 230, RF transceiver 228 with antenna 229, and power supply 226 with power management module 238. Further included can be a multi-functional transceiver 232 which can communicate over Wi-Fi, NFC, Bluetooth, BTLE, and GPS with an antenna 234. As understood by one of skill in the art, these components are electrically and communicatively coupled in a manner to make a functional device.

C. Exemplary Data Receiving Device Architecture

For purpose of illustration and not limitation, reference is made to the exemplary embodiment of a data receiving device 120 for use with the disclosed subject matter as shown in FIG. 2B. The data receiving device 120, and the related multi-purpose data receiving device 130, includes components germane to the discussion of the analyte sensor 110 and its operations and additional components can be included. In particular embodiments, the data receiving device 120 and multi-purpose data receiving device 130 can be or include components provided by a third party and are not necessarily restricted to include devices made by the same manufacturer as the sensor 110.

As illustrated in FIG. 2B, the data receiving device 120 includes an ASIC 4000 including a microcontroller 4010, memory 4020, and storage 4030 and communicatively coupled with a communication module 4040. Power for the components of the data receiving device 120 can be delivered by a power module 4050, which as embodied herein can include a rechargeable battery. The data receiving device 120 can further include a display 4070 for facilitating review of analyte data received from an analyte sensor 110 or other device (e.g., user device 140 or remote application server 150). The data receiving device 120 can include separate user interface components (e.g., physical keys, light sensors, microphones, etc.).

The communication module 4040 can include a BLE module 4041 and an NFC module 4042. The data receiving device 120 can be configured to wirelessly couple with the analyte sensor 110 and transmit commands to and receive data from the analyte sensor 110. As embodied herein, the data receiving device 120 can be configured to operate, with respect to the analyte sensor 110 as described herein, as an NFC scanner and a BLE end point via specific modules (e.g., BLE module 4042 or NFC module 4043) of the communi-

14

cation module 4040. For example, the data receiving device 120 can issue commands (e.g., activation commands for a data broadcast mode of the sensor; pairing commands to identify the data receiving device 120) to the analyte sensor 110 using a first module of the communication module 4040 and receive data from and transmit data to the analyte sensor 110 using a second module of the communication module 4040. The data receiving device 120 can be configured for communication with a user device 140 via a Universal Serial Bus (USB) module 4045 of the communication module 4040.

As another example, the communication module 4040 can include, for example, a cellular radio module 4044. The cellular radio module 4044 can include one or more radio transceivers for communicating using broadband cellular networks, including, but not limited to third generation (3G), fourth generation (4G), and fifth generation (5G) networks. Additionally, the communication module 4040 of the data receiving device 120 can include a Wi-Fi radio module 4043 for communication using a wireless local area network according to one or more of the IEEE 802.11 standards (e.g., 802.11a, 802.11b, 802.11g, 802.11n (aka Wi-Fi 4), 802.11ac (aka Wi-Fi 5), 802.11ax (aka Wi-Fi 6)). Using the cellular radio module 4044 or Wi-Fi radio module 4043, the data receiving device 120 can communicate with the remote application server 150 to receive analyte data or provide updates or input received from a user (e.g., through one or more user interfaces). Although not illustrated, the communication module 5040 of the analyte sensor 120 can similarly include a cellular radio module or Wi-Fi radio module.

As embodied herein, the on-board storage 4030 of the data receiving device 120 can store analyte data received from the analyte sensor 110. Further, the data receiving device 120, multi-purpose data receiving device 130, or a user device 140 can be configured to communicate with a remote application server 150 via a wide area network. As embodied herein, the analyte sensor 110 can provide data to the data receiving device 120 or multi-purpose data receiving device 130. The data receiving device 120 can transmit the data to the user computing device 140. The user computing device 140 (or the multi-purpose data receiving device 130) can in turn transmit that data to a remote application server 150 for processing and analysis.

As embodied herein, the data receiving device 120 can further include sensing hardware 4060 similar to, or expanded from, the sensing hardware 5060 of the analyte sensor 110. In particular embodiments, the data receiving device 120 can be configured to operate in coordination with the analyte sensor 110 and based on analyte data received from the analyte sensor 110. As an example, where the analyte sensor 110 glucose sensor, the data receiving device 120 can be or include an insulin pump or insulin injection pen. In coordination, the compatible device 130 can adjust an insulin dosage for a user based on glucose values received from the analyte sensor.

D. Exemplary Sensor Control Devices

FIGS. 2C and 2D are block diagrams depicting example embodiments of sensor control device 102 having analyte sensor 104 and sensor electronics 160 (including analyte monitoring circuitry) that can have the majority of the processing capability for rendering end-result data suitable for display to the user. In FIG. 2C, a single semiconductor chip 161 is depicted that can be a custom application specific integrated circuit (ASIC). Shown within ASIC 161 are certain high-level functional units, including an analog front end (AFE) 162, power management (or control) circuitry 164, processor 166, and communication circuitry 168

15

(which can be implemented as a transmitter, receiver, transceiver, passive circuit, or otherwise according to the communication protocol). In this embodiment, both AFE 162 and processor 166 are used as analyte monitoring circuitry, but in other embodiments either circuit can perform the analyte monitoring function. Processor 166 can include one or more processors, microprocessors, controllers, and/or microcontrollers, each of which can be a discrete chip or distributed amongst (and a portion of) a number of different chips.

A memory 163 is also included within ASIC 161 and can be shared by the various functional units present within ASIC 161, or can be distributed amongst two or more of them. Memory 163 can also be a separate chip. Memory 163 can be volatile and/or non-volatile memory. In this embodiment, ASIC 161 is coupled with power source 170, which can be a coin cell battery, or the like. AFE 162 interfaces with in vivo analyte sensor 104 and receives measurement data therefrom and outputs the data to processor 166 in digital form, which in turn processes the data to arrive at the end-result glucose discrete and trend values, etc. This data can then be provided to communication circuitry 168 for sending, by way of antenna 171, to reader device 120 (not shown), for example, where minimal further processing is needed by the resident software application to display the data.

FIG. 2D is similar to FIG. 2C but instead includes two discrete semiconductor chips 162 and 174, which can be packaged together or separately. Here, AFE 162 is resident on ASIC 161. Processor 166 is integrated with power management circuitry 164 and communication circuitry 168 on chip 174. AFE 162 includes memory 163 and chip 174 includes memory 165, which can be isolated or distributed within. In one example embodiment, AFE 162 is combined with power management circuitry 164 and processor 166 on one chip, while communication circuitry 168 is on a separate chip. In another example embodiment, both AFE 162 and communication circuitry 168 are on one chip, and processor 166 and power management circuitry 164 are on another chip. It should be noted that other chip combinations are possible, including three or more chips, each bearing responsibility for the separate functions described, or sharing one or more functions for fail-safe redundancy.

For purpose of illustration and not limitation, reference is made to the exemplary embodiment of an analyte sensor 110 for use with the disclosed subject matter as shown in FIG. 2E. FIG. 2E illustrates a block diagram of an example analyte sensor 110 according to exemplary embodiments compatible with the security architecture and communication schemes described herein.

As embodied herein, the analyte sensor 110 can include an Application-Specific Integrated Circuit ("ASIC") 5000 communicatively coupled with a communication module 5040. The ASIC 5000 can include a microcontroller core 5010, on-board memory 5020, and storage memory 5030. The storage memory 5030 can store data used in an authentication and encryption security architecture. The storage memory 5030 can store programming instructions for the sensor 110. As embodied herein, certain communication chipsets can be embedded in the ASIC 5000 (e.g., an NFC transceiver 5025). The ASIC 5000 can receive power from a power module 5050, such as an on-board battery or from an NFC pulse. The storage memory 5030 of the ASIC 5000 can be programmed to include information such as an identifier for the sensor 110 for identification and tracking purposes. The storage memory 5030 can also be programmed with configuration or calibration parameters for

16

use by the sensor 110 and its various components. The storage memory 5030 can include rewritable or one-time programming (OTP) memory. The storage memory 5030 can be updated using techniques described herein to extend the usefulness of the sensor 110.

As embodied herein, the communication module 5040 of the sensor 100 can be or include one or more modules to support the analyte sensor 110 communicating with other devices of the analyte monitoring system 100. As an example only and not by way of limitation, example communication modules 5040 can include a Bluetooth Low-Energy ("BLE") module 5041. As used throughout this disclosure, Bluetooth Low Energy ("BLE") refers to a short-range communication protocol optimized to make pairing of Bluetooth devices simple for end users. The communication module 5040 can transmit and receive data and commands via interaction with similarly-capable communication modules of a data receiving device 120 or user device 140. The communication module 5040 can include additional or alternative chipsets for use with similar short-range communication schemes, such as a personal area network according to IEEE 802.15 protocols, IEEE 802.11 protocols, infrared communications according to the Infrared Data Association standards (IrDA), etc.

To perform its functionalities, the sensor 100 can further include suitable sensing hardware 5060 appropriate to its function. As embodied herein, the sensing hardware 5060 can include an analyte sensor transcutaneously or subcutaneously positioned in contact with a bodily fluid of a subject. The analyte sensor can generate sensor data containing values corresponding to levels of one or more analytes within the bodily fluid.

E. Exemplary Assembly Processes for Sensor Control Devices

The components of sensor control device 102 can be acquired by a user in multiple packages requiring final assembly by the user before delivery to an appropriate user location. FIGS. 3A-3D depict an example embodiment of an assembly process for sensor control device 102 by a user, including preparation of separate components before coupling the components in order to ready the sensor for delivery. FIGS. 3E-3F depict an example embodiment of delivery of sensor control device 102 to an appropriate user location by selecting the appropriate delivery location and applying device 102 to the location.

FIG. 3A is a proximal perspective view depicting an example embodiment of a user preparing a container 810, configured here as a tray (although other packages can be used), for an assembly process. The user can accomplish this preparation by removing lid 812 from tray 810 to expose platform 808, for instance by peeling a non-adhered portion of lid 812 away from tray 810 such that adhered portions of lid 812 are removed. Removal of lid 812 can be appropriate in various embodiments so long as platform 808 is adequately exposed within tray 810. Lid 812 can then be placed aside.

FIG. 3B is a side view depicting an example embodiment of a user preparing an applicator device 150 for assembly. Applicator device 150 can be provided in a sterile package sealed by a cap 708. Preparation of applicator device 150 can include uncoupling housing 702 from cap 708 to expose sheath 704 (FIG. 3C). This can be accomplished by unscrewing (or otherwise uncoupling) cap 708 from housing 702. Cap 708 can then be placed aside.

FIG. 3C is a proximal perspective view depicting an example embodiment of a user inserting an applicator device 150 into a tray 810 during an assembly. Initially, the user can

17

insert sheath **704** into platform **808** inside tray **810** after aligning housing orienting feature **1302** (or slot or recess) and tray orienting feature **924** (an abutment or detent). Inserting sheath **704** into platform **808** temporarily unlocks sheath **704** relative to housing **702** and also temporarily unlocks platform **808** relative to tray **810**. At this stage, removal of applicator device **150** from tray **810** will result in the same state prior to initial insertion of applicator device **150** into tray **810** (i.e., the process can be reversed or aborted at this point and then repeated without consequence).

Sheath **704** can maintain position within platform **808** with respect to housing **702** while housing **702** is distally advanced, coupling with platform **808** to distally advance platform **808** with respect to tray **810**. This step unlocks and collapses platform **808** within tray **810**. Sheath **704** can contact and disengage locking features (not shown) within tray **810** that unlock sheath **704** with respect to housing **702** and prevent sheath **704** from moving (relatively) while housing **702** continues to distally advance platform **808**. At the end of advancement of housing **702** and platform **808**, sheath **704** is permanently unlocked relative to housing **702**. A sharp and sensor (not shown) within tray **810** can be coupled with an electronics housing (not shown) within housing **702** at the end of the distal advancement of housing **702**. Operation and interaction of the applicator device **150** and tray **810** are further described below.

FIG. **3D** is a proximal perspective view depicting an example embodiment of a user removing an applicator device **150** from a tray **810** during an assembly. A user can remove applicator **150** from tray **810** by proximally advancing housing **702** with respect to tray **810** or other motions having the same end effect of uncoupling applicator **150** and tray **810**. The applicator device **150** is removed with sensor control device **102** (not shown) fully assembled (sharp, sensor, electronics) therein and positioned for delivery.

FIG. **3E** is a proximal perspective view depicting an example embodiment of a patient applying sensor control device **102** using applicator device **150** to a target area of skin, for instance, on an abdomen or other appropriate location. Advancing housing **702** distally collapses sheath **704** within housing **702** and applies the sensor to the target location such that an adhesive layer on the bottom side of sensor control device **102** adheres to the skin. The sharp is automatically retracted when housing **702** is fully advanced, while the sensor (not shown) is left in position to measure analyte levels.

FIG. **3F** is a proximal perspective view depicting an example embodiment of a patient with sensor control device **102** in an applied position. The user can then remove applicator **150** from the application site.

System **100**, described with respect to FIGS. **3A-3F** and elsewhere herein, can provide a reduced or eliminated chance of accidental breakage, permanent deformation, or incorrect assembly of applicator components compared to prior art systems. Since applicator housing **702** directly engages platform **808** while sheath **704** unlocks, rather than indirect engagement via sheath **704**, relative angularity between sheath **704** and housing **702** will not result in breakage or permanent deformation of the arms or other components. The potential for relatively high forces (such as in conventional devices) during assembly will be reduced, which in turn reduces the chance of unsuccessful user assembly.

F. Exemplary Sensor Applicator Devices

FIG. **4A** is a side view depicting an example embodiment of an applicator device **150** coupled with screw cap **708**. This is an example of how applicator **150** is shipped to and

18

received by a user, prior to assembly by the user with a sensor. FIG. **4B** is a side perspective view depicting applicator **150** and cap **708** after being decoupled. FIG. **4C** is a perspective view depicting an example embodiment of a distal end of an applicator device **150** with electronics housing **706** and adhesive patch **105** removed from the position they would have retained within sensor carrier **710** of sheath **704**, when cap **708** is in place.

Referring to FIG. **4D-G** for purpose of illustration and not limitation, the applicator device **20150** can be provided to a user as a single integrated assembly. FIGS. **4D** and **4E** provide perspective top and bottom views, respectively, of the applicator device **20150**. FIG. **4F** provides an exploded view of the applicator device **20150** and FIG. **4G** provides a side cut-away view. The perspective views illustrate how applicator **20150** is shipped to and received by a user. The exploded and cut-away views illustrate the components of the applicator device **20150**. The applicator device **20150** can include a housing **20702**, gasket **20701**, sheath **20704**, sharp carrier **201102**, spring **205612**, sensor carrier **20710** (also referred to as a “puck carrier”), sharp hub **205014**, sensor control device (also referred to as a “puck”) **20102**, adhesive patch **20105**, desiccant **20502**, cap **20708**, serial label **20709**, and tamper evidence feature **20712**. As received by a user, only the housing **20702**, cap **20708**, tamper evidence feature **20712**, and label **20709** are visible. The tamper evidence feature **20712** can be, for example, a sticker coupled to each of the housing **20702** and the cap **20708**, and tamper evidence feature **20712** can be damaged, for example, irreparably, by uncoupling housing **20702** and cap **20708**, thereby indicating to a user that the housing **20702** and cap **20708** have been previously uncoupled. These features are described in greater detail below.

G. Exemplary Tray and Sensor Module Assembly

FIG. **5** is a proximal perspective view depicting an example embodiment of a tray **810** with sterilization lid **812** removably coupled thereto, which may be representative of how the package is shipped to and received by a user prior to assembly.

FIG. **6A** is a proximal perspective cutaway view depicting sensor delivery components within tray **810**. Platform **808** is slidably coupled within tray **810**. Desiccant **502** is stationary with respect to tray **810**. Sensor module **504** is mounted within tray **810**.

FIG. **6B** is a proximal perspective view depicting sensor module **504** in greater detail. Here, retention arm extensions **1834** of platform **808** releasably secure sensor module **504** in position. Module **2200** is coupled with connector **2300**, sharp module **2500** and sensor (not shown) such that during assembly they can be removed together as sensor module **504**.

H. Exemplary Applicators and Sensor Control Devices for One Piece Architectures

Referring briefly again to FIGS. **1A** and **3A-3G**, for the two-piece architecture system, the sensor tray **202** and the sensor applicator **102** are provided to the user as separate packages, thus requiring the user to open each package and finally assemble the system. In some applications, the discrete, sealed packages allow the sensor tray **202** and the sensor applicator **102** to be sterilized in separate sterilization processes unique to the contents of each package and otherwise incompatible with the contents of the other. More specifically, the sensor tray **202**, which includes the plug assembly **207**, including the sensor **110** and the sharp **220**, may be sterilized using radiation sterilization, such as electron beam (or “e-beam”) irradiation. Suitable radiation sterilization processes include, but are not limited to, electron

beam (e-beam) irradiation, gamma ray irradiation, X-ray irradiation, or any combination thereof. Radiation sterilization, however, can damage the electrical components arranged within the electronics housing of the sensor control device **102**. Consequently, if the sensor applicator **102**, which contains the electronics housing of the sensor control device **102**, needs to be sterilized, it may be sterilized via another method, such as gaseous chemical sterilization using, for example, ethylene oxide. Gaseous chemical sterilization, however, can damage the enzymes or other chemistry and biologies included on the sensor **110**. Because of this sterilization incompatibility, the sensor tray **202** and the sensor applicator **102** are commonly sterilized in separate sterilization processes and subsequently packaged separately, which requires the user to finally assemble the components for use.

FIGS. 7A and 7B are exploded top and bottom views, respectively, of the sensor control device **3702**, according to one or more embodiments. The shell **3706** and the mount **3708** operate as opposing clamshell halves that enclose or otherwise substantially encapsulate the various electronic components of the sensor control device **3702**. As illustrated, the sensor control device **3702** may include a printed circuit board assembly (PCBA) **3802** that includes a printed circuit board (PCB) **3804** having a plurality of electronic modules **3806** coupled thereto. Example electronic modules **3806** include, but are not limited to, resistors, transistors, capacitors, inductors, diodes, and switches. Prior sensor control devices commonly stack PCB components on only one side of the PCB. In contrast, the PCB components **3806** in the sensor control device **3702** can be dispersed about the surface area of both sides (i.e., top and bottom surfaces) of the PCB **3804**.

Besides the electronic modules **3806**, the PCBA **3802** may also include a data processing unit **3808** mounted to the PCB **3804**. The data processing unit **3808** may comprise, for example, an application specific integrated circuit (ASIC) configured to implement one or more functions or routines associated with operation of the sensor control device **3702**. More specifically, the data processing unit **3808** may be configured to perform data processing functions, where such functions may include but are not limited to, filtering and encoding of data signals, each of which corresponds to a sampled analyte level of the user. The data processing unit **3808** may also include or otherwise communicate with an antenna for communicating with the reader device **106** (FIG. 1A).

A battery aperture **3810** may be defined in the PCB **3804** and sized to receive and seat a battery **3812** configured to power the sensor control device **3702**. An axial battery contact **3814a** and a radial battery contact **3814b** may be coupled to the PCB **3804** and extend into the battery aperture **3810** to facilitate transmission of electrical power from the battery **3812** to the PCB **3804**. As their names suggest, the axial battery contact **3814a** may be configured to provide an axial contact for the battery **3812**, while the radial battery contact **3814b** may provide a radial contact for the battery **3812**. Locating the battery **3812** within the battery aperture **3810** with the battery contacts **3814a,b** helps reduce the height H of the sensor control device **3702**, which allows the PCB **3804** to be located centrally and its components to be dispersed on both sides (i.e., top and bottom surfaces). This also helps facilitate the chamfer **3718** provided on the electronics housing **3704**.

The sensor **3716** may be centrally located relative to the PCB **3804** and include a tail **3816**, a flag **3818**, and a neck **3820** that interconnects the tail **3816** and the flag **3818**. The

tail **3816** may be configured to extend through the central aperture **3720** of the mount **3708** to be transcutaneously received beneath a user's skin. Moreover, the tail **3816** may have an enzyme or other chemistry included thereon to help facilitate analyte monitoring.

The flag **3818** may include a generally planar surface having one or more sensor contacts **3822** (three shown in FIG. 7B) arranged thereon. The sensor contact(s) **3822** may be configured to align with and engage a corresponding one or more circuitry contacts **3824** (three shown in FIG. 7A) provided on the PCB **3804**. In some embodiments, the sensor contact(s) **3822** may comprise a carbon impregnated polymer printed or otherwise digitally applied to the flag **3818**. Prior sensor control devices typically include a connector made of silicone rubber that encapsulates one or more compliant carbon impregnated polymer modules that serve as electrical conductive contacts between the sensor and the PCB. In contrast, the presently disclosed sensor contact(s) **3822** provide a direct connection between the sensor **3716** and the PCB **3804** connection, which eliminates the need for the prior art connector and advantageously reduces the height H. Moreover, eliminating the compliant carbon impregnated polymer modules eliminates a significant circuit resistance and therefor improves circuit conductivity.

The sensor control device **3702** may further include a compliant member **3826**, which may be arranged to interpose the flag **3818** and the inner surface of the shell **3706**. More specifically, when the shell **3706** and the mount **3708** are assembled to one another, the compliant member **3826** may be configured to provide a passive biasing load against the flag **3818** that forces the sensor contact(s) **3822** into continuous engagement with the corresponding circuitry contact(s) **3824**. In the illustrated embodiment, the compliant member **3826** is an elastomeric O-ring, but could alternatively comprise any other type of biasing device or mechanism, such as a compression spring or the like, without departing from the scope of the disclosure.

The sensor control device **3702** may further include one or more electromagnetic shields, shown as a first shield **3828a** and a second shield **3828b**. The shell **3706** may provide or otherwise define a first clocking receptacle **3830a** (FIG. 7B) and a second clocking receptacle **3830b** (FIG. 7B), and the mount **3708** may provide or otherwise define a first clocking post **3832a** (FIG. 7A) and a second clocking post **3832b** (FIG. 7A). Mating the first and second clocking receptacles **3830a,b** with the first and second clocking posts **3832a,b**, respectively, will properly align the shell **3706** to the mount **3708**.

Referring specifically to FIG. 7A, the inner surface of the mount **3708** may provide or otherwise define a plurality of pockets or depressions configured to accommodate various component parts of the sensor control device **3702** when the shell **3706** is mated to the mount **3708**. For example, the inner surface of the mount **3708** may define a battery locator **3834** configured to accommodate a portion of the battery **3812** when the sensor control device **3702** is assembled. An adjacent contact pocket **3836** may be configured to accommodate a portion of the axial contact **3814a**.

Moreover, a plurality of module pockets **3838** may be defined in the inner surface of the mount **3708** to accommodate the various electronic modules **3806** arranged on the bottom of the PCB **3804**. Furthermore, a shield locator **3840** may be defined in the inner surface of the mount **3708** to accommodate at least a portion of the second shield **3828b** when the sensor control device **3702** is assembled. The battery locator **3834**, the contact pocket **3836**, the module pockets **3838**, and the shield locator **3840** all extend a short

distance into the inner surface of the mount 3708 and, as a result, the overall height H of the sensor control device 3702 may be reduced as compared to prior sensor control devices. The module pockets 3838 may also help minimize the diameter of the PCB 3804 by allowing PCB components to be arranged on both sides (i.e., top and bottom surfaces).

Still referring to FIG. 7A, the mount 3708 may further include a plurality of carrier grip features 3842 (two shown) defined about the outer periphery of the mount 3708. The carrier grip features 3842 are axially offset from the bottom 3844 of the mount 3708, where a transfer adhesive (not shown) may be applied during assembly. In contrast to prior sensor control devices, which commonly include conical carrier grip features that intersect with the bottom of the mount, the presently disclosed carrier grip features 3842 are offset from the plane (i.e., the bottom 3844) where the transfer adhesive is applied. This may prove advantageous in helping ensure that the delivery system does not inadvertently stick to the transfer adhesive during assembly. Moreover, the presently disclosed carrier grip features 3842 eliminate the need for a scalloped transfer adhesive, which simplifies the manufacture of the transfer adhesive and eliminates the need to accurately clock the transfer adhesive relative to the mount 3708. This also increases the bond area and, therefore, the bond strength.

Referring to FIG. 7B, the bottom 3844 of the mount 3708 may provide or otherwise define a plurality of grooves 3846, which may be defined at or near the outer periphery of the mount 3708 and equidistantly spaced from each other. A transfer adhesive (not shown) may be coupled to the bottom 3844 and the grooves 3846 may be configured to help convey (transfer) moisture away from the sensor control device 3702 and toward the periphery of the mount 3708 during use. In some embodiments, the spacing of the grooves 3846 may interpose the module pockets 3838 (FIG. 7A) defined on the opposing side (inner surface) of the mount 3708. As will be appreciated, alternating the position of the grooves 3846 and the module pockets 3838 ensures that the opposing features on either side of the mount 3708 do not extend into each other. This may help maximize usage of the material for the mount 3708 and thereby help maintain a minimal height H of the sensor control device 3702. The module pockets 3838 may also significantly reduce mold sink, and improve the flatness of the bottom 3844 that the transfer adhesive bonds to.

Still referring to FIG. 7B, the inner surface of the shell 3706 may also provide or otherwise define a plurality of pockets or depressions configured to accommodate various component parts of the sensor control device 3702 when the shell 3706 is mated to the mount 3708. For example, the inner surface of the shell 3706 may define an opposing battery locator 3848 arrangeable opposite the battery locator 3834 (FIG. 7A) of the mount 3708 and configured to accommodate a portion of the battery 3812 when the sensor control device 3702 is assembled. The opposing battery locator 3848 extends a short distance into the inner surface of the shell 3706, which helps reduce the overall height H of the sensor control device 3702.

A sharp and sensor locator 3852 may also be provided by or otherwise defined on the inner surface of the shell 3706. The sharp and sensor locator 3852 may be configured to receive both the sharp (not shown) and a portion of the sensor 3716. Moreover, the sharp and sensor locator 3852 may be configured to align and/or mate with a corresponding sharp and sensor locator 2054 (FIG. 7A) provided on the inner surface of the mount 3708.

According to embodiments of the present disclosure, an alternative sensor assembly/electronics assembly connection approach is illustrated in FIGS. 8A to 8C. As shown, the sensor assembly 14702 includes sensor 14704, connector support 14706, and sharp 14708. Notably, a recess or receptacle 14710 may be defined in the bottom of the mount of the electronics assembly 14712 and provide a location where the sensor assembly 14702 may be received and coupled to the electronics assembly 14712, and thereby fully assemble the sensor control device. The profile of the sensor assembly 14702 may match or be shaped in complementary fashion to the receptacle 14710, which includes an elastomeric sealing member 14714 (including conductive material coupled to the circuit board and aligned with the electrical contacts of the sensor 14704). Thus, when the sensor assembly 14702 is snap fit or otherwise adhered to the electronics assembly 14712 by driving the sensor assembly 14702 into the integrally formed recess 14710 in the electronics assembly 14712, the on-body device 14714 depicted in FIG. 8C is formed. This embodiment provides an integrated connector for the sensor assembly 14702 within the electronics assembly 14712.

Additional information regarding sensor assemblies is provided in U.S. Publication No. 2013/0150691 and U.S. Publication No. 2021/0204841, each of which is incorporated by reference herein in its entirety.

According to embodiments of the present disclosure, the sensor control device 102 may be modified to provide a one-piece architecture that may be subjected to sterilization techniques specifically designed for a one-piece architecture sensor control device. A one-piece architecture allows the sensor applicator 150 and the sensor control device 102 to be shipped to the user in a single, sealed package that does not require any final user assembly steps. Rather, the user need only open one package and subsequently deliver the sensor control device 102 to the target monitoring location. The one-piece system architecture described herein may prove advantageous in eliminating component parts, various fabrication process steps, and user assembly steps. As a result, packaging and waste are reduced, and the potential for user error or contamination to the system is mitigated.

FIGS. 9A and 9B are side and cross-sectional side views, respectively, of an example embodiment of the sensor applicator 102 with the applicator cap 210 coupled thereto. More specifically, FIG. 9A depicts how the sensor applicator 102 might be shipped to and received by a user, and FIG. 9B depicts the sensor control device 4402 arranged within the sensor applicator 102. Accordingly, the fully assembled sensor control device 4402 may already be assembled and installed within the sensor applicator 102 prior to being delivered to the user, thus removing any additional assembly steps that a user would otherwise have to perform.

The fully assembled sensor control device 4402 may be loaded into the sensor applicator 102, and the applicator cap 210 may subsequently be coupled to the sensor applicator 102. In some embodiments, the applicator cap 210 may be threaded to the housing 208 and include a tamper ring 4702. Upon rotating (e.g., unscrewing) the applicator cap 210 relative to the housing 208, the tamper ring 4702 may shear and thereby free the applicator cap 210 from the sensor applicator 102.

According to the present disclosure, while loaded in the sensor applicator 102, the sensor control device 4402 may be subjected to gaseous chemical sterilization 4704 configured to sterilize the electronics housing 4404 and any other exposed portions of the sensor control device 4402. To accomplish this, a chemical may be injected into a steril-

ization chamber **4706** cooperatively defined by the sensor applicator **102** and the interconnected cap **210**. In some applications, the chemical may be injected into the sterilization chamber **4706** via one or more vents **4708** defined in the applicator cap **210** at its proximal end **610**. Example chemicals that may be used for the gaseous chemical sterilization **4704** include, but are not limited to, ethylene oxide, vaporized hydrogen peroxide, nitrogen oxide (e.g., nitrous oxide, nitrogen dioxide, etc.), and steam.

Since the distal portions of the sensor **4410** and the sharp **4412** are sealed within the sensor cap **4416**, the chemicals used during the gaseous chemical sterilization process do not interact with the enzymes, chemistry, and biologics provided on the tail **4524** and other sensor components, such as membrane coatings that regulate analyte influx.

Once a desired sterility assurance level has been achieved within the sterilization chamber **4706**, the gaseous solution may be removed and the sterilization chamber **4706** may be aerated. Aeration may be achieved by a series of vacuums and subsequently circulating a gas (e.g., nitrogen) or filtered air through the sterilization chamber **4706**. Once the sterilization chamber **4706** is properly aerated, the vents **4708** may be occluded with a seal **4712** (shown in dashed lines).

In some embodiments, the seal **4712** may comprise two or more layers of different materials. The first layer may be made of a synthetic material (e.g., a flash-spun high-density polyethylene fiber), such as Tyvek® available from DuPont®. Tyvek® is highly durable and puncture resistant and allows the permeation of vapors. The Tyvek® layer can be applied before the gaseous chemical sterilization process, and following the gaseous chemical sterilization process, a foil or other vapor and moisture resistant material layer may be sealed (e.g., heat sealed) over the Tyvek® layer to prevent the ingress of contaminants and moisture into the sterilization chamber **4706**. In other embodiments, the seal **4712** may comprise only a single protective layer applied to the applicator cap **210**. In such embodiments, the single layer may be gas permeable for the sterilization process, but may also be capable of protection against moisture and other harmful elements once the sterilization process is complete.

With the seal **4712** in place, the applicator cap **210** provides a barrier against outside contamination, and thereby maintains a sterile environment for the assembled sensor control device **4402** until the user removes (unthreads) the applicator cap **210**. The applicator cap **210** may also create a dust-free environment during shipping and storage that prevents the adhesive patch **4714** from becoming dirty.

FIGS. **10A** and **10B** are isometric and side views, respectively, of another example sensor control device **5002**, according to one or more embodiments of the present disclosure. The sensor control device **5002** may be similar in some respects to the sensor control device **102** of FIG. **1A** and therefore may be best understood with reference thereto. Moreover, the sensor control device **5002** may replace the sensor control device **102** of FIG. **1A** and, therefore, may be used in conjunction with the sensor applicator **102** of FIG. **1A**, which may deliver the sensor control device **5002** to a target monitoring location on a user's skin.

Unlike the sensor control device **102** of FIG. **1A**, however, the sensor control device **5002** may comprise a one-piece system architecture not requiring a user to open multiple packages and finally assemble the sensor control device **5002** prior to application. Rather, upon receipt by the user, the sensor control device **5002** may already be fully assembled and properly positioned within the sensor applicator **150** (FIG. **1A**). To use the sensor control device **5002**,

the user need only open one barrier (e.g., the applicator cap **708** of FIG. **3B**) before promptly delivering the sensor control device **5002** to the target monitoring location for use.

As illustrated, the sensor control device **5002** includes an electronics housing **5004** that is generally disc-shaped and may have a circular cross-section. In other embodiments, however, the electronics housing **5004** may exhibit other cross-sectional shapes, such as ovoid or polygonal, without departing from the scope of the disclosure. The electronics housing **5004** may be configured to house or otherwise contain various electrical components used to operate the sensor control device **5002**. In at least one embodiment, an adhesive patch (not shown) may be arranged at the bottom of the electronics housing **5004**. The adhesive patch may be similar to the adhesive patch **105** of FIG. **1A**, and may thus help adhere the sensor control device **5002** to the user's skin for use.

As illustrated, the sensor control device **5002** includes an electronics housing **5004** that includes a shell **5006** and a mount **5008** that is matable with the shell **5006**. The shell **5006** may be secured to the mount **5008** via a variety of ways, such as a snap fit engagement, an interference fit, sonic welding, one or more mechanical fasteners (e.g., screws), a gasket, an adhesive, or any combination thereof. In some cases, the shell **5006** may be secured to the mount **5008** such that a sealed interface is generated therebetween.

The sensor control device **5002** may further include a sensor **5010** (partially visible) and a sharp **5012** (partially visible), used to help deliver the sensor **5010** transcutaneously under a user's skin during application of the sensor control device **5002**. As illustrated, corresponding portions of the sensor **5010** and the sharp **5012** extend distally from the bottom of the electronics housing **5004** (e.g., the mount **5008**). The sharp **5012** may include a sharp hub **5014** configured to secure and carry the sharp **5012**. As best seen in FIG. **10B**, the sharp hub **5014** may include or otherwise define a mating member **5016**. To couple the sharp **5012** to the sensor control device **5002**, the sharp **5012** may be advanced axially through the electronics housing **5004** until the sharp hub **5014** engages an upper surface of the shell **5006** and the mating member **5016** extends distally from the bottom of the mount **5008**. As the sharp **5012** penetrates the electronics housing **5004**, the exposed portion of the sensor **5010** may be received within a hollow or recessed (arcuate) portion of the sharp **5012**. The remaining portion of the sensor **5010** is arranged within the interior of the electronics housing **5004**.

The sensor control device **5002** may further include a sensor cap **5018**, shown exploded or detached from the electronics housing **5004** in FIGS. **10A-10B**. The sensor cap **5018** may be removably coupled to the sensor control device **5002** (e.g., the electronics housing **5004**) at or near the bottom of the mount **5008**. The sensor cap **5018** may help provide a sealed barrier that surrounds and protects the exposed portions of the sensor **5010** and the sharp **5012** from gaseous chemical sterilization. As illustrated, the sensor cap **5018** may comprise a generally cylindrical body having a first end **5020a** and a second end **5020b** opposite the first end **5020a**. The first end **5020a** may be open to provide access into an inner chamber **5022** defined within the body. In contrast, the second end **5020b** may be closed and may provide or otherwise define an engagement feature **5024**. As described herein, the engagement feature **5024** may help mate the sensor cap **5018** to the cap (e.g., the applicator cap **708** of FIG. **3B**) of a sensor applicator (e.g., the sensor applicator **150** of FIGS. **1** and **3A-3G**), and may help remove

25

the sensor cap **5018** from the sensor control device **5002** upon removing the cap from the sensor applicator.

The sensor cap **5018** may be removably coupled to the electronics housing **5004** at or near the bottom of the mount **5008**. More specifically, the sensor cap **5018** may be removably coupled to the mating member **5016**, which extends distally from the bottom of the mount **5008**. In at least one embodiment, for example, the mating member **5016** may define a set of external threads **5026a** (FIG. 10B) matable with a set of internal threads **5026b** (FIG. 10A) defined by the sensor cap **5018**. In some embodiments, the external and internal threads **5026a, b** may comprise a flat thread design (e.g., lack of helical curvature), which may prove advantageous in molding the parts. Alternatively, the external and internal threads **5026a, b** may comprise a helical threaded engagement. Accordingly, the sensor cap **5018** may be threadably coupled to the sensor control device **5002** at the mating member **5016** of the sharp hub **5014**. In other embodiments, the sensor cap **5018** may be removably coupled to the mating member **5016** via other types of engagements including, but not limited to, an interference or friction fit, or a frangible member or substance that may be broken with minimal separation force (e.g., axial or rotational force).

In some embodiments, the sensor cap **5018** may comprise a monolithic (singular) structure extending between the first and second ends **5020a, b**. In other embodiments, however, the sensor cap **5018** may comprise two or more component parts. In the illustrated embodiment, for example, the sensor cap **5018** may include a seal ring **5028** positioned at the first end **5020a** and a desiccant cap **5030** arranged at the second end **5020b**. The seal ring **5028** may be configured to help seal the inner chamber **5022**, as described in more detail below. In at least one embodiment, the seal ring **5028** may comprise an elastomeric O-ring. The desiccant cap **5030** may house or comprise a desiccant to help maintain preferred humidity levels within the inner chamber **5022**. The desiccant cap **5030** may also define or otherwise provide the engagement feature **5024** of the sensor cap **5018**.

FIGS. 11A-11C are progressive cross-sectional side views showing assembly of the sensor applicator **102** with the sensor control device **5002**, according to one or more embodiments. Once the sensor control device **5002** is fully assembled, it may then be loaded into the sensor applicator **102**. With reference to FIG. 11A, the sharp hub **5014** may include or otherwise define a hub snap pawl **5302** configured to help couple the sensor control device **5002** to the sensor applicator **102**. More specifically, the sensor control device **5002** may be advanced into the interior of the sensor applicator **102** and the hub snap pawl **5302** may be received by corresponding arms **5304** of a sharp carrier **5306** positioned within the sensor applicator **102**.

In FIG. 11B, the sensor control device **5002** is shown received by the sharp carrier **5306** and, therefore, secured within the sensor applicator **102**. Once the sensor control device **5002** is loaded into the sensor applicator **102**, the applicator cap **210** may be coupled to the sensor applicator **102**. In some embodiments, the applicator cap **210** and the housing **208** may have opposing, matable sets of threads **5308** that enable the applicator cap **210** to be screwed onto the housing **208** in a clockwise (or counter-clockwise) direction and thereby secure the applicator cap **210** to the sensor applicator **102**.

As illustrated, the sheath **212** is also positioned within the sensor applicator **102**, and the sensor applicator **102** may include a sheath locking mechanism **5310** configured to ensure that the sheath **212** does not prematurely collapse

26

during a shock event. In the illustrated embodiment, the sheath locking mechanism **5310** may comprise a threaded engagement between the applicator cap **210** and the sheath **212**. More specifically, one or more internal threads **5312a** may be defined or otherwise provided on the inner surface of the applicator cap **210**, and one or more external threads **5312b** may be defined or otherwise provided on the sheath **212**. The internal and external threads **5312a, b** may be configured to threadably mate as the applicator cap **210** is threaded to the sensor applicator **102** at the threads **5308**. The internal and external threads **5312a, b** may have the same thread pitch as the threads **5308** that enable the applicator cap **210** to be screwed onto the housing **208**.

In FIG. 11C, the applicator cap **210** is shown fully threaded (coupled) to the housing **208**. As illustrated, the applicator cap **210** may further provide and otherwise define a cap post **5314** centrally located within the interior of the applicator cap **210** and extending proximally from the bottom thereof. The cap post **5314** may be configured to receive at least a portion of the sensor cap **5018** as the applicator cap **210** is screwed onto the housing **208**.

With the sensor control device **5002** loaded within the sensor applicator **102** and the applicator cap **210** properly secured, the sensor control device **5002** may then be subjected to a gaseous chemical sterilization configured to sterilize the electronics housing **5004** and any other exposed portions of the sensor control device **5002**. Since the distal portions of the sensor **5010** and the sharp **5012** are sealed within the sensor cap **5018**, the chemicals used during the gaseous chemical sterilization process are unable to interact with the enzymes, chemistry, and biologies provided on the tail **5104**, and other sensor components, such as membrane coatings that regulate analyte influx.

FIGS. 12A-12C are progressive cross-sectional side views showing assembly and disassembly of an alternative embodiment of the sensor applicator **102** with the sensor control device **5002**, according to one or more additional embodiments. A fully assembled sensor control device **5002** may be loaded into the sensor applicator **102** by coupling the hub snap pawl **5302** into the arms **5304** of the sharp carrier **5306** positioned within the sensor applicator **102**, as generally described above.

In the illustrated embodiment, the sheath arms **5604** of the sheath **212** may be configured to interact with a first detent **5702a** and a second detent **5702b** defined within the interior of the housing **208**. The first detent **5702a** may alternately be referred to a “locking” detent, and the second detent **5702b** may alternately be referred to as a “firing” detent. When the sensor control device **5002** is initially installed in the sensor applicator **102**, the sheath arms **5604** may be received within the first detent **5702a**. As discussed below, the sheath **212** may be actuated to move the sheath arms **5604** to the second detent **5702b**, which places the sensor applicator **102** in firing position.

In FIG. 12B, the applicator cap **210** is aligned with the housing **208** and advanced toward the housing **208** so that the sheath **212** is received within the applicator cap **210**. Instead of rotating the applicator cap **210** relative to the housing **208**, the threads of the applicator cap **210** may be snapped onto the corresponding threads of the housing **208** to couple the applicator cap **210** to the housing **208**. Axial cuts or slots **5703** (one shown) defined in the applicator cap **210** may allow portions of the applicator cap **210** near its threading to flex outward to be snapped into engagement with the threading of the housing **208**. As the applicator cap **210** is snapped to the housing **208**, the sensor cap **5018** may correspondingly be snapped into the cap post **5314**.

Similar to the embodiment of FIGS. 11A-11C, the sensor applicator 102 may include a sheath locking mechanism configured to ensure that the sheath 212 does not prematurely collapse during a shock event. In the illustrated embodiment, the sheath locking mechanism includes one or more ribs 5704 (one shown) defined near the base of the sheath 212 and configured to interact with one or more ribs 5706 (two shown) and a shoulder 5708 defined near the base of the applicator cap 210. The ribs 5704 may be configured to inter-lock between the ribs 5706 and the shoulder 5708 while attaching the applicator cap 210 to the housing 208. More specifically, once the applicator cap 210 is snapped onto the housing 208, the applicator cap 210 may be rotated (e.g., clockwise), which locates the ribs 5704 of the sheath 212 between the ribs 5706 and the shoulder 5708 of the applicator cap 210 and thereby “locks” the applicator cap 210 in place until the user reverse rotates the applicator cap 210 to remove the applicator cap 210 for use. Engagement of the ribs 5704 between the ribs 5706 and the shoulder 5708 of the applicator cap 210 may also prevent the sheath 212 from collapsing prematurely.

In FIG. 12C, the applicator cap 210 is removed from the housing 208. As with the embodiment of FIGS. 21A-21C, the applicator cap 210 can be removed by reverse rotating the applicator cap 210, which correspondingly rotates the cap post 5314 in the same direction and causes sensor cap 5018 to unthread from the mating member 5016, as generally described above. Moreover, detaching the sensor cap 5018 from the sensor control device 5002 exposes the distal portions of the sensor 5010 and the sharp 5012.

As the applicator cap 210 is unscrewed from the housing 208, the ribs 5704 defined on the sheath 212 may slidably engage the tops of the ribs 5706 defined on the applicator cap 210. The tops of the ribs 5706 may provide corresponding ramped surfaces that result in an upward displacement of the sheath 212 as the applicator cap 210 is rotated, and moving the sheath 212 upward causes the sheath arms 5604 to flex out of engagement with the first detent 5702a to be received within the second detent 5702b. As the sheath 212 moves to the second detent 5702b, the radial shoulder 5614 moves out of radial engagement with the carrier arm(s) 5608, which allows the passive spring force of the spring 5612 to push upward on the sharp carrier 5306 and force the carrier arm(s) 5608 out of engagement with the groove(s) 5610. As the sharp carrier 5306 moves upward within the housing 208, the mating member 5016 may correspondingly retract until it becomes flush, substantially flush, or sub-flush with the bottom of the sensor control device 5002. At this point, the sensor applicator 102 is in firing position. Accordingly, in this embodiment, removing the applicator cap 210 correspondingly causes the mating member 5016 to retract.

FIGS. 13A-13F illustrate example details of embodiments of the internal device mechanics of “firing” the applicator 216 to apply sensor control device 222 to a user and including retracting sharp 1030 safely back into used applicator 216. All together, these drawings represent an example sequence of driving sharp 1030 (supporting a sensor coupled to sensor control device 222) into the skin of a user, withdrawing the sharp while leaving the sensor behind in operative contact with interstitial fluid of the user, and adhering the sensor control device to the skin of the user with an adhesive. Modification of such activity for use with the alternative applicator assembly embodiments and components can be appreciated in reference to the same by those with skill in the art. Moreover, applicator 216 may be a

sensor applicator having one-piece architecture or a two-piece architecture as disclosed herein.

Turning now to FIG. 13A, a sensor 1102 is supported within sharp 1030, just above the skin 1104 of the user. Rails 1106 (optionally three of them) of an upper guide section 1108 may be provided to control applicator 216 motion relative to sheath 318. The sheath 318 is held by detent features 1110 within the applicator 216 such that appropriate downward force along the longitudinal axis of the applicator 216 will cause the resistance provided by the detent features 1110 to be overcome so that sharp 1030 and sensor control device 222 can translate along the longitudinal axis into (and onto) skin 1104 of the user. In addition, catch arms 1112 of sensor carrier 1022 engage the sharp retraction assembly 1024 to maintain the sharp 1030 in a position relative to the sensor control device 222.

In FIG. 13B, user force is applied to overcome or override detent features 1110 and sheath 318 collapses into housing 314 driving the sensor control device 222 (with associated parts) to translate down as indicated by the arrow L along the longitudinal axis. An inner diameter of the upper guide section 1108 of the sheath 318 constrains the position of carrier arms 1112 through the full stroke of the sensor/sharp insertion process. The retention of the stop surfaces 1114 of carrier arms 1112 against the complimentary faces 1116 of the sharp retraction assembly 1024 maintains the position of the members with return spring 1118 fully energized. According to embodiments, rather than employing user force to drive the sensor control device 222 to translate down as indicated by the arrow L along the longitudinal axis, housing 314 can include a button (for example, not limitation, a push button) which activates a drive spring (for example, not limitation, a coil spring) to drive the sensor control device 222.

In FIG. 13C, sensor 1102 and sharp 1030 have reached full insertion depth. In so doing, the carrier arms 1112 clear the upper guide section 1108 inner diameter. Then, the compressed force of the coil return spring 1118 drives angled stop surfaces 1114 radially outward, releasing force to drive the sharp carrier 1102 of the sharp retraction assembly 1024 to pull the (slotted or otherwise configured) sharp 1030 out of the user and off of the sensor 1102 as indicated by the arrow R in FIG. 13D.

With the sharp 1030 fully retracted as shown in FIG. 13E, the upper guide section 1108 of the sheath 318 is set with a final locking feature 1120. As shown in FIG. 13F, the spent applicator assembly 216 is removed from the insertion site, leaving behind the sensor control device 222, and with the sharp 1030 secured safely inside the applicator assembly 216. The spent applicator assembly 216 is now ready for disposal.

Operation of the applicator 216 when applying the sensor control device 222 is designed to provide the user with a sensation that both the insertion and retraction of the sharp 1030 is performed automatically by the internal mechanisms of the applicator 216. In other words, the present invention avoids the user experiencing the sensation that he is manually driving the sharp 1030 into his skin. Thus, once the user applies sufficient force to overcome the resistance from the detent features of the applicator 216, the resulting actions of the applicator 216 are perceived to be an automated response to the applicator being “triggered.” The user does not perceive that he is supplying additional force to drive the sharp 1030 to pierce his skin despite that all the driving force is provided by the user and no additional biasing/driving means are used to insert the sharp 1030. As detailed above

in FIG. 13C, the retraction of the sharp 1030 is automated by the coil return spring 1118 of the applicator 216.

With respect to any of the applicator embodiments described herein, as well as any of the components thereof, including but not limited to the sharp, sharp module and sensor module embodiments, those of skill in the art will understand that said embodiments can be dimensioned and configured for use with sensors configured to sense an analyte level in a bodily fluid in the epidermis, dermis, or subcutaneous tissue of a subject. In some embodiments, for example, sharps and distal portions of analyte sensors disclosed herein can both be dimensioned and configured to be positioned at a particular end-depth (i.e., the furthest point of penetration in a tissue or layer of the subject's body, e.g., in the epidermis, dermis, or subcutaneous tissue). With respect to some applicator embodiments, those of skill in the art will appreciate that certain embodiments of sharps can be dimensioned and configured to be positioned at a different end-depth in the subject's body relative to the final end-depth of the analyte sensor. In some embodiments, for example, a sharp can be positioned at a first end-depth in the subject's epidermis prior to retraction, while a distal portion of an analyte sensor can be positioned at a second end-depth in the subject's dermis. In other embodiments, a sharp can be positioned at a first end-depth in the subject's dermis prior to retraction, while a distal portion of an analyte sensor can be positioned at a second end-depth in the subject's subcutaneous tissue. In still other embodiments, a sharp can be positioned at a first end-depth prior to retraction and the analyte sensor can be positioned at a second end-depth, wherein the first end-depth and second end-depths are both in the same layer or tissue of the subject's body.

Additionally, with respect to any of the applicator embodiments described herein, those of skill in the art will understand that an analyte sensor, as well as one or more structural components coupled thereto, including but not limited to one or more spring-mechanisms, can be disposed within the applicator in an off-center position relative to one or more axes of the applicator. In some applicator embodiments, for example, an analyte sensor and a spring mechanism can be disposed in a first off-center position relative to an axis of the applicator on a first side of the applicator, and the sensor electronics can be disposed in a second off-center position relative to the axis of the applicator on a second side of the applicator. In other applicator embodiments, the analyte sensor, spring mechanism, and sensor electronics can be disposed in an off-center position relative to an axis of the applicator on the same side. Those of skill in the art will appreciate that other permutations and configurations in which any or all of the analyte sensor, spring mechanism, sensor electronics, and other components of the applicator are disposed in a centered or off-centered position relative to one or more axes of the applicator are possible and fully within the scope of the present disclosure.

Additional details of suitable devices, systems, methods, components and the operation thereof along with related features are set forth in International Publication No. WO2018/136898 to Rao et al., International Publication No. WO2019/236850 to Thomas et al., International Publication No. WO2019/236859 to Thomas et al., International Publication No. WO2019/236876 to Thomas et al., and U.S. Patent Publication No. 2020/0196919, filed Jun. 6, 2019, each of which is incorporated by reference in its entirety herein. Further details regarding embodiments of applicators, their components, and variants thereof, are described in U.S. Patent Publication Nos. 2013/0150691, 2016/0331283, and 2018/0235520, all of which are incorporated by refer-

ence herein in their entireties and for all purposes. Further details regarding embodiments of sharp modules, sharps, their components, and variants thereof, are described in U.S. Patent Publication No. 2014/0171771, which is incorporated by reference herein in its entirety and for all purposes.

J. Exemplary Methods of Calibrating Analyte Sensors

Biochemical sensors can be described by one or more sensing characteristics. A common sensing characteristic is referred to as the biochemical sensor's sensitivity, which is a measure of the sensor's responsiveness to the concentration of the chemical or composition it is designed to detect. For electrochemical sensors, this response can be in the form of an electrical current (amperometric) or electrical charge (coulometric). For other types of sensors, the response can be in a different form, such as a photonic intensity (e.g., optical light). The sensitivity of a biochemical analyte sensor can vary depending on a number of factors, including whether the sensor is in an in vitro state or an in vivo state.

FIG. 14 is a graph depicting the in vitro sensitivity of an amperometric analyte sensor. The in vitro sensitivity can be obtained by in vitro testing the sensor at various analyte concentrations and then performing a regression (e.g., linear or non-linear) or other curve fitting on the resulting data. In this example, the analyte sensor's sensitivity is linear, or substantially linear, and can be modeled according to the equation $y=mx+b$, where y is the sensor's electrical output current, x is the analyte level (or concentration), m is the slope of the sensitivity and b is the intercept of the sensitivity, where the intercept generally corresponds to a background signal (e.g., noise). For sensors with a linear or substantially linear response, the analyte level that corresponds to a given current can be determined from the slope and intercept of the sensitivity. Sensors with a non-linear sensitivity require additional information to determine the analyte level resulting from the sensor's output current, and those of ordinary skill in the art are familiar with manners by which to model non-linear sensitivities. In certain embodiments of in vivo sensors, the in vitro sensitivity can be the same as the in vivo sensitivity, but in other embodiments a transfer (or conversion) function is used to translate the in vitro sensitivity into the in vivo sensitivity that is applicable to the sensor's intended in vivo use.

Calibration is a technique for improving or maintaining accuracy by adjusting a sensor's measured output to reduce the differences with the sensor's expected output. One or more parameters that describe the sensor's sensing characteristics, like its sensitivity, are established for use in the calibration adjustment.

Certain in vivo analyte monitoring systems require calibration to occur after implantation of the sensor into the user or patient, either by user interaction or by the system itself in an automated fashion. For example, when user interaction is required, the user performs an in vitro measurement (e.g., a blood glucose (BG) measurement using a finger stick and an in vitro test strip) and enters this into the system, while the analyte sensor is implanted. The system then compares the in vitro measurement with the in vivo signal and, using the differential, determines an estimate of the sensor's in vivo sensitivity. The in vivo sensitivity can then be used in an algorithmic process to transform the data collected with the sensor to a value that indicates the user's analyte level. This and other processes that require user action to perform calibration are referred to as "user calibration." Systems can require user calibration due to instability of the sensor's sensitivity, such that the sensitivity drifts or changes over time. Thus, multiple user calibrations (e.g., according to a periodic (e.g., daily) schedule, variable schedule, or on an

as-needed basis) can be required to maintain accuracy. While the embodiments described herein can incorporate a degree of user calibration for a particular implementation, generally this is not preferred as it requires the user to perform a painful or otherwise burdensome BG measurement, and can introduce user error.

Some in vivo analyte monitoring systems can regularly adjust the calibration parameters through the use of automated measurements of characteristics of the sensor made by the system itself (e.g., processing circuitry executing software). The repeated adjustment of the sensor's sensitivity based on a variable measured by the system (and not the user) is referred to generally as "system" (or automated) calibration, and can be performed with user calibration, such as an early BG measurement, or without user calibration. Like the case with repeated user calibrations, repeated system calibrations are typically necessitated by drift in the sensor's sensitivity over time. Thus, while the embodiments described herein can be used with a degree of automated system calibration, preferably the sensor's sensitivity is relatively stable over time such that post-implantation calibration is not required.

Some in vivo analyte monitoring systems operate with a sensor that is factory calibrated. Factory calibration refers to the determination or estimation of the one or more calibration parameters prior to distribution to the user or healthcare professional (HCP). The calibration parameter can be determined by the sensor manufacturer (or the manufacturer of the other components of the sensor control device if the two entities are different). Many in vivo sensor manufacturing processes fabricate the sensors in groups or batches referred to as production lots, manufacturing stage lots, or simply lots. A single lot can include thousands of sensors.

Sensors can include a calibration code or parameter which can be derived or determined during one or more sensor manufacturing processes and coded or programmed, as part of the manufacturing process, in the data processing device of the analyte monitoring system or provided on the sensor itself, for example, as a bar code, a laser tag, an RFID tag, or other machine readable information provided on the sensor. User calibration during in vivo use of the sensor can be obviated, or the frequency of in vivo calibrations during sensor wear can be reduced if the code is provided to a receiver (or other data processing device). In embodiments where the calibration code or parameter is provided on the sensor itself, prior to or at the start of the sensor use, the calibration code or parameter can be automatically transmitted or provided to the data processing device in the analyte monitoring system.

Some in vivo analyte monitoring system operate with a sensor that can be one or more of factory calibrated, system calibrated, and/or user calibrated. For example, the sensor can be provided with a calibration code or parameter which can allow for factory calibration. If the information is provided to a receiver (for example, entered by a user), the sensor can operate as a factory calibrated sensor. If the information is not provided to a receiver, the sensor can operate as a user calibrated sensor and/or a system calibrated sensor.

In a further aspect, programming or executable instructions can be provided or stored in the data processing device of the analyte monitoring system, and/or the receiver/controller unit, to provide a time varying adjustment algorithm to the in vivo sensor during use. For example, based on a retrospective statistical analysis of analyte sensors used in vivo and the corresponding glucose level feedback, a pre-determined or analytical curve or a database can be gener-

ated which is time based, and configured to provide additional adjustment to the one or more in vivo sensor parameters to compensate for potential sensor drift in stability profile, or other factors.

In accordance with the disclosed subject matter, the analyte monitoring system can be configured to compensate or adjust for the sensor sensitivity based on a sensor drift profile. A time varying parameter $\beta(t)$ can be defined or determined based on analysis of sensor behavior during in vivo use, and a time varying drift profile can be determined. In certain aspects, the compensation or adjustment to the sensor sensitivity can be programmed in the receiver unit, the controller or data processor of the analyte monitoring system such that the compensation or the adjustment or both can be performed automatically and/or iteratively when sensor data is received from the analyte sensor. In accordance with the disclosed subject matter, the adjustment or compensation algorithm can be initiated or executed by the user (rather than self-initiating or executing) such that the adjustment or the compensation to the analyte sensor sensitivity profile is performed or executed upon user initiation or activation of the corresponding function or routine, or upon the user entering the sensor calibration code.

In accordance with the disclosed subject matter, each sensor in the sensor lot (in some instances not including sample sensors used for in vitro testing) can be examined non-destructively to determine or measure its characteristics such as membrane thickness at one or more points of the sensor, and other characteristics including physical characteristics such as the surface area/volume of the active area can be measured or determined. Such measurement or determination can be performed in an automated manner using, for example, optical scanners or other suitable measurement devices or systems, and the determined sensor characteristics for each sensor in the sensor lot is compared to the corresponding mean values based on the sample sensors for possible correction of the calibration parameter or code assigned to each sensor. For example, for a calibration parameter defined as the sensor sensitivity, the sensitivity is approximately inversely proportional to the membrane thickness, such that, for example, a sensor having a measured membrane thickness of approximately 4% greater than the mean membrane thickness for the sampled sensors from the same sensor lot as the sensor, the sensitivity assigned to that sensor in one embodiment is the mean sensitivity determined from the sampled sensors divided by 1.04. Likewise, since the sensitivity is approximately proportional to active area of the sensor, a sensor having measured active area of approximately 3% lower than the mean active area for the sampled sensors from the same sensor lot, the sensitivity assigned to that sensor is the mean sensitivity multiplied by 0.97. The assigned sensitivity can be determined from the mean sensitivity from the sampled sensors, by multiple successive adjustments for each examination or measurement of the sensor. In certain embodiments, examination or measurement of each sensor can additionally include measurement of membrane consistency or texture in addition to the membrane thickness and/or surface area or volume of the active sensing area.

Additional information regarding sensor calibration is provided in U.S. Publication No. 2010/00230285 and U.S. Publication No. 2019/0274598, each of which is incorporated by reference herein in its entirety.

K. Exemplary Bluetooth Communication Protocols

The storage memory **5030** of the sensor **110** can include the software blocks related to communication protocols of the communication module. For example, the storage

memory **5030** can include a BLE services software block with functions to provide interfaces to make the BLE module **5041** available to the computing hardware of the sensor **110**. These software functions can include a BLE logical interface and interface parser. BLE services offered by the communication module **5040** can include the generic access profile service, the generic attribute service, generic access service, device information service, data transmission services, and security services. The data transmission service can be a primary service used for transmitting data such as sensor control data, sensor status data, analyte measurement data (historical and current), and event log data. The sensor status data can include error data, current time active, and software state. The analyte measurement data can include information such as current and historical raw measurement values, current and historical values after processing using an appropriate algorithm or model, projections and trends of measurement levels, comparisons of other values to patient-specific averages, calls to action as determined by the algorithms or models and other similar types of data.

According to aspects of the disclosed subject matter, and as embodied herein, a sensor **110** can be configured to communicate with multiple devices concurrently by adapting the features of a communication protocol or medium supported by the hardware and radios of the sensor **110**. As an example, the BLE module **5041** of the communication module **5040** can be provided with software or firmware to enable multiple concurrent connections between the sensor **110** as a central device and the other devices as peripheral devices, or as a peripheral device where another device is a central device.

Connections, and ensuing communication sessions, between two devices using a communication protocol such as BLE can be characterized by a similar physical channel operated between the two devices (e.g., a sensor **110** and data receiving device **120**). The physical channel can include a single channel or a series of channels, including for example and without limitation using an agreed upon series of channels determined by a common clock and channel- or frequency-hopping sequence. Communication sessions can use a similar amount of the available communication spectrum, and multiple such communication sessions can exist in proximity. In certain embodiment, each collection of devices in a communication session uses a different physical channel or series of channels, to manage interference of devices in the same proximity.

For purpose of illustration and not limitation, reference is made to an exemplary embodiment of a procedure for a sensor-receiver connection for use with the disclosed subject matter. First, the sensor **110** repeatedly advertises its connection information to its environment in a search for a data receiving device **120**. The sensor **110** can repeat advertising on a regular basis until a connection established. The data receiving device **120** detects the advertising packet and scans and filters for the sensor **120** to connect to through the data provided in the advertising packet. Next, data receiving device **120** sends a scan request command and the sensor **110** responds with a scan response packet providing additional details. Then, the data receiving device **120** sends a connection request using the Bluetooth device address associated with the data receiving device **120**. The data receiving device **120** can also continuously request to establish a connection to a sensor **110** with a specific Bluetooth device address. Then, the devices establish an initial connection allowing them to begin to exchange data. The devices begin

a process to initialize data exchange services and perform a mutual authentication procedure.

During a first connection between the sensor **110** and data receiving device **120**, the data receiving device **120** can initialize a service, characteristic, and attribute discovery procedure. The data receiving device **120** can evaluate these features of the sensor **110** and store them for use during subsequent connections. Next, the devices enable a notification for a customized security service used for mutual authentication of the sensor **110** and data receiving device **120**. The mutual authentication procedure can be automated and require no user interaction. Following the successful completion of the mutual authentication procedure, the sensor **110** sends a connection parameter update to request the data receiving device **120** to use connection parameter settings preferred by the sensor **110** and configured to maximum longevity.

The data receiving device **120** then performs sensor control procedures to backfill historical data, current data, event log, and factory data. As an example, for each type of data, the data receiving device **120** sends a request to initiate a backfill process. The request can specify a range of records defined based on, for example, the measurement value, timestamp, or similar, as appropriate. The sensor **110** responds with requested data until all previously unsent data in the memory of the sensor **110** is delivered to the data receiving device **120**. The sensor **110** can respond to a backfill request from the data receiving device **120** that all data has already been sent. Once backfill is completed, the data receiving device **120** can notify sensor **110** that it is ready to receive regular measurement readings. The sensor **110** can send readings across multiple notifications result on a repeating basis. As embodied herein, the multiple notifications can be redundant notifications to ensure that data is transmitted correctly. Alternatively, multiple notifications can make up a single payload.

For purpose of illustration and not limitation, reference is made to an exemplary embodiment of a procedure to send a shutdown command to the sensor **110**. The shutdown operation is executed if the sensor **110** is in, for example, an error state, insertion failed state, or sensor expired state. If the sensor **110** is not in those states, the sensor **110** can log the command and execute the shutdown when sensor **110** transitions into the error state or sensor expired state. The data receiving device **120** sends a properly formatted shutdown command to the sensor **110**. If the sensor **110** is actively processing another command, the sensor **110** will respond with a standard error response indicating that the sensor **110** is busy. Otherwise, the sensor **110** sends a response as the command is received. Additionally, the sensor **110** sends a success notification through the sensor control characteristic to acknowledge the sensor **110** has received the command. The sensor **110** registers the shutdown command. At the next appropriate opportunity (e.g., depending on the current sensor state, as described herein), the sensor **110** will shut down.

L. Exemplary Sensor States and Activation

For purpose of illustration and not limitation, reference is made to the exemplary embodiment of a high-level depiction of a state machine representation **6000** of the actions that can be taken by the sensor **110** as shown in FIG. 15. After initialization, the sensor enters state **6005**, which relates to the manufacture of the sensor **110**. In the manufacture state **6005** the sensor **110** can be configured for operation, for example, the storage memory **5030** can be written. At various times while in state **6005**, the sensor **110** checks for a received command to go to the storage state

6015. Upon entry to the storage state **6015**, the sensor performs a software integrity check. While in the storage state **6015**, the sensor can also receive an activation request command before advancing to the insertion detection state **6025**.

Upon entry to state **6025**, the sensor **110** can store information relating to devices authenticated to communicate with the sensor as set during activation or initialize algorithms related to conducting and interpreting measurements from the sensing hardware **5060**. The sensor **110** can also initialize a lifecycle timer, responsible for maintaining an active count of the time of operation of the sensor **110** and begin communication with authenticated devices to transmit recorded data. While in the insertion detection state **6025**, the sensor can enter state **6030**, where the sensor **110** checks whether the time of operation is equal to a predetermined threshold. This time of operation threshold can correspond to a timeout function for determining whether an insertion has been successful. If the time of operation has reached the threshold, the sensor **110** advances to state **6035**, in which the sensor **110** checks whether the average data reading is greater than a threshold amount corresponding to an expected data reading volume for triggering detection of a successful insertion. If the data reading volume is lower than the threshold while in state **6035**, the sensor advances to state **6040**, corresponding to a failed insertion. If the data reading volume satisfies the threshold, the sensor advances to the active paired state **6055**.

The active paired state **6055** of the sensor **110** reflects the state while the sensor **110** is operating as normal by recording measurements, processing the measurements, and reporting them as appropriate. While in the active paired state **6055**, the sensor **110** sends measurement results or attempts to establish a connection with a receiving device **120**. The sensor **110** also increments the time of operation. Once the sensor **110** reaches a predetermined threshold time of operation (e.g., once the time of operation reaches a predetermined threshold), the sensor **110** transitions to the active expired state **6065**. The active expired state **6065** of the sensor **110** reflects the state while the sensor **110** has operated for its maximum predetermined amount of time.

While in the active expired state **6065**, the sensor **110** can generally perform operations relating to winding down operation and ensuring that the collected measurements have been securely transmitted to receiving devices as needed. For example, while in the active expired state **6065**, the sensor **110** can transmit collected data and, if no connection is available, can increase efforts to discover authenticated devices nearby and establish a connection therewith. While in the active expired state **6065**, the sensor **110** can receive a shutdown command at state **6070**. If no shutdown command is received, the sensor **110** can also, at state **6075**, check if the time of operation has exceeded a final operation threshold. The final operation threshold can be based on the battery life of the sensor **110**. The normal termination state **6080** corresponds to the final operations of the sensor **110** and ultimately shutting down the sensor **110**.

Before a sensor is activated, the ASIC **5000** resides in a low power storage mode state. The activation process can begin, for example, when an incoming RF field (e.g., NFC field) drives the voltage of the power supply to the ASIC **5000** above a reset threshold, which causes the sensor **110** to enter a wake-up state. While in the wake-up state, the ASIC **5000** enters an activation sequence state. The ASIC **5000** then wakes the communication module **5040**. The communication module **5040** is initialized, triggering a power on self-test. The power on self-test can include the ASIC **5000**

communicating with the communication module **5040** using a prescribed sequence of reading and writing data to verify the memory and one-time programmable memory are not corrupted.

When the ASIC **5000** enters the measurement mode for the first time, an insertion detection sequence is performed to verify that the sensor **110** has been properly installed onto the patient's body before a proper measurement can take place. First, the sensor **110** interprets a command to activate the measurement configuration process, causing the ASIC **5000** to enter measurement command mode. The sensor **110** then temporarily enters the measurement lifecycle state to run a number of consecutive measurements to test whether the insertion has been successful. The communication module **5040** or ASIC **5000** evaluates the measurement results to determine insertion success. When insertion is deemed successful, the sensor **110** enters a measurement state, in which the sensor **110** begins taking regular measurements using sensing hardware **5060**. If the sensor **110** determines that the insertion was not successful, sensor **110** is triggered into an insertion failure mode, in which the ASIC **5000** is commanded back to storage mode while the communication module **5040** disables itself.

M. Exemplary Over-the-Air Updates

FIG. 1B further illustrates an example operating environment for providing over-the-air ("OTA") updates for use with the techniques described herein. An operator of the analyte monitoring system **100** can bundle updates for the data receiving device **120** or sensor **110** into updates for an application executing on the multi-purpose data receiving device **130**. Using available communication channels between the data receiving device **120**, the multi-purpose data receiving device **130**, and the sensor **110**, the multi-purpose data receiving device **130** can bundle updates for the data receiving device **120** or sensor **110** and initiate installation of the updates on the data receiving device **120** or sensor **110**. The multi-purpose data receiving device **130** acts as an installation or update platform for the data receiving device **120** or sensor **110** because the application that enables the multi-purpose data receiving device **130** to communicate with an analyte sensor **110**, data receiving device **120** and/or remote application server **150** can update software or firmware on a data receiving device **120** or sensor **110** without wide-area networking capabilities.

As embodied herein, a remote application server **150** operated by the manufacturer of the analyte sensor **110** and/or the operator of the analyte monitoring system **100** can provide software and firmware updates to the devices of the analyte monitoring system **100**. In particular embodiments, the remote application server **150** can provide the updated software and firmware to a user device **140** or directly to a multi-purpose data receiving device. As embodied herein, the remote application server **150** can also provide application software updates to an application storefront server **160** using interfaces provided by the application storefront. The multi-purpose data receiving device **130** can contact the application storefront server **160** periodically to download and install the updates.

After the multi-purpose data receiving device **130** downloads an application update including a firmware or software update for a data receiving device **120** or sensor **110**, the data receiving device **120** or sensor **110** and multi-purpose data receiving device **130** establish a connection. The multi-purpose data receiving device **130** determines that a firmware or software update is available for the data receiving device **120** or sensor **110**. The multi-purpose data receiving device **130** can prepare the software or firmware update for

delivery to the data receiving device **120** or sensor **110**. As an example, the multi-purpose data receiving device **130** can compress or segment the data associated with the software or firmware update, can encrypt or decrypt the firmware or software update, or can perform an integrity check of the firmware or software update. The multi-purpose data receiving device **130** sends the data for the firmware or software update to the data receiving device **120** or sensor **110**. The multi-purpose data receiving device **130** can also send a command to the data receiving device **120** or sensor **110** to initiate the update. Additionally or alternatively, the multi-purpose data receiving device **130** can provide a notification to the user of the multi-purpose data receiving device **130** and include instructions for facilitating the update, such as instructions to keep the data receiving device **120** and the multi-purpose data receiving device **130** connected to a power source and in close proximity until the update is complete.

The data receiving device **120** or sensor **110** receives the data for the update and the command to initiate the update from the multi-purpose data receiving device **130**. The data receiving device **120** can then install the firmware or software update. To install the update, the data receiving device **120** or sensor **110** can place or restart itself in a so-called "safe" mode with limited operational capabilities. Once the update is completed, the data receiving device **120** or sensor **110** re-enters or resets into a standard operational mode. The data receiving device **120** or sensor **110** can perform one or more self-tests to determine that the firmware or software update was installed successfully. The multi-purpose data receiving device **130** can receive the notification of the successful update. The multi-purpose data receiving device **130** can then report a confirmation of the successful update to the remote application server **150**.

In particular embodiments, the storage memory **5030** of the sensor **110** includes one-time programmable (OTP) memory. The term OTP memory can refer to memory that includes access restrictions and security to facilitate writing to particular addresses or segments in the memory a predetermined number of times. The memory **5030** can be pre-arranged into multiple pre-allocated memory blocks or containers. The containers are pre-allocated into a fixed size. If storage memory **5030** is one-time programming memory, the containers can be considered to be in a non-programmable state. Additional containers which have not yet been written to can be placed into a programmable or writable state. Containerizing the storage memory **5030** in this fashion can improve the transportability of code and data to be written to the storage memory **5030**. Updating the software of a device (e.g., the sensor device described herein) stored in an OTP memory can be performed by superseding only the code in a particular previously-written container or containers with updated code written to a new container or containers, rather than replacing the entire code in the memory. In a second embodiment, the memory is not prearranged. Instead, the space allocated for data is dynamically allocated or determined as needed. Incremental updates can be issued, as containers of varying sizes can be defined where updates are anticipated.

FIG. **16** is a diagram illustrating an example operational and data flow for over-the-air (OTA) programming of a storage memory **5030** in a sensor device **100** as well as use of the memory after the OTA programming in execution of processes by the sensor device **110** according to the disclosed subject matter. In the example OTA programming **500** illustrated in FIG. **5**, a request is sent from an external device (e.g., the data receiving device **130**) to initiate OTA

programming (or re-programming). At **511**, a communication module **5040** of a sensor device **110** receives an OTA programming command. The communication module **5040** sends the OTA programming command to the microcontroller **5010** of the sensor device **110**.

At **531**, after receiving the OTA programming command, the microcontroller **5010** validates the OTA programming command. The microcontroller **5010** can determine, for example, whether the OTA programming command is signed with an appropriate digital signature token. Upon determining that the OTA programming command is valid, the microcontroller **5010** can set the sensor device into an OTA programming mode. At **532**, the microcontroller **5010** can validate the OTA programming data. At **533**, The microcontroller **5010** can reset the sensor device **110** to re-initialize the sensor device **110** in a programming state. Once the sensor device **110** has transitioned into the OTA programming state, the microcontroller **5010** can begin to write data to the rewriteable memory **540** (e.g., memory **5020**) of the sensor device at **534** and write data to the OTP memory **550** of the sensor device at **535** (e.g., storage memory **5030**). The data written by the microcontroller **5010** can be based on the validated OTA programming data. The microcontroller **5010** can write data to cause one or more programming blocks or regions of the OTP memory **550** to be marked invalid or inaccessible. The data written to the free or unused portion of the OTP memory can be used to replace invalidated or inaccessible programming blocks of the OTP memory **550**. After the microcontroller **5010** writes the data to the respective memories at **534** and **535**, the microcontroller **5010** can perform one or more software integrity checks to ensure that errors were not introduced into the programming blocks during the writing process. Once the microcontroller **5010** is able to determine that the data has been written without errors, the microcontroller **5010** can resume standard operations of the sensor device.

In execution mode, at **536**, the microcontroller **5010** can retrieve a programming manifest or profile from the rewriteable memory **540**. The programming manifest or profile can include a listing of the valid software programming blocks and can include a guide to program execution for the sensor **110**. By following the programming manifest or profile, the microcontroller **5010** can determine which memory blocks of the OTP memory **550** are appropriate to execute and avoid execution of out-of-date or invalidated programming blocks or reference to out-of-date data. At **537**, the microcontroller **5010** can selectively retrieve memory blocks from the OTP memory **550**. At **538**, the microcontroller **5010** can use the retrieved memory blocks, by executing programming code stored or using variable stored in the memory.

N. Exemplary Security and Other Architecture Features

As embodied herein a first layer of security for communications between the analyte sensor **110** and other devices can be established based on security protocols specified by and integrated in the communication protocols used for the communication. Another layer of security can be based on communication protocols that necessitate close proximity of communicating devices. Furthermore certain packets and/or certain data included within packets can be encrypted while other packets and/or data within packets is otherwise encrypted or not encrypted. Additionally or alternatively, application layer encryption can be used with one or more block ciphers or stream ciphers to establish mutual authentication and communication encryption with other devices in the analyte monitoring system **100**.

The ASIC **5000** of the analyte sensor **110** can be configured to dynamically generate authentication and encryption

keys using data retained within the storage memory **5030**. The storage memory **5030** can also be pre-programmed with a set of valid authentication and encryption keys to use with particular classes of devices. The ASIC **5000** can be further configured to perform authentication procedures with other devices using received data and apply the generated key to sensitive data prior to transmitting the sensitive data. The generated key can be unique to the analyte sensor **110**, unique to a pair of devices, unique to a communication session between an analyte sensor **110** and other device, unique to a message sent during a communication session, or unique to a block of data contained within a message.

Both the sensor **110** and a data receiving device **120** can ensure the authorization of the other party in a communication session to, for example, issue a command or receive data. In particular embodiments, identity authentication can be performed through two features. First, the party asserting its identity provides a validated certificate signed by the manufacturer of the device or the operator of the analyte monitoring system **100**. Second, authentication can be enforced through the use of public keys and private keys, and shared secrets derived therefrom, established by the devices of the analyte monitoring system **100** or established by the operator of the analyte monitoring system **100**. To confirm the identity of the other party, the party can provide proof that the party has control of its private key.

The manufacturer of the analyte sensor **110**, data receiving device **120**, or provider of the application for multi-purpose data receiving device **130** can provide information and programming necessary for the devices to securely communicate through secured programming and updates. For example, the manufacturer can provide information that can be used to generate encryption keys for each device, including secured root keys for the analyte sensor **110** and optionally for the data receiving device **120** that can be used in combination with device-specific information and operational data (e.g., entropy-based random values) to generate encryption values unique to the device, session, or data transmission as need.

Analyte data associated with a user is sensitive data at least in part because this information can be used for a variety of purposes, including for health monitoring and medication dosing decisions. In addition to user data, the analyte monitoring system **100** can enforce security hardening against efforts by outside parties to reverse-engineering. Communication connections can be encrypted using a device-unique or session-unique encryption key. Encrypted communications or unencrypted communications between any two devices can be verified with transmission integrity checks built into the communications. Analyte sensor **110** operations can be protected from tampering by restricting access to read and write functions to the memory **5020** via a communication interface. The sensor can be configured to grant access only to known or "trusted" devices, provided in a "whitelist" or only to devices that can provide a predetermined code associated with the manufacturer or an otherwise authenticated user. A whitelist can represent an exclusive range, meaning that no connection identifiers besides those included in the whitelist will be used, or a preferred range, in which the whitelist is searched first, but other devices can still be used. The sensor **110** can further deny and shut down connection requests if the requestor cannot complete a login procedure over a communication interface within a predetermined period of time (e.g., within four seconds). These characteristics safeguard against specific denial of service attacks, and in particular against denial of service attacks on a BLE interface.

As embodied herein, the analyte monitoring system **100** can employ periodic key rotation to further reduce the likelihood of key compromise and exploitation. A key rotation strategy employed by the analyte monitoring system **100** can be designed to support backward compatibility of field-deployed or distributed devices. As an example, the analyte monitoring system **100** can employ keys for downstream devices (e.g., devices that are in the field or cannot be feasibly provided updates) that are designed to be compatible with multiple generations of keys used by upstream devices.

For purpose of illustration and not limitation, reference is made to the exemplary embodiment of a message sequence diagram **600** for use with the disclosed subject matter as shown in FIG. **17** and demonstrating an example exchange of data between a pair of devices, particularly a sensor **110** and a data receiving device **120**. The data receiving device **120** can, as embodied herein, be a data receiving device **120** or a multi-purpose data receiving device **130**. At step **605**, the data receiving device **120** can transmit a sensor activation command **605** to the sensor **110**, for example via a short-range communication protocol. The sensor **110** can, prior to step **605** be in a primarily dormant state, preserving its battery until full activation is needed. After activation during step **610**, the sensor **110** can collect data or perform other operations as appropriate to the sensing hardware **5060** of the sensor **110**. At step **615** the data receiving device **120** can initiate an authentication request command **615**. In response to the authentication request command **615**, both the sensor **110** and data receiving device **120** can engage in a mutual authentication process **620**. The mutual authentication process **620** can involve the transfer of data, including challenge parameters that allow the sensor **110** and data receiving device **120** to ensure that the other device is sufficiently capable of adhering to an agreed-upon security framework described herein. Mutual authentication can be based on mechanisms for authentication of two or more entities to each other with or without on-line trusted third parties to verify establishment of a secret key via challenge-response. Mutual authentication can be performed using two-, three-, four-, or five-pass authentication, or similar versions thereof.

Following a successful mutual authentication process **620**, at step **625** the sensor **110** can provide the data receiving device **120** with a sensor secret **625**. The sensor secret can contain sensor-unique values and be derived from random values generated during manufacture. The sensor secret can be encrypted prior to or during transmission to prevent third-parties from accessing the secret. The sensor secret **625** can be encrypted via one or more of the keys generated by or in response to the mutual authentication process **620**. At step **630**, the data receiving device **120** can derive a sensor-unique encryption key from the sensor secret. The sensor-unique encryption key can further be session-unique. As such, the sensor-unique encryption key can be determined by each device without being transmitted between the sensor **110** or data receiving device **120**. At step **635**, the sensor **110** can encrypt data to be included in payload. At step **640**, the sensor **110** can transmit the encrypted payload **640** to the data receiving device **120** using the communication link established between the appropriate communication models of the sensor **110** and data receiving device **120**. At step **645**, the data receiving device **120** can decrypt the payload using the sensor-unique encryption key derived during step **630**. Following step **645**, the sensor **110** can deliver additional (including newly

collected) data and the data receiving device **120** can process the received data appropriately.

As discussed herein, the sensor **110** can be a device with restricted processing power, battery supply, and storage. The encryption techniques used by the sensor **110** (e.g., the cipher algorithm or the choice of implementation of the algorithm) can be selected based at least in part on these restrictions. The data receiving device **120** can be a more powerful device with fewer restrictions of this nature. Therefore, the data receiving device **120** can employ more sophisticated, computationally intense encryption techniques, such as cipher algorithms and implementations.

O. Exemplary Payload/Communication Frequencies

The analyte sensor **110** can be configured to alter its discoverability behavior to attempt to increase the probability of the receiving device receiving an appropriate data packet and/or provide an acknowledgement signal or otherwise reduce restrictions that can be causing an inability to receive an acknowledgement signal. Altering the discoverability behavior of the analyte sensor **110** can include, for example and without limitation, altering the frequency at which connection data is included in a data packet, altering how frequently data packets are transmitted generally, lengthening or shortening the broadcast window for data packets, altering the amount of time that the analyte sensor **110** listens for acknowledgement or scan signals after broadcasting, including directed transmissions to one or more devices (e.g., through one or more attempted transmissions) that have previously communicated with the analyte sensor **110** and/or to one or more devices on a whitelist, altering a transmission power associated with the communication module when broadcasting the data packets (e.g., to increase the range of the broadcast or decrease energy consumed and extend the life of the battery of the analyte sensor), altering the rate of preparing and broadcasting data packets, or a combination of one or more other alterations. Additionally, or alternatively, the receiving device can similarly adjust parameters relating to the listening behavior of the device to increase the likelihood of receiving a data packet including connection data.

As embodied herein, the analyte sensor **110** can be configured to broadcast data packets using two types of windows. The first window refers to the rate at which the analyte sensor **110** is configured to operate the communication hardware. The second window refers to the rate at which the analyte sensor **110** is configured to be actively transmitting data packets (e.g., broadcasting). As an example, the first window can indicate that the analyte sensor **110** operates the communication hardware to send and/or receive data packets (including connection data) during the first 2 seconds of each 60 second period. The second window can indicate that, during each 2 second window, the analyte sensor **110** transmits a data packet every 60 milliseconds. The rest of the time during the 2 second window, the analyte sensor **110** is scanning. The analyte sensor **110** can lengthen or shorten either window to modify the discoverability behavior of the analyte sensor **110**.

In particular embodiments, the discoverability behavior of the analyte sensor can be stored in a discoverability profile, and alterations can be made based on one or more factors, such as the status of the analyte sensor **110** and/or by applying rules based on the status of the analyte sensor **110**. For example, when the battery level of the analyte sensor **110** is below a certain amount, the rules can cause the analyte sensor **110** to decrease the power consumed by the broadcast process. As another example, configuration settings associated with broadcasting or otherwise transmitting

packets can be adjusted based on the ambient temperature, the temperature of the analyte sensor **110**, or the temperature of certain components of communication hardware of the analyte sensor **110**. In addition to modifying the transmission power, other parameters associated with the transmission capabilities or processes of the communication hardware of the analyte sensor **110** can be modified, including, but not limited to, transmission rate, frequency, and timing. As another example, when the analyte data indicates that the subject is, or is about to be, experiencing a negative health event, the rules can cause the analyte sensor **110** to increase its discoverability to alert the receiving device of the negative health event.

P. Exemplary Sensor Sensitivity Initialization/Adjustment Features

As embodied herein, certain calibration features for the sensing hardware **5060** of the analyte sensor **110** can be adjusted based on external or interval environment features as well as to compensate for the decay of the sensing hardware **5060** during expended period of disuse (e.g., a “shelf time” prior to use). The calibration features of the sensing hardware **5060** can be autonomously adjusted by the sensor **110** (e.g., by operation of the ASIC **5000** to modify features in the memory **5020** or storage **5030**) or can be adjusted by other devices of the analyte monitoring system **100**.

As an example, sensor sensitivity of the sensing hardware **5060** can be adjusted based on external temperature data or the time since manufacture. When external temperatures are monitored during the storage of the sensors, the disclosed subject matter can adaptively change the compensation to sensor sensitivity over time when the device experiences changing storage conditions. For purpose of illustration not limitations, adaptive sensitivity adjustment can be performed in an “active” storage mode where the analyte sensor **110** wakes up periodically to measure temperature. These features can save the battery of the analyte device and extend the lifespan of the analyte sensors. At each temperature measurement, the analyte sensor **110** can calculate a sensitivity adjustment for that time period based on the measured temperature. Then, the temperature-weighted adjustments can be accumulated over the active storage mode period to calculate a total sensor sensitivity adjustment value at the end of the active storage mode (e.g., at insertion). Similarly, at insertion, the sensor **110** can determine the time difference between manufacture of the sensor **110** (which can be written to the storage **5030** of the ASIC **5000**) or the sensing hardware **5060** and modify sensor sensitivity or other calibration features according to one or more known decay rates or formulas.

Additionally, for purpose of illustration and not limitation, as embodied herein, sensor sensitivity adjustments can account for other sensor conditions, such as sensor drift. Sensor sensitivity adjustments can be hardcoded into the sensor **110** during manufacture, for example in the case of sensor drift, based on an estimate of how much an average sensor would drift. Sensor **110** can use a calibration function that has time-varying functions for sensor offset and gain, which can account for drift over a wear period of the sensor. Thus, sensor **110** can utilize a function used to transform an interstitial current to interstitial glucose utilizing device-dependent functions describing sensor **110** drift over time, and which can represent sensor sensitivity, and can be device specific, combined with a baseline of the glucose profile. Such functions to account for sensor sensitivity and drift can improve sensor **110** accuracy over a wear period and without involving user calibration.

Q. Exemplary Model-Based Analyte Measurements

The sensor **110** detects raw measurement values from sensing hardware **5060**. On-sensor processing can be performed, such as by one or more models trained to interpret the raw measurement values. Models can be machine learned models trained off-device to detect, predict, or interpret the raw measurement values to detect, predict, or interpret the levels of one or more analytes. Additional trained models can operate on the output of the machine learning models trained to interact with raw measurement values. As an example, models can be used to detect, predict, or recommend events based on the raw measurements and type of analyte(s) detected by the sensing hardware **5060**. Events can include, initiation or completion of physical activity, meals, application of medical treatment or medication, emergent health events, and other events of a similar nature.

Models can be provided to the sensor **110**, data receiving device **120**, or multi-purpose data receiving device **130** during manufacture or during firmware or software updates. Models can be periodically refined, such as by the manufacturer of the sensor **110** or the operator of the analyte monitoring system **100**, based on data received from the sensor **110** and data receiving devices of an individual user or multiple users collectively. In certain embodiments, the sensor **110** includes sufficient computational components to assist with further training or refinement of the machine learned models, such as based on unique features of the user to which the sensor **110** is attached. Machine learning models can include, by way of example and not limitation, models trained using or encompassing decision tree analysis, gradient boosting, ada boosting, artificial neural networks or variants thereof, linear discriminant analysis, nearest neighbor analysis, support vector machines, supervised or unsupervised classification, and others. The models can also include algorithmic or rules-based models in addition to machine learned models. Model-based processing can be performed by other devices, including the data receiving device **120** or multi-purpose data receiving device **130**, upon receiving data from the sensor **110** (or other downstream devices).

R. Exemplary Alarm Features

Data transmitted between the sensor **110** and a data receiving device **120** can include raw or processed measurement values. Data transmitted between the sensor **110** and data receiving device **120** can further include alarms or notification for display to a user. The data receiving device **120** can display or otherwise convey notifications to the user based on the raw or processed measurement values or can display alarms when received from the sensor **110**. Alarms that may be triggered for display to the user include alarms based on direct analyte values (e.g., one-time reading exceeding a threshold or failing to satisfy a threshold), analyte value trends (e.g., average reading over a set period of time exceeding a threshold or failing to satisfy a threshold; slope); analyte value predictions (e.g., algorithmic calculation based on analyte values exceeds a threshold or fails to satisfy a threshold), sensor alerts (e.g., suspected malfunction detected), communication alerts (e.g., no communication between sensor **110** and data receiving device **120** for a threshold period of time; unknown device attempting or failing to initiate a communication session with the sensor **110**), reminders (e.g., reminder to charge data receiving device **120**; reminder to take a medication or perform other activity), and other alerts of a similar nature. For purpose of illustration and not limitation, as embodied herein, the alarm parameters described herein can be con-

figurable by a user or can be fixed during manufacture, or combinations of user-settable and non-user-settable parameters.

S. Exemplary Electrode Configurations

Sensor configurations featuring a single active area that is configured for detection of a corresponding single analyte can employ two-electrode or three-electrode detection motifs, as described further herein in reference to FIGS. **18A-18C**. Sensor configurations featuring two different active areas for detection of separate analytes, either upon separate working electrodes or upon the same working electrode, are described separately thereafter in reference to FIGS. **19A-21C**. Sensor configurations having multiple working electrodes can be particularly advantageous for incorporating two different active areas within the same sensor tail, since the signal contribution from each active area can be determined more readily.

When a single working electrode is present in an analyte sensor, three-electrode sensor configurations can include a working electrode, a counter electrode, and a reference electrode. Related two-electrode sensor configurations can include a working electrode and a second electrode, in which the second electrode can function as both a counter electrode and a reference electrode (i.e., a counter/reference electrode). The various electrodes can be at least partially stacked (layered) upon one another and/or laterally spaced apart from one another upon the sensor tail. Suitable sensor configurations can be substantially flat in shape or substantially cylindrical in shape, or any other suitable shape. In any of the sensor configurations disclosed herein, the various electrodes can be electrically isolated from one another by a dielectric material or similar insulator.

Analyte sensors featuring multiple working electrodes can similarly include at least one additional electrode. When one additional electrode is present, the one additional electrode can function as a counter/reference electrode for each of the multiple working electrodes. When two additional electrodes are present, one of the additional electrodes can function as a counter electrode for each of the multiple working electrodes and the other of the additional electrodes can function as a reference electrode for each of the multiple working electrodes.

FIG. **18A** shows a diagram of an illustrative two-electrode analyte sensor configuration, which is compatible for use in the disclosure herein. As shown, analyte sensor **200** includes substrate **30212** disposed between working electrode **214** and counter/reference electrode **30216**. Alternately, working electrode **214** and counter/reference electrode **30216** can be located upon the same side of substrate **30212** with a dielectric material interposed in between (configuration not shown). Active area **218** is disposed as at least one layer upon at least a portion of working electrode **214**. Active area **218** can include multiple spots or a single spot configured for detection of an analyte, as discussed further herein.

Referring still to FIG. **18A**, membrane **220** overcoats at least active area **218**. In certain embodiments, membrane **220** can also overcoat some or all of working electrode **214** and/or counter/reference electrode **30216**, or the entirety of analyte sensor **200**. One or both faces of analyte sensor **200** can be overcoated with membrane **220**. Membrane **220** can include one or more polymeric membrane materials having capabilities of limiting analyte flux to active area **218** (i.e., membrane **220** is a mass transport limiting membrane having some permeability for the analyte of interest). According to the disclosure herein, membrane **220** can be crosslinked with a branched crosslinker in certain particular sensor configurations. The composition and thickness of membrane

220 can vary to promote a desired analyte flux to active area 218, thereby providing a desired signal intensity and stability. Analyte sensor 200 can be operable for assaying an analyte by any of coulometric, amperometric, voltammetric, or potentiometric electrochemical detection techniques.

FIGS. 18B and 18C show diagrams of illustrative three-electrode analyte sensor configurations, which are also compatible for use in the disclosure herein. Three-electrode analyte sensor configurations can be similar to that shown for analyte sensor 200 in FIG. 18A, except for the inclusion of additional electrode 217 in analyte sensors 201 and 202 (FIGS. 18B and 18C). With additional electrode 217, counter/reference electrode 30216 can then function as either a counter electrode or a reference electrode, and additional electrode 217 fulfills the other electrode function not otherwise accounted for. Working electrode 214 continues to fulfill its original function. Additional electrode 217 can be disposed upon either working electrode 214 or electrode 30216, with a separating layer of dielectric material in between. For example, and not by the way of limitation, as depicted in FIG. 18B, dielectric layers 219a, 219b and 219c separate electrodes 214, 30216 and 217 from one another and provide electrical isolation. Alternatively, at least one of electrodes 214, 30216 and 217 can be located upon opposite faces of substrate 30212, as shown in FIG. 18C. Thus, in certain embodiments, electrode 214 (working electrode) and electrode 30216 (counter electrode) can be located upon opposite faces of substrate 30212, with electrode 217 (reference electrode) being located upon one of electrodes 214 or 30216 and spaced apart therefrom with a dielectric material. Reference material layer 30230 (e.g., Ag/AgCl) can be present upon electrode 217, with the location of reference material layer 30230 not being limited to that depicted in FIGS. 18B and 18C. As with sensor 200 shown in FIG. 18A, active area 218 in analyte sensors 201 and 202 can include multiple spots or a single spot. Additionally, analyte sensors 201 and 202 can be operable for assaying an analyte by any of coulometric, amperometric, voltammetric, or potentiometric electrochemical detection techniques.

Like analyte sensor 200, membrane 220 can also overcoat active area 218, as well as other sensor components, in analyte sensors 201 and 202, thereby serving as a mass transport limiting membrane. In certain embodiments, the additional electrode 217 can be overcoated with membrane 220. Although FIGS. 18B and 18C have depicted electrodes 214, 30216 and 217 as being overcoated with membrane 220, it is to be recognized that in certain embodiments only working electrode 214 is overcoated. Moreover, the thickness of membrane 220 at each of electrodes 214, 30216 and 217 can be the same or different. As in two-electrode analyte sensor configurations (FIG. 18A), one or both faces of analyte sensors 201 and 202 can be overcoated with membrane 220 in the sensor configurations of FIGS. 18B and 18C, or the entirety of analyte sensors 201 and 202 can be overcoated. Accordingly, the three-electrode sensor configurations shown in FIGS. 18B and 18C should be understood as being non-limiting of the embodiments disclosed herein, with alternative electrode and/or layer configurations remaining within the scope of the present disclosure.

FIG. 19A shows an illustrative configuration for sensor 203 having a single working electrode with two different active areas disposed thereon. FIG. 19A is similar to FIG. 19A, except for the presence of two active areas upon working electrode 214: first active area 218a and second active area 218b, which are responsive to different analytes and are laterally spaced apart from one another upon the surface of working electrode 214. Active areas 218a and

218b can include multiple spots or a single spot configured for detection of each analyte. The composition of membrane 220 can vary or be compositionally the same at active areas 218a and 218b. First active area 218a and second active area 218b can be configured to detect their corresponding analytes at working electrode potentials that differ from one another, as discussed further below. In certain embodiments, any one of active areas 218a and 218b, or both, can be configured to detect an analyte using an NAD(P)-dependent enzyme. In certain embodiments, any one of active areas 218a and 218b, or both, can be configured to detect an analyte using an NAD(P)-dependent enzyme, e.g., ketones by using an enzyme system comprising NADH oxidase and β -hydroxybutyrate dehydrogenase. In certain embodiments, only one active area of 218a and 218b is configured to detect an analyte using an NAD(P)-dependent enzyme. In certain embodiments, the other active area is configured to detect a second analyte not using an NAD(P)-dependent enzyme.

FIGS. 19B and 19C show cross-sectional diagrams of illustrative three-electrode sensor configurations for sensors 204 and 205, respectively, each featuring a single working electrode having first active area 218a and second active area 218b disposed thereon. FIGS. 19B and 19C are otherwise similar to FIGS. 18B and 18C and can be better understood by reference thereto. As with FIG. 19A, the composition of membrane 220 can vary or be compositionally the same at active areas 218a and 218b.

Illustrative sensor configurations having multiple working electrodes, specifically two working electrodes, are described in further detail in reference to FIGS. 20-21C. Although the following description is primarily directed to sensor configurations having two working electrodes, it is to be appreciated that more than two working electrodes can be incorporated through extension of the disclosure herein. Additional working electrodes can be used to impart additional sensing capabilities to the analyte sensors beyond just a first analyte and a second analyte, e.g., for the detection of a third and/or fourth analyte.

FIG. 20 shows a cross-sectional diagram of an illustrative analyte sensor configuration having two working electrodes, a reference electrode and a counter electrode, which is compatible for use in the disclosure herein. As shown, analyte sensor 300 includes working electrodes 304 and 306 disposed upon opposite faces of substrate 302. First active area 310a is disposed upon the surface of working electrode 304, and second active area 310b is disposed upon the surface of working electrode 306. Counter electrode 320 is electrically isolated from working electrode 304 by dielectric layer 322, and reference electrode 321 is electrically isolated from working electrode 306 by dielectric layer 323. Outer dielectric layers 330 and 332 are positioned upon reference electrode 321 and counter electrode 320, respectively. Membrane 340 can overcoat at least active areas 310a and 310b, according to various embodiments, with other components of analyte sensor 300 or the entirety of analyte sensor 300 optionally being overcoated with first membrane portion 340a and/or second membrane portion 340b as well. Again, membrane 340 can be continuous but vary compositionally within first membrane portion 340a and second membrane portion 340b (i.e., upon active areas 310a and 310b) in order to afford different permeability values for differentially regulating the analyte flux at each location. For example, different membrane formulations can be sprayed and/or printed onto the opposing faces of analyte sensor 300. Dip coating techniques can also be appropriate, particularly for depositing at least a portion of a bilayer membrane upon one of active areas 310a and 310b. Accordingly, one of first

47

membrane portion **340a** and second membrane portion **340b** can comprise a bilayer membrane and the other of first membrane portion **340a** and second membrane portion **340b** can comprise a single membrane polymer, according to particular embodiments of the present disclosure. Like analyte sensors **200**, **201** and **202**, analyte sensor **300** can be operable for assaying ketones (and/or a second analyte) by any of coulometric, amperometric, voltammetric, or potentiometric electrochemical detection techniques. In certain embodiments, an analyte sensor can include more than one membrane **340**, e.g., two or more membranes. For example, but not by way of limitation, an analyte sensor can include a membrane that overcoats the one or more active areas, e.g., **310a** and **310a**, and an additional membrane that overcoats the entire sensor as shown in FIG. **20**. In certain embodiments, any one of active areas **310a** and **310b**, or both, can be configured to detect an analyte using an NAD(P)-dependent enzyme, e.g., ketones by using an enzyme system comprising NADH oxidase and β -hydroxybutyrate dehydrogenase or β -hydroxybutyrate dehydrogenase and diaphorase. In certain embodiments, only one active area of **310a** and **310b** is configured to detect an analyte using an NAD(P)-dependent enzyme, e.g., ketones by using an enzyme system comprising NADH oxidase and β -hydroxybutyrate dehydrogenase or β -hydroxybutyrate dehydrogenase and diaphorase. In certain embodiments, the other active area is configured to detect a second analyte, e.g., that is not detected using an NAD(P)-dependent enzyme.

Alternative sensor configurations having multiple working electrodes and differing from the configuration shown in FIG. **20** can feature a counter/reference electrode instead of separate counter and reference electrodes **320**, **321**, and/or feature layer and/or membrane arrangements varying from those expressly depicted. For example, and not by the way of limitation the positioning of counter electrode **320** and reference electrode **321** can be reversed from that depicted in FIG. **20**. In addition, working electrodes **304** and **306** need not necessarily reside upon opposing faces of substrate **302** in the manner shown in FIG. **20**.

Although suitable sensor configurations can feature electrodes that are substantially planar in character, it is to be appreciated that sensor configurations featuring non-planar electrodes can be advantageous and particularly suitable for use in the disclosure herein. In particular, substantially cylindrical electrodes that are disposed concentrically with respect to one another can facilitate deposition of a mass transport limiting membrane, as described hereinbelow. In particular, concentric working electrodes that are spaced apart along the length of a sensor tail can facilitate membrane deposition through sequential dip coating operations, in a similar manner to that described above for substantially planar sensor configurations. FIGS. **21A-21C** show perspective views of analyte sensors featuring two working electrodes that are disposed concentrically with respect to one another. It is to be appreciated that sensor configurations having a concentric electrode disposition but lacking a second working electrode are also possible in the present disclosure.

FIG. **21A** shows a perspective view of an illustrative sensor configuration in which multiple electrodes are substantially cylindrical and are disposed concentrically with respect to one another about a central substrate. As shown, analyte sensor **400** includes central substrate **402** about which all electrodes and dielectric layers are disposed concentrically with respect to one another. In particular, working electrode **410** is disposed upon the surface of central substrate **402**, and dielectric layer **412** is disposed upon a

48

portion of working electrode **410** distal to sensor tip **404**. Working electrode **420** is disposed upon dielectric layer **412**, and dielectric layer **422** is disposed upon a portion of working electrode **420** distal to sensor tip **404**. Counter electrode **430** is disposed upon dielectric layer **422**, and dielectric layer **432** is disposed upon a portion of counter electrode **430** distal to sensor tip **404**. Reference electrode **440** is disposed upon dielectric layer **432**, and dielectric layer **442** is disposed upon a portion of reference electrode **440** distal to sensor tip **404**. As such, exposed surfaces of working electrode **410**, working electrode **420**, counter electrode **430**, and reference electrode **440** are spaced apart from one another along longitudinal axis B of analyte sensor **400**.

Referring still to FIG. **21A**, first active areas **414a** and second active areas **414b**, which are responsive to different analytes or the same analyte, are disposed upon the exposed surfaces of working electrodes **410** and **420**, respectively, thereby allowing contact with a fluid to take place for sensing. In certain embodiments, any one of active areas **414a** and **414b**, or both, can be configured to detect an analyte using an NAD(P)-dependent enzyme. In certain embodiments, any one of active areas **414a** and **414b**, or both, can be configured to detect ketones, e.g., by using an enzyme system comprising NADH oxidase and β -hydroxybutyrate dehydrogenase. In certain embodiments, only one active area of **414a** and **414b** is configured to detect ketones, e.g., by using an enzyme system comprising NADH oxidase and β -hydroxybutyrate dehydrogenase. In certain embodiments, the other active area is configured to detect a second analyte. In certain embodiments, any one of active areas **414a** and **414b**, or both, can be configured to detect an analyte using an NAD(P)-dependent enzyme, e.g., ketones by using an enzyme system comprising NADH oxidase and β -hydroxybutyrate dehydrogenase or β -hydroxybutyrate dehydrogenase and diaphorase. In certain embodiments, only one active area of **414a** and **414b** is configured to detect an analyte using an NAD(P)-dependent enzyme, e.g., ketones by using an enzyme system comprising NADH oxidase and β -hydroxybutyrate dehydrogenase or β -hydroxybutyrate dehydrogenase and diaphorase. In certain embodiments, the other active area is configured to detect a second analyte, e.g., that is not detected using an NAD(P)-dependent enzyme. Although active areas **414a** and **414b** have been depicted as three discrete spots in FIG. **21A**, it is to be appreciated that fewer or greater than three spots, including a continuous layer of active area, can be present in alternative sensor configurations.

In FIG. **21A**, sensor **400** is partially coated with membrane **450** upon working electrodes **410** and **420** and active areas **414a** and **414b** disposed thereon. FIG. **21B** shows an alternative sensor configuration in which the substantial entirety of sensor **401** is overcoated with membrane **450**. Membrane **450** can be the same or vary compositionally at active areas **414a** and **414b**. For example, membrane **450** can include a bilayer overcoating active areas **414a** and be a homogeneous membrane overcoating active areas **414b**. In certain embodiments, one or more membranes are deposited over the exposed electroactive surface, e.g., platinum surface, of a working electrode, including an interference domain and a mass transport limiting membrane. For example, but not by way of limitation, an interference domain can be disposed upon the working electrode, an active area can be disposed upon the interference domain and a mass transport limiting membrane can be disposed upon the active area.

It is to be further appreciated that the positioning of the various electrodes in FIGS. 21A and 21B can differ from that expressly depicted. For example, the positions of counter electrode 430 and reference electrode 440 can be reversed from the depicted configurations in FIGS. 21A and 21B. Similarly, the positions of working electrodes 410 and 420 are not limited to those that are expressly depicted in FIGS. 21A and 21B. FIG. 21C shows an alternative sensor configuration to that shown in FIG. 21B, in which sensor 405 contains counter electrode 430 and reference electrode 440 that are located more proximal to sensor tip 404 and working electrodes 410 and 420 that are located more distal to sensor tip 404. Sensor configurations in which working electrodes 410 and 420 are located more distal to sensor tip 404 can be advantageous by providing a larger surface area for deposition of active areas 414a and 414b (five discrete sensing spots illustratively shown in FIG. 21C), thereby facilitating an increased signal strength in some cases. Similarly, central substrate 402 can be omitted in any concentric sensor configuration disclosed herein, wherein the innermost electrode can instead support subsequently deposited layers.

In certain embodiments, one or more electrodes of an analyte sensor described herein is a wire electrode, e.g., a permeable wire electrode. In certain embodiments, the sensor tail comprises a working electrode and a reference electrode helically wound around the working electrode. In certain embodiments, an insulator is disposed between the working and reference electrodes. In certain embodiments, portions of the electrodes are exposed to allow reaction of the one or more enzymes with an analyte on the electrode. In certain embodiments, each electrode is formed from a fine wire with a diameter of from about 0.001 inches or less to about 0.010 inches or more. In certain embodiments, the working electrode has a diameter of from about 0.001 inches or less to about 0.010 inches or more, e.g., from about 0.002 inches to about 0.008 inches, and more preferably from about 0.004 inches to about 0.005 inches. In certain embodiments, an electrode is formed from a plated insulator, a plated wire or bulk electrically conductive material. In certain embodiments, the working electrode comprises a wire formed from a conductive material, such as platinum, platinum-iridium, palladium, graphite, gold, carbon, conductive polymer, alloys or the like. In certain embodiments, the conductive material is a permeable conductive material. In certain embodiments, the electrodes can be formed by a variety of manufacturing techniques (e.g., bulk metal processing, deposition of metal onto a substrate or the like), the electrodes can be formed from plated wire (e.g., platinum on steel wire) or bulk metal (e.g., platinum wire). In certain embodiments, the electrode is formed from tantalum wire covered with platinum.

In certain embodiments, the reference electrode, which can function as a reference electrode alone, or as a dual reference and counter electrode, is formed from silver, silver/silver chloride or the like. In certain embodiments, the reference electrode is juxtaposed and/or twisted with or around the working electrode. In certain embodiments, the reference electrode is helically wound around the working electrode. In certain embodiments, the assembly of wires can be coated or adhered together with an insulating material so as to provide an insulating attachment.

In certain embodiments, additional electrodes can be included in the sensor tail. For example, but not by way of limitation, a three-electrode system (a working electrode, a reference electrode and a counter electrode) and/or an additional working electrode (e.g., an electrode for detecting a second analyte). In certain embodiments where the sensor

comprises two working electrodes, the two working electrodes can be juxtaposed around which the reference electrode is disposed upon (e.g., helically wound around the two or more working electrodes). In certain embodiments, the two or more working electrodes can extend parallel to each other. In certain embodiments, the reference electrode is coiled around the working electrode and extends towards the distal end (i.e., in vivo end) of the sensor tail. In certain embodiments, the reference electrode extends (e.g., helically) to the exposed region of the working electrode.

In certain embodiments, one or more working electrodes are helically wound around a reference electrode. In certain embodiments where two or more working electrodes are provided, the working electrodes can be formed in a double-, triple-, quad-, etc. helix configuration along the length of the sensor tail (for example, surrounding a reference electrode, insulated rod or other support structure). In certain embodiments, the electrodes, e.g., two or more working electrodes, are coaxially formed. For example, but not by way of limitation, the electrodes all share the same central axis.

In certain embodiments, the working electrode comprises a tube with a reference electrode disposed or coiled inside, including an insulator therebetween. Alternatively, the reference electrode comprises a tube with a working electrode disposed or coiled inside, including an insulator therebetween. In certain embodiments, a polymer (e.g., insulating) rod is provided, wherein the one or more electrodes (e.g., one or more electrode layers) are disposed upon (e.g., by electro-plating). In certain embodiments, a metallic (e.g., steel or tantalum) rod or wire is provided, coated with an insulating material (described herein), onto which the one or more working and reference electrodes are disposed upon. For example, but not by way of limitation, the present disclosure provides a sensor, e.g., a sensor tail, that comprises one or more tantalum wires, where platinum is disposed upon a portion of the one or more tantalum wires to function as a working electrode. In certain embodiments, the platinum-clad tantalum wire is covered with an insulating material, where the insulating material is partially covered with a silver/silver chloride composition to function as a reference and/or counter electrode.

In certain embodiments where an insulator is disposed upon the working electrode (e.g., upon the platinum surface of the electrode), a portion of the insulator can be stripped or otherwise removed to expose the electroactive surface of the working electrode. For example, but not by way of limitation, a portion of the insulator can be removed by hand, excimer laser, chemical etching, laser ablation, grit-blasting or the like. Alternatively, a portion of the electrode can be masked prior to depositing the insulator to maintain an exposed electroactive surface area. In certain embodiments, the portion of the insulator that is stripped and/or removed can be from about 0.1 mm (about 0.004 inches) or less to about 2 mm (about 0.078 inches) or more in length, e.g., from about 0.5 mm (about 0.02 inches) to about 0.75 mm (0.03 inches) in length. In certain embodiments, the insulator is a non-conductive polymer. In certain embodiments, the insulator comprises parylene, fluorinated polymers, polyethylene terephthalate, polyvinylpyrrolidone, polyurethane, polyimide and other non-conducting polymers. In certain embodiments, glass or ceramic materials can also be used in the insulator layer. In certain embodiments, the insulator comprises parylene. In certain embodi-

ments, the insulator comprises a polyurethane. In certain embodiments, the insulator comprises a polyurethane and polyvinylpyrrolidone.

Several parts of the sensor are further described below.

2. NAD(P) Depot

The present disclosure provides analyte sensors that can include an internal supply of a cofactor. For example, but not by way of limitation, the present disclosure provides analyte sensors that can include an internal supply of a cofactor that allows the controlled release of the cofactor over an extended period of the time.

In certain embodiments, the cofactor internal supply can be coated with or distributed within a permeable layer that controls diffusion of the cofactor from the cofactor supply to maintain a sufficient concentration of the cofactor in an active area, e.g., sensing chemistry layer, during use of the analyte sensor. The exact nature, size and configuration of the cofactor depot present within an analyte sensor can vary based on the particular application of the analyte sensor, e.g., which analyte is being detected, the duration of analyte detection and the conditions under which the detection of the analyte occurs.

In certain embodiments, the cofactor is NAD or NADP (both of which are referred to herein collectively as “NAD(P)”). In certain embodiments, the NAD(P) is a derivative of NAD(P). Non-limiting examples of NAD(P) derivatives are disclosed in WO 2007/012494 and WO 1998/033936, the contents of each which are disclosed herein in their entireties. In certain embodiments, the present disclosure provides analyte sensors that can include an internal supply of NAD(P) that allows the controlled release of NAD(P) or derivative thereof over an extended period of the time. In certain embodiments, the NAD(P) internal supply can be coated with or distributed within a permeable layer that controls diffusion of NAD(P) from the NAD(P) supply to maintain a sufficient concentration of NAD(P) in an active area, e.g., sensing chemistry layer, comprising one or more NAD(P)-dependent enzymes during use of the analyte sensor.

Non-limiting embodiments of analyte sensors that include an NAD(P) depot are provided in FIGS. 22 and 23A. For example, but not by way of limitation, NAD(P) can be deposited onto a substrate, e.g., a plastic substrate, as shown in FIGS. 22 and 23A. In certain embodiments, an NAD(P) depot can be disposed on substrate 30212. In certain embodiments, an NAD(P) depot can be deposited between dielectric material, e.g., two dielectric layers, as shown in FIG. 23A.

In certain embodiments, the deposited NAD(P) can be overlaid with a permeable layer. As shown in FIGS. 22 and 23A, the NAD(P) depot is at least partially coated with a permeable layer. For example, but not by way of limitation, at least about 10%, at least about 20%, at least about 30%, at least about 40%, at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90% or at least about 95% of the NAD(P) depot is coated with a permeable layer. In certain embodiments, the NAD(P) depot is entirely coated with a permeable layer. In certain embodiments, the permeable layer provides sustained NAD(P) release over time. The composition of the permeable layer can vary depending on the desired release kinetics of the NAD(P), e.g., rate of NAD(P) release, from the internal supply.

Alternatively or additionally, NAD(P) can be present within the permeable layer. For example, but not by way of limitation, NAD(P) can be mixed directly into the permeable layer, e.g., polymer permeable layer, rather than added as a separate layer coated by the permeable layer. In certain

embodiments, an analyte sensor of the present disclosure can include a permeable layer comprising NAD(P) disposed on substrate 30212. In certain embodiments, an analyte sensor of the present disclosure can include an NAD(P) depot disposed on substrate 30212 that is overlaid with a permeable layer that includes a separate supply of NAD(P).

In certain embodiments, the analyte sensor further includes a working electrode, e.g., 214, 30216 and/or 217, that is permeable. In certain embodiments, the permeable working electrode is disposed over the permeable layer as shown in FIGS. 22 and 23A. In certain embodiments, at least one active area (which contains the sensing chemistry) is disposed upon the working electrode as described herein. In certain embodiments, two or more active areas are disposed upon the working electrode as described herein. In the non-limiting exemplary embodiments shown in FIGS. 22 and 23A, NAD(P) diffuses through the permeable layer, e.g., polymer layer, and the permeable working electrode to contact the active area to maintain a sufficient NAD(P) concentration in the active area over time.

In certain embodiments, the amount of NAD(P) present within a NAD(P) depot can vary depending on the duration of use of the analyte sensor. For example, but not by way of limitation, NAD(P) can be present in an NAD(P) depot from about 0.1 μg to about 1,000 μg . In certain embodiments, from about 0.1 μg to about 900 μg , from about 0.1 μg to about 800 μg , from about 0.1 μg to about 700 μg , from about 0.1 μg to about 600 μg , from about 0.1 μg to about 500 μg , from about 0.1 μg to about 400 μg , from about 0.1 μg to about 300 μg , from about 0.1 μg to about 200 μg , from about 0.1 μg to about 100 μg , from about 0.1 μg to about 90 μg , from about 0.1 μg to about 80 μg , from about 0.1 μg to about 70 μg , from about 0.1 μg to about 60 μg , from about 0.1 μg to about 50 μg , from about 0.1 μg to about 40 μg , from about 0.1 μg to about 30 μg , from about 0.1 μg to about 20 μg , from about 0.1 μg to about 10 μg , from about 0.1 μg to about 9 μg , from about 0.1 μg to about 8 μg , from about 0.1 μg to about 7 μg , from about 0.1 μg to about 6 μg , from about 0.1 μg to about 5 μg , from about 0.1 μg to about 4 μg , from about 0.1 μg to about 3 μg , from about 0.1 μg to about 2 μg , from about 0.1 μg to about 1 μg , from about 0.1 μg to about 0.9 μg , from about 0.1 μg to about 0.8 μg , from about 0.1 μg to about 0.7 μg , from about 0.1 μg to about 0.6 μg , from about 0.1 μg to about 0.5 μg , from about 0.1 μg to about 0.4 μg , from about 0.1 μg to about 0.3 μg , from about 0.1 μg to about 0.2 μg , from about 0.2 μg to about 1,000 μg , from about 0.3 μg to about 1,000 μg , from about 0.4 μg to about 1,000 μg , from about 0.5 μg to about 1,000 μg , from about 0.6 μg to about 1,000 μg , from about 0.7 μg to about 1,000 μg , from about 0.8 μg to about 1,000 μg , from about 0.9 μg to about 1,000 μg , from about 1 μg to about 1,000 μg , from about 2 μg to about 1,000 μg , from about 3 μg to about 1,000 μg , from about 4 μg to about 1,000 μg , from about 5 μg to about 1,000 μg , from about 6 μg to about 1,000 μg , from about 7 μg to about 1,000 μg , from about 8 μg to about 1,000 μg , from about 9 μg to about 1,000 μg , from about 10 μg to about 1,000 μg , from about 11 μg to about 1,000 μg , from about 12 μg to about 1,000 μg , from about 13 μg to about 1,000 μg , from about 14 μg to about 1,000 μg , from about 15 μg to about 1,000 μg , from about 16 μg to about 1,000 μg , from about 17 μg to about 1,000 μg , from about 18 μg to about 1,000 μg , from about 19 μg to about 1,000 μg , from about 20 μg to about 1,000 μg , from about 30 μg to about 1,000 μg , from about 40 μg to about 1,000 μg , from about 50 μg to about 1,000 μg , from about 60 μg to about 1,000 μg , from about 70 μg to about 1,000 μg , from about 80 μg to about 1,000 μg , from about 90 μg to about 1,000 μg , from about

100 µg to about 1,000 µg, from about 200 µg to about 1,000 µg, from about 300 µg to about 1,000 µg, from about 400 µg to about 1,000 µg, from about 500 µg to about 1,000 µg, from about 600 µg to about 1,000 µg, from about 700 µg to about 1,000 µg, from about 800 µg to about 1,000 µg, from about 900 µg to about 1,000 µg, from about 0.1 µg to about 100 µg, from about 1 µg to about 100 µg, from about 1 µg to about 90 µg, from about 1 µg to about 80 µg, from about 1 µg to about 70 µg, from about 1 µg to about 60 µg, from about 1 µg to about 50 µg, from about 1 µg to about 40 µg, from about 1 µg to about 30 µg, from about 1 µg to about 20 µg, from about 1 µg to about 15 µg, from about 1 µg to about 10 µg or from about 5 µg to about 15 µg NAD(P) can be present in an NAD(P) depot. In certain embodiments, NAD(P) can be present in an NAD(P) depot from about 0.1 µg to about 100 µg.

In certain embodiments, the amount of NAD(P) present in the NAD(P) depot varies depending on the lifetime of the analyte sensor. For example, but not by way of limitation, the amount of NAD(P) in the NAD(P) depot allows the analyte sensor to detect an analyte using an NAD(P)-dependent enzyme for at least about 7 days, for at least about 8 days, for at least about 9 days, for at least about 10 days, for at least about 11 days, for at least about 12 days, for at least about 13 days, for at least about 14 days, for at least about 15 days, for at least about 16 days, for at least about 17 days, for at least about 18 days, for at least about 19 days, for at least about 20 days, for at least about 25 days, for at least about 30 days, for at least about 35 days or for at least about 40 days. In certain embodiments, the amount of NAD(P) in the NAD(P) depot allows the analyte sensor to detect an analyte using an NAD(P)-dependent enzyme for at least about 14 days. In certain embodiments, the amount of NAD(P) in the NAD(P) depot allows the analyte sensor to detect an analyte using an NAD(P)-dependent enzyme for greater than about two weeks, for greater than about three weeks, for greater than about four weeks, for greater than about five weeks, for greater than about six weeks, for greater than about seven weeks or for greater than about eight weeks.

In certain embodiments, the permeable layer can include a polymer. In certain embodiments, the permeable polymer layer can include a diffusion-controlling polymer. In certain embodiments, the permeable polymer layer can include any polymer that allows controlled diffusion of a cofactor. In certain embodiments, the permeable polymer layer can include any polymer that allows controlled diffusion of NAD(P) thereof.

In certain embodiments, the permeable polymer layer can include hyaluronic acid (HA), poly(ethylene glycol) (PEG), phosphoryl choline based polymers and other hydrophilic polymers having a hydrophilicity comparable to HA, PEG, or phosphoryl choline, ethylene vinyl alcohol copolymer, polyhydroxyalkanoate, poly(hydroxyvalerate), polycaprolactone, poly(lactide-co-glycolide), poly(hydroxybutyrate), poly(hydroxybutyrate-co-valerate), polydioxanone, polyorthoester, polyanhydride, poly(glycolic acid), poly(D,L-lactic acid) (DLPLA), poly(ortho esters), poly(glycolic acid-co-trimethylene carbonate), polyphosphoester, polyphosphoester urethane, poly(amino acids), cyanoacrylates, poly(trimethylene carbonate), poly(iminocarbonate), polyurethanes, copoly(ether-esters) (e.g., PEO/PLA), polyalkylene oxalates, polyphosphazenes, biomolecules, such as fibrin, fibrinogen, cellulose, starch and collagen, polyurethanes, silicones, polyesters, polyolefins, polyisobutylene and ethylene-alphaolefin copolymers, acrylic polymers and copolymers, vinyl halide polymers and copolymers, poly

(amide ester) (PEA), polycaprolactone (PCL), poly(hexafluoro propylene) (HFP), poly(ethylene vinyl alcohol) (EVAL), polyvinyl ethers such as polyvinyl methyl ether, polyvinylidene halides such as polyvinylidene fluoride (PVDF) and polyvinylidene chloride, polyacrylonitrile, polyvinyl ketones, polyvinyl aromatics such as polystyrene, polyvinyl esters such as polyvinyl acetate, copolymers of vinyl monomers with each other and olefins such as ethylene-methyl methacrylate copolymers, acrylonitrile-styrene copolymers, ABS resins, and ethylene-vinyl acetate copolymers, polyamides such as Nylon 66 and polycaprolactam, alkyd resins, polycarbonates, polyoxymethylenes, polyimides, polyethers, epoxy resins, polyurethanes, rayon, rayon-triacetate, cellulose acetate, cellulose butyrate, cellulose acetate butyrate, cellophane, cellulose nitrate, cellulose propionate, cellulose ethers and carboxymethyl cellulose. In certain embodiments, a suitable polymer is a copolymer comprising a poly(ethylene glycol terephthalate) and poly(butylene terephthalate) (PEGT/PBT) segments.

Further non-limiting examples of polymers that can be present in the permeable layer include polycarboxylic acids, cellulosic polymers, gelatin, polyvinylpyrrolidone, cross-linked polyvinylpyrrolidone, polyanhydrides including maleic anhydride polymers, polyvinyl alcohols, polyvinyl aromatics such as copolymers of polystyrene with other vinyl monomers such as isobutylene, isoprene and butadiene, for example, styrene-isobutylene-styrene (SIBS), styrene-isoprene-styrene (SIS) copolymers, styrene-butadiene-styrene (SBS) copolymers, polyethylene oxides, glycosaminoglycans, polysaccharides, polyesters including polyethylene terephthalate, polyacrylamides, polyether sulfone, polyalkylenes including polypropylene, polyethylene and high molecular weight polyethylene, halogenated polyalkylenes including polytetrafluoroethylene, natural and synthetic rubbers including polyisoprene, polybutadiene, polyisobutylene and copolymers thereof with other vinyl monomers such as polyorthoesters, proteins, polypeptides, siloxane polymers, polylactic acid, polyglycolic acid, polyhydroxybutyrate valerate and blends and copolymers thereof as well as other biodegradable, bioabsorbable and biostable polymers and copolymers. In certain embodiments, suitable polymers include polyacrylic acid and a copolymer of polylactic acid and polycaprolactone.

In certain embodiments, the permeable layer can include a polyether-based polymer. In certain embodiments, the permeable layer can include a poly(ethylene glycol). In certain embodiments, the permeable layer can include a poly(ethylene glycol)-based polymer. In certain embodiments, the permeable layer can include a poly(propylene glycol). In certain embodiments, the permeable layer can include a poly(propylene glycol)-based polymer. In certain embodiments, the permeable layer can include poly(propylene glycol) methacrylate (POMA). In certain embodiments, the permeable layer can include 2-hydroxyethyl methacrylate (HEMA). In certain embodiments, the permeable layer can include a mixture of POMA and HEMA. In certain embodiments, the permeable layer can include a mixture of POMA and HEMA to generate the permeable polymer. For example, but not by way of limitation, the permeable layer can include a ratio of POMA to HEMA from about 10:1 to about 1:10, e.g., from about 9:1 to about 1:9, from about 8:1 to about 1:8, from about 7:1 to about 1:7, from about 6:1 to about 1:6, from about 5:1 to about 1:5, from about 4:1 to about 1:4, from about 3:1 to about 1:3, from about 2:1 to about 1:2. In certain embodiments, the ratio of POMA to HEMA can be from about 2:1 to about 1:2. In certain embodiments, the ratio of POMA to HEMA can be about

1:1. In certain embodiments, the permeable layer can include from about 20% to about 80% by weight of POMA, e.g., from about 30% to about 70% or from about 40% to about 60% by weight. In certain embodiments, the permeable layer can from about 40% to about 60% by weight of POMA. In certain embodiments, the permeable layer can from about 20% to about 80% by weight of HEMA, e.g., from about 30% to about 70% or from about 40% to about 60% by weight. In certain embodiments, the permeable layer can from about 40% to about 60% by weight of HEMA.

In certain embodiments, the permeable polymer is a hydrogel. In certain embodiments, a permeable polymer, e.g., hydrogel, for use in the present disclosure is capable of absorbing from about 30% to about 95% of its weight in water, e.g., from about 30% to about 70% or from about 40% to about 60%. In certain embodiments, the permeable polymer, e.g., hydrogel, is capable of absorbing at least about 30% of its weight in water. In certain embodiments, the permeable polymer, e.g., hydrogel, is capable of absorbing at least about 40% of its weight in water. In certain embodiments, the permeable polymer, e.g., hydrogel, is capable of absorbing at least about 50% of its weight in water. In certain embodiments, the permeable polymer, e.g., hydrogel, is capable of absorbing at least about 60% of its weight in water. In certain embodiments, the permeable polymer, e.g., hydrogel, is capable of absorbing at least about 70% of its weight in water. In certain embodiments, the permeable polymer, e.g., hydrogel, is capable of absorbing from about 30% to about 60% of its weight in water.

In certain embodiments, the NAD(P) depot has a thickness, e.g., dry thickness, ranging from about 0.1 μm to about 1,000 μm , e.g., from about 1 μm to about 500 μm , about 10 μm to about 100 μm or about 10 μm to about 100 μm . In certain embodiments, the NAD(P) depot can have a thickness from about 0.1 μm to about 10 μm , e.g., from about 0.5 μm to about 10 μm , from about 1 μm to about 10 μm , from about 1 μm to about 5 μm or from about 0.1 μm to about 5 μm .

In certain embodiments, the permeable electrode can include any material that is permeable to NAD(P). In certain embodiments, the permeable electrode can include any conductive material, e.g., conductive ink or polymer, that is permeable to NAD(P). For example, but not by way of limitation, the permeable electrode can include carbon, silver, amorphous carbon, graphite, graphene, glassy carbon, platinized carbon, gold, platinum and/or palladium. In certain embodiments, the permeable electrode can include a carbon material. In certain embodiments, the permeable electrode can include a carbon material that includes an additive such as, but not limited to, silver, amorphous carbon, graphite, graphene, glassy carbon, platinized carbon, gold, platinum and/or palladium. In certain embodiments, the permeable electrode can be at least partially composed of a carbon nanotube. In certain embodiments, the permeable electrode can include a conductive polymer, e.g., poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT: PSS). In certain embodiments, the conductive material can be present in a polymer, e.g., a polymeric carrier. In certain embodiments, the electrode comprises a polymer comprising a conductive material and/or conductive particles.

3. Enzymes

The sensors of the present disclosure include one or more enzymes for detecting one or more analytes in at least one active area. Suitable enzymes for use in a sensor of the present disclosure include, but are not limited to, any NAD(P)-dependent enzyme. For example, an NAD(P)-dependent enzyme for use in the present disclosure can be used

for detecting glucose, ketones, lactate, oxygen, hemoglobin A1C, albumin, alcohol, alkaline phosphatase, alanine transaminase, aspartate aminotransferase, bilirubin, blood urea nitrogen, calcium, carbon dioxide, chloride, creatinine, hematocrit, lactate, magnesium, oxygen, pH, phosphorus, potassium, sodium, total protein, uric acid, etc. In certain embodiments, the analyte to be detected using an NAD(P)-dependent enzyme is glucose, lactate, ketones, creatinine, alcohol, e.g., ethanol, or the like. In certain embodiments, an active area can include multiple enzymes, e.g., an enzyme system, that are collectively responsive to the analyte.

In certain embodiments, an active area of a presently disclosed analyte sensor includes at least one NAD(P)-dependent enzyme. In certain embodiments, an active area of a presently disclosed analyte sensor includes two or more NAD(P)-dependent enzymes. In certain embodiments, an analyte sensor of the present disclosure includes two active sites that each include at least one NAD(P)-dependent enzyme. Alternatively, an analyte sensor of the present disclosure in certain embodiments includes two active sites, where only one active site includes an NAD(P)-dependent enzyme. Non-limiting examples of NAD(P)-dependent enzymes are disclosed in Vidal et al., *Biochimica et Biophysica Acta—Proteins and Proteomics* 1866(2):327-347 (2018) (see Tables 1-2), the contents of which are incorporated by reference herein.

In certain embodiments, an analyte sensor of the present disclosure includes one or more internal supplies of NAD(P) for an NAD(P)-dependent enzyme included in one or more active sites of the analyte sensor. For example, but not by way of limitation, an NAD(P) depot can be disposed underneath each electrode configured for detecting an analyte. Alternatively, an NAD(P) depot can be disposed underneath only one electrode that is configured for detecting an analyte.

In certain embodiments, an active site can include an NAD(P)-dependent dehydrogenase. Non-limiting examples of NAD(P)-dependent dehydrogenases include glucose dehydrogenase (E.C.1.1.1.47), lactate dehydrogenase (EC1.1.1.27 and EC1.1.1.28), malate dehydrogenase (EC1.1.1.37), glycerol dehydrogenase (EC1.1.1.6), alcohol dehydrogenase (EC1.1.1.1), alpha-hydroxybutyrate dehydrogenase, sorbitol dehydrogenase, amino acid dehydrogenase such as L-amino acid dehydrogenase (EC1.4.1.5), diaphorase (EC1.8.1.4) and combinations thereof.

In certain embodiments, the NAD(P)-dependent dehydrogenase can include diaphorase, glucose dehydrogenase, alcohol dehydrogenase, lactate dehydrogenase and β -hydroxybutyrate dehydrogenase. In certain embodiments, the enzyme system can include two or more NAD(P)-dependent dehydrogenases, e.g., a first NAD(P)-dependent dehydrogenase and diaphorase. For example, but not by way of limitation, the NAD(P)-dependent dehydrogenase can convert the analyte and oxidized nicotinamide adenine dinucleotide (NAD⁺) into an oxidized analyte and reduced nicotinamide adenine dinucleotide (NADH), respectively. The enzyme cofactors NAD⁺ and NADH aid in promoting the concerted enzymatic reactions disclosed herein. The NADH can then undergo reduction under diaphorase mediation, with the electrons transferred during this process providing the basis for analyte detection at the working electrode.

In certain embodiments, an analyte sensor of the present disclosure can include a glucose-responsive active area, a ketones-responsive active area, a lactate-responsive active area, a creatinine-responsive active area, an alcohol-responsive active area or a combination thereof. In certain embodiments, a glucose-responsive active area can include one or

more NAD(P)-dependent enzymes for detecting glucose. In certain embodiments, a ketones-responsive active area can include one or more NAD(P)-dependent enzymes for detecting ketones. In certain embodiments, a lactate-responsive active area can include one or more NAD(P)-dependent enzymes for detecting lactate. In certain embodiments, a creatinine-responsive active area can include one or more NAD(P)-dependent enzymes for detecting creatinine. In certain embodiments, an alcohol-responsive active area can include one or more NAD(P)-dependent enzymes for detecting alcohol. In certain embodiments, an active area can include an enzyme system comprising two or more enzymes that are collectively responsive to the analyte. For example, but not by way of limitation, a ketones-responsive active area can include an enzyme system comprising at least one NAD(P)-dependent enzyme.

In certain embodiments, an active site can be a glucose-responsive active site that includes at least one NAD(P)-dependent enzyme for detecting glucose. In certain embodiments, a glucose-responsive active site can include a glucose dehydrogenase. For example, but not by way of limitation, an analyte sensor of the present disclosure for detecting glucose can include an NAD(P) depot and an active area comprising a glucose dehydrogenase.

In certain embodiments, an active site can be an alcohol-responsive active site that includes at least one NAD(P)-dependent enzyme for detecting one or more alcohols. In certain embodiments, an alcohol-responsive active site can include an alcohol dehydrogenase. For example, but not by way of limitation, an analyte sensor of the present disclosure for detecting an alcohol can include an NAD(P) depot and an active area comprising an alcohol dehydrogenase.

In certain embodiments, an active site can be a ketones-responsive active site that includes at least one NAD(P)-dependent enzyme for detecting one or more ketones. In certain embodiments, a ketones-responsive active site can include β -hydroxybutyrate dehydrogenase. For example, but not by way of limitation, an analyte sensor of the present disclosure for detecting ketones can include an NAD(P) depot and an active area comprising an enzyme system that includes β -hydroxybutyrate dehydrogenase.

In certain embodiments, an active site can be a lactate-responsive active site that includes at least one NAD(P)-dependent enzyme for detecting lactate. For example, but not by way of limitation, a lactate-responsive active site can include a lactate dehydrogenase. In certain embodiments an analyte sensor of the present disclosure for detecting lactate can include an NAD(P) depot and an active area comprising a lactate dehydrogenase.

In certain embodiments, an analyte sensor disclosed herein can include at least one active site that includes one or more NAD(P)-dependent enzymes, as disclosed herein, for detecting an analyte. Alternatively, an analyte sensor disclosed herein can include two or more active sites, with each active site contains one or more enzymes, e.g., where at least one of the active sites includes one or more NAD(P)-dependent enzymes. For example, but not by way of limitation, an analyte sensor of the present disclosure can include a first active area that comprises a first enzyme (or enzyme system) for use in detecting a first analyte and a second active site that includes a second enzyme (or second enzyme system) for detecting a second analyte, where at least the first active area or second active area includes an NAD(P)-dependent enzyme.

In certain embodiments, the analyte-responsive active area can include from about 10% to about 80% by weight, e.g., from about 15% to about 75%, from about 20% to about

70%, from about 25% to about 65% or from about 30% to about 60% by weight, of one or more enzymes (e.g., one or more NAD(P)-dependent enzymes) disclosed herein. In certain embodiments, the analyte-responsive active area can include from about 10% to about 80% by weight, e.g., from about 15% to about 75%, from about 20% to about 70%, from about 25% to about 65%, from about 30% to about 60% by weight, from about 20% to about 60% or from about 20% to about 50%, of one or more enzymes (e.g., one or more NAD(P)-dependent enzymes) disclosed herein. In certain embodiments, the analyte-responsive active area can include from about 10% to about 80% by weight, e.g., from about 15% to about 75%, from about 20% to about 70%, from about 25% to about 65% or from about 30% to about 60% by weight, of one or more enzymes (e.g., one or more NAD(P)-dependent enzymes) disclosed herein. In certain embodiments, the analyte-responsive active area can include from about 10% to about 80% by weight, e.g., from about 15% to about 75%, from about 20% to about 70%, from about 25% to about 65% or from about 30% to about 60% by weight, of one or more enzymes (e.g., one or more NAD(P)-dependent enzymes) disclosed herein.

In certain embodiments, the analyte-responsive active area can further include a stabilizer, e.g., for stabilizing the enzyme. For example, but not by way of limitation, the stabilizer can be an albumin, e.g., a serum albumin. Non-limiting examples of serum albumins include bovine serum albumin and human serum albumin. In certain embodiments, the stabilizer is a human serum albumin. In certain embodiments, the stabilizer is a bovine serum albumin. In certain embodiments, the stabilizer can be catalase. In certain embodiments, the analyte-responsive active area can include a ratio of stabilizer to the one or more enzymes present in the analyte-responsive active area, e.g., NAD(P)-dependent enzyme, from about 40:1 to about 1:40, e.g., from about 35:1 to about 1:35, from about 30:1 to about 1:30, from about 25:1 to about 1:25, from about 20:1 to about 1:20, from about 15:1 to about 1:15, from about 10:1 to about 1:10, from about 9:1 to about 1:9, from about 8:1 to about 1:8, from about 7:1 to about 1:7, from about 6:1 to about 1:6, from about 5:1 to about 1:5, from about 4:1 to about 1:4, from about 3:1 to about 1:3, from about 2:1 to about 1:2 or about 1:1. In certain embodiments, the analyte-responsive active area can include a ratio of stabilizer to the one or more enzymes present in the analyte-responsive active area, e.g., NAD(P)-dependent enzyme, from about 2:1 to about 1:2. In certain embodiments, the analyte-responsive active area can include a ratio of stabilizer to the NAD(P)-dependent enzyme, e.g., NAD(P)-dependent dehydrogenase, from about 40:1 to about 1:40, e.g., from about 35:1 to about 1:35, from about 30:1 to about 1:30, from about 25:1 to about 1:25, from about 20:1 to about 1:20, from about 15:1 to about 1:15, from about 10:1 to about 1:10, from about 9:1 to about 1:9, from about 8:1 to about 1:8, from about 7:1 to about 1:7, from about 6:1 to about 1:6, from about 5:1 to about 1:5, from about 4:1 to about 1:4, from about 3:1 to about 1:3, from about 2:1 to about 1:2 or about 1:1. In certain embodiments, the analyte-responsive active area can include a ratio of stabilizer to the NAD(P)-dependent enzyme, e.g., NAD(P)-dependent dehydrogenase, from about 2:1 to about 1:2. In certain embodiments, the analyte-responsive active area can include from about 10% to about 50%, e.g., from about 15% to about 45%, from about 20% to about 40%, from about 20% to about 35% or from about 20% to about 30% by weight of the stabilizer. In certain embodiments, the analyte-responsive active area can include from about 15% to about 35% of the stabilizer by weight.

In certain embodiments, in addition to the presence of an NAD(P) depot, the analyte-responsive active area can further include a cofactor for one or more enzyme present in the analyte-responsive active area. In certain embodiments, the cofactor is NAD(P). In certain embodiments, the cofactor is a cofactor different from NAD(P). In certain embodiments, the analyte-responsive active area can include a ratio of cofactor to enzyme from about 40:1 to about 1:40, e.g., from about 35:1 to about 1:35, from about 30:1 to about 1:30, from about 25:1 to about 1:25, from about 20:1 to about 1:20, from about 15:1 to about 1:15, from about 10:1 to about 1:10, from about 9:1 to about 1:9, from about 8:1 to about 1:8, from about 7:1 to about 1:7, from about 6:1 to about 1:6, from about 5:1 to about 1:5, from about 4:1 to about 1:4, from about 3:1 to about 1:3, from about 2:1 to about 1:2 or about 1:1. In certain embodiments, the analyte-responsive active area can include a ratio of cofactor to enzyme from about 2:1 to about 1:2. In certain embodiments, the analyte-responsive active area can include from about 10% to about 50% by weight, e.g., from about 15% to about 45%, from about 20% to about 40%, from about 20% to about 35%, from about 20% to about 30% by weight of the cofactor. In certain embodiments, the analyte-responsive active area can include from about 15% to about 35% by weight of the cofactor. In certain embodiments, the cofactor, e.g., NAD(P), can be physically retained within the analyte-responsive active area. For example, but not by way of limitation, a membrane overcoating the analyte-responsive active area can aid in retaining the cofactor within the analyte-responsive active area while still permitting sufficient inward diffusion of the analyte to permit detection thereof.

In certain embodiments, an analyte sensor of the present disclosure can include a sensor tail comprising at least one NAD(P) depot, at least one working electrode, e.g., permeable electrode, and an analyte-responsive active area disposed upon the surface of the working electrode, where the analyte-responsive active area includes at least one NAD(P)-dependent enzyme. In certain embodiments, an analyte sensor of the present disclosure can include a sensor tail comprising a substrate, at least one NAD(P) depot disposed upon a surface of the substrate, at least one working electrode, e.g., permeable electrode, and an analyte-responsive active area disposed upon the surface of the working electrode, where the analyte-responsive active area includes at least one NAD(P)-dependent enzyme. In certain embodiments, the NAD(P)-dependent enzyme is an NAD(P)-dependent dehydrogenase. For example, but not by way of limitation, a sensor of the present disclosure can include a sensor tail comprising at least one NAD(P) depot, a permeable layer disposed upon the NAD(P) depot, at least one permeable working electrode and an analyte-responsive active area disposed upon the surface of the permeable working electrode, where the analyte-responsive active area includes an enzyme system comprising an NAD(P) dependent-dehydrogenase.

In certain embodiments, a sensor of the present disclosure can include a sensor tail comprising at least one NAD(P) depot, a permeable layer disposed on top of the NAD(P) depot, at least one permeable working electrode and a ketones-responsive active area disposed upon the surface of the permeable working electrode, where the ketones-responsive active area includes an enzyme system comprising an NAD(P)-dependent dehydrogenase, e.g., β -hydroxybutyrate dehydrogenase. In certain embodiments, the enzyme system further includes diaphorase.

In certain embodiments, an analyte sensor of the present disclosure can include a second active area, e.g., for detecting an analyte different from the analyte detected by the first active area. In certain embodiments, the second active area is disposed upon the same working electrode as the first active area or on a second working electrode. In certain embodiments, the second active area is a glucose-responsive active area, a lactate-responsive active area, a creatinine-responsive active area or an alcohol-responsive active area.

In certain embodiments, the second active area of an analyte sensor of the present disclosure can include one or more enzymes for detecting glucose. For example, but not by way of limitation, an analyte sensor of the present disclosure can include an active area (e.g., a second active area) that comprises one or more enzymes for detecting glucose, e.g., disposed on a second working electrode. In certain embodiments, the analyte sensor can include an active site comprising a glucose oxidase and/or a glucose dehydrogenase for detecting glucose.

In certain embodiments, the second active area can include one or more enzymes for detecting lactate. For example, but not by way of limitation, an analyte sensor of the present disclosure can include an active area (e.g., a second active area) that comprises one or more enzymes, e.g., an enzyme system, for detecting lactate, e.g., disposed on a second working electrode. In certain embodiments, the analyte sensor can include an active site comprising a lactate dehydrogenase and/or a lactate oxidase.

In certain embodiments, the second enzyme-responsive active area, e.g., present on a second working electrode, of an analyte sensor of the present disclosure can include one or more enzymes for detecting alcohol. For example, but not by way of limitation, an analyte sensor of the present disclosure can include an active area (e.g., a second active area) that comprises one or more enzymes, e.g., an enzyme system, for detecting alcohol, e.g., disposed on a second working electrode. In certain embodiments, the analyte sensor can include an active site comprising an alcohol dehydrogenase.

In certain embodiments, the second enzyme-responsive active area, e.g., present on a second working electrode, of an analyte sensor of the present disclosure can include one or more enzymes for detecting creatinine. For example, but not by way of limitation, an analyte sensor of the present disclosure can include an active area (e.g., a second active area) that comprises one or more enzymes, e.g., an enzyme system, for detecting creatinine, e.g., disposed on a second working electrode. In certain embodiments, the analyte sensor can include an active site comprising an amidohydrolase, creatine amidinohydrolase and/or sarcosine oxidase.

In certain embodiments, an analyte sensor can include two working electrodes, e.g., a first active area disposed on a first working electrode (e.g., permeable electrode) and a second active area disposed on a second working electrode. For example, but not by way of limitation, an analyte sensor disclosed herein can feature at least one NAD(P) depot, a first analyte-responsive active area disposed on a first working electrode and a second analyte-responsive active area disposed upon the surface of a different working electrode, e.g., second working electrode, where at least one of the analyte-responsive active areas includes an NAD(P)-dependent enzyme. In certain embodiments, the second analyte-responsive active area can be configured to detect a different analyte or the same analyte detected by first analyte-responsive active area. In certain embodiments, such analyte sensors can include a sensor tail with at least one NAD(P)

depot, a first working electrode and a second working electrode, a first analyte-responsive active area disposed upon a surface of the first working electrode and a second analyte-responsive active area disposed upon a surface of the second working electrode, where at least one of the analyte-responsive active areas includes an NAD(P)-dependent enzyme and at least one of the working electrodes is permeable. For example, but not by way of limitation, the analyte-responsive active area that includes an NAD(P)-dependent enzyme is disposed upon the permeable working electrode.

In certain embodiments, when the sensor is configured to detect two or more analytes using two working electrodes, detection of each analyte can include applying a potential to each working electrode separately, such that separate signals are obtained from each analyte. The signal obtained from each analyte can then be correlated to an analyte concentration through use of a calibration curve or function, or by employing a lookup table. In certain particular embodiments, correlation of the analyte signal to an analyte concentration can be conducted through use of a processor.

In certain other analyte sensor configurations, the first active area and the second active area can be disposed upon a single working electrode. For example, but not by way of limitation, an analyte sensor disclosed herein can feature at least one NAD(P) depot, a first analyte-responsive active area and a second analyte-responsive active area disposed upon the surface of a single permeable working electrode, where at least one of the analyte-responsive active areas includes an NAD(P)-dependent enzyme. In certain embodiments, a first signal can be obtained from the first active area, e.g., at a low potential, and a second signal containing a signal contribution from both active areas can be obtained at a higher potential. Subtraction of the first signal from the second signal can then allow the signal contribution arising from the second analyte to be determined. The signal contribution from each analyte can then be correlated to an analyte concentration in a similar manner to that described for sensor configurations having multiple working electrodes. In certain embodiments, when a ketones-responsive active area and the second active area configured to detect a different analyte, e.g., a glucose-responsive active area, are arranged upon a single working electrode in this manner, one of the active areas can be configured such that it can be interrogated separately to facilitate detection of each analyte. For example, either the ketones-responsive active area or glucose-responsive active area can produce a signal independently of the other active area.

It is also to be appreciated that the sensitivity (output current) of the analyte sensors toward each analyte can be varied by changing the coverage (area or size) of the active areas, the area ratio of the active areas with respect to one another, the identity, thickness and/or composition of a mass transport limiting membrane overcoating the active areas. Variation of these parameters can be conducted readily by one having ordinary skill in the art once granted the benefit of the disclosure herein.

4. Redox Mediators

In certain embodiments, an analyte sensor disclosed herein can include an electron transfer agent. For example, but not by way of limitation, one or more active sites of an analyte sensor can include an electron transfer agent. In certain embodiments, an analyte sensor can include one active site that includes an electron transfer agent and a second active site that does not include an electron transfer agent. In certain embodiments, the presence of an electron transfer agent in an active area can depend on the enzyme or

enzyme system used to detect the analyte and/or the composition of the working electrode. Alternatively, an analyte sensor can include two active sites, where both active sites include an electron transfer agent.

Suitable electron transfer agents can facilitate conveyance of electrons to the adjacent working electrode after an analyte undergoes an enzymatic oxidation-reduction reaction within the corresponding active area, thereby generating a current that is indicative of the presence of that particular analyte. The amount of current generated is proportional to the quantity of analyte that is present. In certain embodiments, suitable electron transfer agents can include electroreducible and electrooxidizable ions, complexes or molecules (e.g., quinones) having oxidation-reduction potentials that are a few hundred millivolts above or below the oxidation-reduction potential of the standard calomel electrode (SCE). In certain embodiments, the redox mediators can include osmium complexes and other transition metal complexes, such as those described in U.S. Pat. Nos. 6,134,461 and 6,605,200, which are incorporated herein by reference in their entirety. Additional examples of suitable redox mediators include those described in U.S. Pat. Nos. 6,736,957, 7,501,053 and 7,754,093, the disclosures of each of which are also incorporated herein by reference in their entirety. Other examples of suitable redox mediators include metal compounds or complexes of ruthenium, osmium, iron (e.g., polyvinylferrocene or hexacyanoferrate), or cobalt, including metallocene compounds thereof, for example. Suitable ligands for the metal complexes can also include, for example, bidentate or higher denticity ligands such as, for example, bipyridine, biimidazole, phenanthroline, or pyridyl(imidazole). Other suitable bidentate ligands can include, for example, amino acids, oxalic acid, acetylacetone, diaminoalkanes, or o-diaminoarenes. Any combination of monodentate, bidentate, tridentate, tetradentate, or higher denticity ligands can be present in a metal complex, e.g., osmium complex, to achieve a full coordination sphere.

In certain embodiments, electron transfer agents disclosed herein can comprise suitable functionality to promote covalent bonding to a polymer (also referred to herein as a polymeric backbone) within the active areas as discussed further below. For example, but not by way of limitation, an electron transfer agent for use in the present disclosure can include a polymer-bound electron transfer agent. Suitable non-limiting examples of polymer-bound electron transfer agents include those described in U.S. Pat. Nos. 8,444,834, 8,268,143 and 6,605,201, the disclosures of which are incorporated herein by reference in their entirety. In certain embodiments, the electron transfer agent is a bidentate osmium complex bound to a polymer described herein, e.g., a polymeric backbone described in Section 5 below. In certain embodiments, the polymer-bound electron transfer agent shown in FIG. 3 of U.S. Pat. No. 8,444,834 can be used in a sensor of the present disclosure.

In certain embodiments of the present disclosure, an analyte sensor can include at least one NAD(P) depot, at least one permeable layer disposed upon the NAD(P) depot, at least one working electrode, e.g., permeable electrode, and at least one analyte-responsive active area disposed upon the surface of the working electrode, where the analyte-responsive active area includes at least one NAD(P)-dependent enzyme and at least one redox mediator, e.g., an osmium complex. In certain embodiments, the analyte-responsive active area includes an enzyme system comprising a diaphorase, an NAD(P)-dependent dehydrogenase, e.g., β -hydroxybutyrate dehydrogenase, and a redox mediator, e.g., an osmium complex.

5. Polymeric Backbone

In certain embodiments, one or more active sites for promoting analyte detection can include a polymer to which an enzyme and/or redox mediator is covalently bound. Any suitable polymeric backbone can be present in the active area for facilitating detection of an analyte through covalent bonding of the enzyme and/or redox mediator thereto. Non-limiting examples of suitable polymers within the active area include polyvinylpyridines, e.g., poly(-vinylpyridine) and/or poly(-vinylpyridine), and polyvinylimidazoles, e.g., poly(N-vinylimidazole) and poly(l-vinylimidazole), or a copolymer thereof, for example, in which quaternized pyridine groups serve as a point of attachment for the redox mediator or enzyme thereto. Illustrative copolymers that can be suitable for inclusion in the active areas include those containing monomer units such as styrene, acrylamide, methacrylamide, or acrylonitrile, for example. In certain embodiments, polymers that can be present in an active area include a polyurethane or a copolymer thereof, and/or polyvinylpyrrolidone. In certain embodiments, polymers that can be present in the active area include, but are not limited to, those described in U.S. Pat. No. 6,605,200, incorporated herein by reference in its entirety, such as poly(acrylic acid), styrene/maleic anhydride copolymer, methylvinylether/maleic anhydride copolymer (GANTREZ polymer), poly(vinylbenzylchloride), poly(allylamine), polylysine, poly(4-vinylpyridine) quaternized with carboxypentyl groups, and poly(sodium 4-styrene sulfonate). In certain embodiments where the analyte sensor includes two active sites, the polymer within each active area can be the same or different.

In certain embodiments, the polymer is polyvinylpyridine or a copolymer thereof. In certain embodiments, the polymer is a co-polymer of vinylpyridine and styrene.

In certain embodiments, when an enzyme system with multiple enzymes is present in a given active area, all of the multiple enzymes can be covalently bonded to the polymer. In certain other embodiments, only a subset of the multiple enzymes is covalently bonded to the polymer. For example, and not by the way of limitation, one or more enzymes within an enzyme system can be covalently bonded to the polymer and at least one enzyme can be non-covalently associated with the polymer, such that the non-covalently bonded enzyme is physically retained within the polymer. In certain embodiments, the NAD(P)-dependent enzyme can be covalently bonded to the polymer. Alternatively, the NAD(P)-dependent enzyme can be non-covalently associated with the polymer. In certain embodiments, the NAD(P)-dependent dehydrogenase and the diaphorase can be covalently bonded to a polymer within an active area of the disclosed analyte sensors. In certain embodiments, the NAD(P)-dependent dehydrogenase can be covalently bonded to the polymer and diaphorase can be non-covalently associated with the polymer. Alternatively, diaphorase can be covalently bonded to the polymer and the NAD(P)-dependent dehydrogenase can be non-covalently associated with the polymer.

In certain embodiments, when a stabilizer is present in an active area, one or more enzymes within the area can be covalently bonded to the stabilizer. For example, and not by the way of limitation, one or more enzymes within an enzyme system, e.g., one or more NAD(P)-dependent enzymes, can be covalently bonded to the stabilizer, e.g., albumin, present in the active area.

In certain particular embodiments, covalent bonding of the one or more enzymes and/or redox mediators to the polymer and/or stabilizer in a given active area can take place via crosslinking introduced by a suitable crosslinking

agent. In certain embodiments, crosslinking of the polymer to the one or more enzymes and/or redox mediators can reduce the occurrence of delamination of the enzyme compositions from an electrode. Suitable crosslinking agents can include one or more crosslinkable functionalities such as, but not limited to, vinyl, alkoxy, acetoxy, enoxy, oxime, amino, hydroxyl, cyano, halo, acrylate, epoxide and isocyanato groups. In certain embodiments, the crosslinking agent comprises one or more, two or more, three or more or four or more epoxide groups. For example, but not by way of limitation, a crosslinker for use in the present disclosure can include mono-, di-, tri- and tetra-ethylene oxides. In certain embodiments, crosslinking agents for reaction with free amino groups in the enzyme (e.g., with the free side chain amine in lysine) can include crosslinking agents such as, for example, polyethylene glycol dibutyl ethers, polypropylene glycol dimethyl ethers, polyalkylene glycol allyl methyl ethers, polyethylene glycol diglycidyl ether (PEGDGE) or other polyepoxides, cyanuric chloride, N-hydroxysuccinimide, imidoesters, epichlorohydrin, or derivatized variants thereof. In certain embodiments, the crosslinking agent is PEGDGE, e.g., having an average molecular weight (M_n) from about 200 to 1,000, e.g., about 400. In certain embodiments, the crosslinking agent is PEGDGE 400. In certain embodiments, the crosslinking agent can be glutaraldehyde. Suitable crosslinking agents for reaction with free carboxylic acid groups in the enzyme can include, for example, carbodiimides. In certain embodiments, the crosslinking agent is polyethylene glycol diglycidyl ether. In certain embodiments, the crosslinking of the enzyme to the polymer is generally intermolecular. In certain embodiments, the crosslinking of the enzyme to the polymer is generally intramolecular.

6. Mass Transport Limiting Membrane

In certain embodiments, the analyte sensors disclosed herein further include a membrane permeable to an analyte that overcoats at least an active area, e.g., a first active area and/or a second active area.

In certain embodiments, the analyte sensors disclosed herein further include a membrane permeable to an analyte that overcoats at least one active area, e.g., a first active area and/or a second active area. In certain embodiments, the membrane overcoats each of the active areas of an analyte sensor. Alternatively, a first membrane overcoats one of the active areas and a second membrane overcoats the second active area. Alternatively, a first membrane overcoats one of the active areas and a second membrane overcoats both the first and second active areas.

In certain embodiments, a membrane overcoating an analyte-responsive active area can function as a mass transport limiting membrane and/or to improve biocompatibility. A mass transport limiting membrane can act as a diffusion-limiting barrier to reduce the rate of mass transport of the analyte, e.g., glucose, an alcohol, a ketone, lactate or O-hydroxybutyrate, when the sensor is in use. For example, but not by way of limitation, limiting access of an analyte, e.g., a ketone, to the analyte-responsive active area with a mass transport limiting membrane can aid in avoiding sensor overload (saturation), thereby improving detection performance and accuracy. In certain embodiments, the mass transport limiting layers limit the flux of an analyte to the electrode in an electrochemical sensor so that the sensor is linearly responsive over a large range of analyte concentrations.

In certain embodiments, the mass transport limiting membrane can be homogeneous and can be single-component (contain a single membrane polymer). Alternatively, the

mass transport limiting membrane can be multi-component (contain two or more different membrane polymers). In certain embodiments, the mass transport limiting membrane can include two or more layers, e.g., a bilayer or trilayer membrane. In certain embodiments, each layer can comprise a different polymer or the same polymer at different concentrations or thicknesses. In certain embodiments, the first analyte-responsive active area can be covered by a multi-layered membrane, e.g., a bilayer membrane, and the second analyte-responsive active area can be covered by a single membrane. In certain embodiments, the first analyte-responsive active area can be covered by a multi-layered membrane, e.g., a bilayer membrane, and the second analyte-responsive active area can be covered by a multi-layered membrane, e.g., a bilayer membrane. In certain embodiments, the first analyte-responsive active area can be covered by a single membrane and the second analyte-responsive active area can be covered by a multi-layered membrane, e.g., a bilayer membrane. In certain embodiments, the first analyte-responsive active area can be covered by a single membrane and the second analyte-responsive active area can be covered by a single membrane.

In certain embodiments, a mass transport limiting membrane can include crosslinked polymers containing heterocyclic nitrogen groups. In certain embodiments, a mass transport limiting membrane can include a polyvinylpyridine-based polymer. Non-limiting examples of polyvinylpyridine-based polymers are disclosed in U.S. Patent Publication No. 2003/0042137 (e.g., at Formula 2b), the contents of which are incorporated by reference herein in its entirety.

In certain embodiments, a mass transport limiting membrane can include a polyvinylpyridine (e.g., poly(4-vinylpyridine) or poly(4-vinylpyridine)), a polyvinylimidazole, a polyvinylpyridine copolymer (e.g., a copolymer of vinylpyridine and styrene), a polyacrylate, a polyurethane, a polyether urethane, a silicone, a polytetrafluoroethylene, a polyethylene-co-tetrafluoroethylene, a polyolefin, a polyester, a polycarbonate, a biostable polytetrafluoroethylene, homopolymers, copolymers or terpolymers of polyurethanes, a polypropylene, a polyvinylchloride, a polyvinylidene difluoride, a polybutylene terephthalate, a polymethylmethacrylate, a polyether ether ketone, cellulosic polymers, polysulfones and block copolymers thereof including, for example, di-block, tri-block, alternating, random and graft copolymers or a chemically related material and the like.

In certain embodiments, a membrane for use in the present disclosure, e.g., a single-component membrane, can include a polyvinylpyridine (e.g., poly(-vinylpyridine) and/or poly(2-vinylpyridine)). In certain embodiments, a membrane for use in the present disclosure, e.g., a single-component membrane, can include poly(-vinylpyridine). In certain embodiments, a membrane for use in the present disclosure, e.g., a single-component membrane, can include a copolymer of vinylpyridine and styrene. In certain embodiments, the membrane can comprise a polyvinylpyridine-co-styrene copolymer. For example, but not by way of limitation, a polyvinylpyridine-co-styrene copolymer for use in the present disclosure can include a polyvinylpyridine-co-styrene copolymer in which a portion of the pyridine nitrogen atoms were functionalized with a non-crosslinked polyethylene glycol tail and a portion of the pyridine nitrogen atoms were functionalized with an alkylsulfonic acid group. In certain embodiments, a derivatized polyvinylpyridine-co-styrene copolymer for use as a membrane

polymer can be the 10Q5 polymer as described in U.S. Pat. No. 8,761,857, the contents of which are incorporated by reference herein in its entirety. In certain embodiments, the polyvinylpyridine-based polymer has a molecular weight from about 50 Da to about 500 kDa.

In certain embodiments, the membrane can comprise polymers such as, but not limited to, poly(styrene co-maleic anhydride), dodecylamine and poly(propylene glycol)-block-polyethylene glycol)-block-poly(propylene glycol) (2-aminopropyl ether) crosslinked with poly(propylene glycol)-block-poly(ethylene glycol)-block-poly(propylene glycol) bis(2-aminopropyl ether); poly(N-isopropyl acrylamide); a copolymer of poly(ethylene oxide) and poly(propylene oxide); or a combination thereof.

In certain embodiments, the membrane includes a polyurethane membrane that includes both hydrophilic and hydrophobic regions. In certain embodiments, a hydrophobic polymer component is a polyurethane, a polyurethane urea or poly(ether-urethane-urea). In certain embodiments, a polyurethane is a polymer produced by the condensation reaction of a diisocyanate and a difunctional hydroxyl-containing material. In certain embodiments, a polyurethane urea is a polymer produced by the condensation reaction of a diisocyanate and a difunctional amine-containing material. In certain embodiments, diisocyanates for use herein include aliphatic diisocyanates, e.g., containing from about 4 to about 8 methylene units, or diisocyanates containing cycloaliphatic moieties. Additional non-limiting examples of polymers that can be used for the generation of a membrane of a presently disclose sensor include vinyl polymers, polyethers, polyesters, polyamides, inorganic polymers (e.g., polysiloxanes and polycarbosiloxanes), natural polymers (e.g., cellulosic and protein based materials) and mixtures (e.g., admixtures or layered structures) or combinations thereof. In certain embodiments, the hydrophilic polymer component is polyethylene oxide and/or polyethylene glycol. In certain embodiments, the hydrophilic polymer component is a polyurethane copolymer. For example, but not by way of limitation, a hydrophobic-hydrophilic copolymer component for use in the present disclosure is a polyurethane polymer that comprises about 10% to about 50%, e.g., 20%, hydrophilic polyethylene oxide.

In certain embodiments, the membrane includes a silicone polymer/hydrophobic-hydrophilic polymer blend. In certain embodiments, the hydrophobic-hydrophilic polymer for use in the blend can be any suitable hydrophobic-hydrophilic polymer such as, but not limited to, polyvinylpyrrolidone, polyhydroxyethyl methacrylate, polyvinylalcohol, polyacrylic acid, polyethers such as polyethylene glycol or polypropylene oxide, and copolymers thereof, including, for example, di-block, tri-block, alternating, random, comb, star, dendritic and graft copolymers. In certain embodiments, the hydrophobic-hydrophilic polymer is a copolymer of poly(ethylene oxide) (PEO) and poly(propylene oxide) (PPO). Non-limiting examples of PEO and PPO copolymers include PEO-PPO diblock copolymers, PPO-PEO-PPO triblock copolymers, PEO-PPO-PEO triblock copolymers, alternating block copolymers of PEO-PPO, random copolymers of ethylene oxide and propylene oxide and blends thereof. In certain embodiments, the copolymers can be substituted with hydroxy substituents.

In certain embodiments, hydrophilic or hydrophobic modifiers can be used to "fine-tune" the permeability of the resulting membrane to an analyte of interest. In certain embodiments, hydrophilic modifiers such as poly(ethylene glycol), hydroxyl or polyhydroxyl modifiers and the like, and

any combinations thereof, can be used to enhance the biocompatibility of the polymer or the resulting membrane.

In certain embodiments where multiple active areas are present, the mass transport limiting membrane can overcoat each active area, including the option of compositional variation upon differing active areas, which can be achieved through sequential dip coating operations to produce a bilayer membrane portion upon a working electrode located closer to the sensor tip.

In certain embodiments where multiple active areas are present, a separate mass transport limiting membrane can overcoat each active area. For example, but not by way of limitation, a mass transport limiting membrane can be disposed on the first active area, e.g., the ketones-responsive active area, and a separate, second mass transport limiting membrane can overcoat the second active area, e.g., glucose-responsive active area. In certain embodiments, the two mass transport limiting membranes are spatially separated and do not overlap each other. In certain embodiments, the first mass transport limiting membrane does not overlap the second mass transport limiting membrane and the second mass transport limiting membrane does not overlap the first mass transport limiting membrane. In certain embodiments, the first mass transport limiting membrane comprises different polymers than the second mass transport limiting membrane. Alternatively, the first mass transport limiting membrane comprises the same polymers as the second mass transport limiting membrane. In certain embodiments, the first mass transport limiting membrane comprises the same polymers as the second mass transport limiting membrane but comprise different crosslinking agents.

In certain embodiments, the composition of the mass transport limiting membrane disposed on an analyte sensor that has two active areas can be the same or different where the mass transport limiting membrane overcoats each active area. For example, but not by way of limitation, the portion of the mass transport limiting membrane overcoating the ketones-responsive active area can be multi-component and/or the portion of the mass transport limiting membrane overcoating the glucose-responsive active area can be single-component. Alternatively, the portion of the mass transport limiting membrane overcoating the ketones-responsive active area can be single-component and/or the portion of the mass transport limiting membrane overcoating the glucose-responsive active area can be multi-component.

In certain embodiments, a glucose-responsive active area can be overcoated with a membrane comprising a polyurethane, a polyurethane urea or poly(ether-urethane-urea). In certain embodiments, a glucose-responsive active area can be overcoated with a membrane comprising a polyurethane. In certain embodiments of the present disclosure, the ketones-responsive active area and the second analyte-responsive area, e.g., glucose-responsive active area, can be overcoated with a membrane comprising a polyvinylpyridine-co-styrene copolymer.

In certain embodiments, a membrane, e.g., a single-component membrane, can include a polyvinylpyridine. In certain embodiments, a membrane, e.g., a single-component membrane, can include a copolymer of vinylpyridine and styrene (or a derivative thereof).

In certain embodiments, the multi-component membrane can be present as a bilayer membrane or as a homogeneous admixture of two or more membrane polymers. A homogeneous admixture can be deposited by combining the two or more membrane polymers in a solution and then depositing the solution upon a working electrode, e.g., dipping. In certain embodiments of the present disclosure, a first ana-

lyte-responsive active area, e.g., a ketones-responsive active area, can be overcoated with a multi-component membrane comprising a polyvinylpyridine and a polyvinylpyridine-co-styrene copolymer, either as a bilayer membrane or a homogeneous admixture, and a second analyte-responsive active area, e.g., a glucose-responsive active area, can be overcoated with a membrane comprising a polyvinylpyridine-co-styrene copolymer.

A suitable copolymer of vinylpyridine and styrene can have a styrene content ranging from about 0.01% to about 50% mole percent, or from about 0.05% to about 45% mole percent, or from about 0.1% to about 40% mole percent, or from about 0.5% to about 35% mole percent, or from about 1% to about 30% mole percent, or from about 2% to about 25% mole percent, or from about 5% to about 20% mole percent. Substituted styrenes can be used similarly and in similar amounts. A suitable copolymer of vinylpyridine and styrene can have a molecular weight of 5 kDa or more, or about 10 kDa or more, or about 15 kDa or more, or about 20 kDa or more, or about 25 kDa or more, or about 30 kDa or more, or about 40 kDa or more, or about 50 kDa or more, or about 75 kDa or more, or about 90 kDa or more, or about 100 kDa or more. In non-limiting examples, a suitable copolymer of vinylpyridine and styrene can have a molecular weight ranging from about 5 kDa to about 150 kDa, or from about 10 kDa to about 125 kDa, or from about 15 kDa to about 100 kDa, or from about 20 kDa to about 80 kDa, or from about 25 kDa to about 75 kDa, or from about 30 kDa to about 60 kDa.

Polydimethylsiloxane (PDMS) can be incorporated in any of the mass transport limiting membranes disclosed herein.

In certain embodiments, an analyte sensor described herein can comprise a sensor tail comprising at least an NAD(P) depot, a permeable polymer that overcoats the NAD(P) depot, a first permeable working electrode, a first active area disposed upon a surface of the first working electrode and a mass transport limiting membrane permeable to the first analyte that overcoats at least the first active area.

In certain embodiments, the first active area comprises at least one NAD(P)-dependent-enzyme (optionally, covalently bonded to a first polymer present within the active area) that is responsive to a first analyte. For example, but not by way of limitation, an analyte sensor described herein can comprise a sensor tail comprising at least an NAD(P) depot, a permeable polymer that overcoats the NAD(P) depot, a first working electrode, an analyte-responsive active area comprising at least one NAD(P)-dependent-enzyme disposed upon a surface of the first working electrode and a mass transport limiting membrane permeable to the analyte that overcoats the analyte-responsive active area.

In certain embodiments, the first active area comprises a first polymer and an enzyme responsive, e.g., an NAD(P)-dependent enzyme, to a first analyte, e.g., glucose, that is, optionally, covalently bonded to a first polymer. For example, but not by way of limitation, an analyte sensor described herein can comprise a sensor tail comprising at least an NAD(P) depot, a first working electrode, a glucose-responsive active area comprising a glucose dehydrogenase (optionally, covalently bonded to a first polymer) disposed upon a surface of the first working electrode and a mass transport limiting membrane permeable to glucose that overcoats the glucose-responsive active area.

In certain embodiments, the first active area comprises a first polymer and an enzyme system responsive to a first analyte, e.g., ketones, that comprises at least one enzyme, e.g., an NAD-dependent enzyme, that is, optionally, cova-

lently bonded to the first polymer. For example, but not by way of limitation, an analyte sensor described herein can comprise a sensor tail comprising at least an NAD(P) depot, a first working electrode, a ketones-responsive active area comprising an enzyme system comprising β -hydroxybutyrate dehydrogenase and diaphorase (where one or both enzymes are covalently bonded to a polymer) disposed upon a surface of the first working electrode and a mass transport limiting membrane permeable to ketones that overcoats the ketones-responsive active area.

In certain embodiments when a first active area and a second active area configured for assaying different analytes are disposed on separate working electrodes, the mass transport limiting membrane can have differing permeability values for the first analyte and the second analyte. For example, but not by way of limitation, the mass transport limiting membrane overcoating at least one of the active areas can include an admixture of a first membrane polymer and a second membrane polymer or a bilayer of the first membrane polymer and the second membrane polymer. A homogeneous membrane can overcoat the active area not overcoated with the admixture or the bilayer, wherein the homogeneous membrane includes only one of the first membrane polymer or the second membrane polymer. Advantageously, the architectures of the analyte sensors disclosed herein readily allow a continuous membrane having a homogenous membrane portion to be disposed upon a first active area and a multi-component membrane portion to be disposed upon a second active area of the analyte sensors, thereby levelizing the permeability values for each analyte concurrently to afford improved sensitivity and detection accuracy. Continuous membrane deposition can take place through sequential dip coating operations in particular embodiments.

In certain embodiments, when multiple active areas are present, the mass transport limiting membrane can overcoat each active area. In certain embodiments, a mass transport limiting layer is a membrane composed of crosslinked polymers containing heterocyclic nitrogen groups, such as polymers of polyvinylpyridine and polyvinylimidazole. Embodiments also include membranes that are made of a polyurethane, or polyether urethane, or chemically related material, or membranes that are made of silicone, and the like. In certain embodiments, the mass transport limiting membrane can include a membrane polymer, such as a polyvinylpyridine or polyvinylimidazole homopolymer or copolymer, which can be further crosslinked with a suitable crosslinking agent. In certain particular embodiments, the membrane polymer can include a copolymer of vinylpyridine and styrene.

In certain embodiments, the mass transport limiting membrane can comprise a membrane polymer crosslinked with a crosslinking agent disclosed herein and above in section 5. In certain embodiments where there are two mass transport limiting membranes, e.g., a first mass transport limiting membrane and a second mass transport limiting membrane, each membrane can be crosslinked with a different crosslinking agent. For example, but not by way of limitation, the crosslinking agent can result in a membrane that is more restrictive to diffusion of certain compounds, e.g., analytes within the membrane, or less restrictive to diffusion of certain compounds, e.g., by affecting the size of the pores within the membrane. For example, but not by way of limitation, in a sensor that is configured to detect ketones and glucose, the mass transport limiting membrane overcoating the ketones-responsive area can have a pore size that

restricts the diffusion of compounds larger than ketones, e.g., glucose, through the membrane.

In certain embodiments, crosslinking agents for use in the present disclosure can include polyepoxides, carbodiimide, cyanuric chloride, triglycidyl glycerol, N-hydroxysuccinimide, imidoesters, epichlorohydrin or derivatized variants thereof. In certain embodiments, a membrane polymer overcoating one or more active areas can be crosslinked with a branched crosslinker, e.g., which can decrease the amount of extractables obtainable from the mass transport limiting membrane. Non-limiting examples of a branched crosslinker include branched glycidyl ether crosslinkers, e.g., including branched glycidyl ether crosslinkers that include two or three or more crosslinkable groups. In certain embodiments, the branched crosslinker can include two or more crosslinkable groups, such as polyethylene glycol diglycidyl ether. In certain embodiments, the branched crosslinker can include three or more crosslinkable groups, such as polyethylene glycol tetraglycidyl ether. In certain embodiments, the mass transport limiting membrane can include polyvinylpyridine or a copolymer of vinylpyridine and styrene crosslinked with a branched glycidyl ether crosslinker including two or three crosslinkable groups, such as polyethylene glycol tetraglycidyl ether or polyethylene glycol diglycidyl ether. In certain embodiments, the epoxide groups of a polyepoxides, e.g., polyethylene glycol tetraglycidyl ether or polyethylene glycol diglycidyl ether, can form a covalent bond with pyridine or an imidazole via epoxide ring opening resulting in a hydroxyalkyl group bridging a body of the crosslinker to the heterocycle of the membrane polymer.

In certain embodiments, the crosslinking agent is polyethylene glycol diglycidyl ether (PEGDGE). In certain embodiments, the PEGDGE used to promote crosslinking (e.g., intermolecular crosslinking) between two or more membrane polymer backbones can exhibit a broad range of suitable molecular weights. In certain embodiments, the molecular weight of the PEGDGE can range from about 100 g/mol to about 5,000 g/mol. The number of ethylene glycol repeat units in each arm of the PEGDGE can be the same or different, and can typically vary over a range within a given sample to afford an average molecular weight. In certain embodiments, the PEGDGE for use in the present disclosure has an average molecular weight (M_n) from about 200 to 1,000, e.g., about 400. In certain embodiments, the crosslinking agent is PEGDGE 400.

In certain embodiments, the polyethylene glycol tetraglycidyl ether used to promote crosslinking (e.g., intermolecular crosslinking) between two or more membrane polymer backbones can exhibit a broad range of suitable molecular weights. Up to four polymer backbones may crosslinked with a single molecule of the polyethylene glycol tetraglycidyl ether crosslinker. In certain embodiments, the molecular weight of the polyethylene glycol tetraglycidyl ether can range from about 1,000 g/mol to about 5,000 g/mol. The number of ethylene glycol repeat units in each arm of the polyethylene glycol tetraglycidyl ether can be the same or different, and can typically vary over a range within a given sample to afford an average molecular weight. In certain embodiments, the mass transport limiting membrane can be deposited directly onto the active area.

In certain embodiments, the mass transport limiting membrane has a thickness, e.g., dry thickness, ranging from about 0.1 μm to about 1,000 μm , e.g., from about 1 μm to about 500 μm , about 10 μm to about 100 μm or about 10 μm to about 100 μm . In certain embodiments, the mass transport limiting membrane can have a thickness from about 0.1 μm to about 10 μm , e.g., from about 0.5 μm to about 10 μm , from

about 1 μm to about 10 μm , from about 1 μm to about 5 μm or from about 0.1 μm to about 5 μm . In certain embodiments, the sensor can be dipped in the mass transport limiting membrane solution more than once. For example, but not by way of limitation, a sensor (or working electrode) of the present disclosure can be dipped in an interference domain solution at least twice, at least three times, at least four times or at least five times to obtain the desired interference domain thickness.

7. Interference Domain

In certain embodiments, the sensor of the present disclosure, e.g., sensor tail, can further comprise an interference domain. In certain embodiments, the interference domain can include a polymer domain that restricts the flow of one or more interferants, e.g., to the surface of the working electrode. In certain embodiments, the interference domain can function as a molecular sieve that allows analytes and other substances that are to be measured by the working electrode to pass through, while preventing passage of other substances such as interferants. In certain embodiments, the interferants can affect the signal obtained at the working electrode. Non-limiting examples of interferants include acetaminophen, ascorbate, ascorbic acid, bilirubin, cholesterol, creatinine, dopamine, ephedrine, ibuprofen, L-dopa, methyl dopa, salicylate, tetracycline, tolazamide, tolbutamide, triglycerides, urea and uric acid.

In certain embodiments, the interference domain is located between the working electrode and one or more active areas, e.g., ketones-responsive active area. In certain embodiments, non-limiting examples of polymers that can be used in the interference domain include polyurethanes, polymers having pendant ionic groups and polymers having controlled pore size. In certain embodiments, the interference domain is formed from one or more cellulosic derivatives. Non-limiting examples of cellulosic derivatives include polymers such as cellulose acetate, cellulose acetate butyrate, 2-hydroxyethyl cellulose, cellulose acetate phthalate, cellulose acetate propionate, cellulose acetate trimellitate and the like.

In certain embodiments, the interference domain is part of the mass transport limiting membrane and not a separate membrane.

In certain embodiments, the interference domain includes a thin, hydrophobic membrane that is non-swelling and restricts diffusion of high molecular weight species. For example, but not by way of limitation, the interference domain can be permeable to relatively low molecular weight substances, such as hydrogen peroxide, while restricting the passage of higher molecular weight substances, such as ketones, glucose, acetaminophen and/or ascorbic acid.

In certain embodiments, the interference domain can be deposited directly onto the working electrode, e.g., onto the surface of the permeable working electrode. In certain embodiments, the interference domain has a thickness, e.g., dry thickness, ranging from about 0.1 μm to about 1,000 μm , e.g., from about 1 μm to about 500 μm , about 10 μm to about 100 μm or about 10 μm to about 100 μm . In certain embodiments, the interference domain can have a thickness from about 0.1 μm to about 10 μm , e.g., from about 0.5 μm to about 10 μm , from about 1 μm to about 10 μm , from about 1 μm to about 5 μm or from about 0.1 μm to about 5 μm . In certain embodiments, the sensor can be dipped in the interference domain solution more than once. For example, but not by way of limitation, a sensor (or working electrode) of the present disclosure can be dipped in an interference domain solution at least twice, at least three times, at least

four times or at least five times to obtain the desired interference domain thickness.

8. Manufacturing

The present disclosure further provides methods for manufacturing the presently disclosed analyte sensors that includes one or more active areas, one or more NAD(P) depots and one or more working electrodes.

In certain embodiments, the method includes depositing a composition containing NAD(P) on a substrate to generate an NAD(P) depot. For example, but not by way of limitation, the composition can be NAD and/or NADP, which depends on the enzyme present in the active area. In certain embodiments, the method can further include adding a permeable layer on top of the NAD(P) depot. In certain embodiments, the permeable layer can include a polymer that controls the release of the NAD(P) from the depot. Alternatively, a composition comprising a polymer and NAD(P) can be deposited onto a substrate to generate an NAD(P) depot. In certain embodiments, the polymer of the NAD(P) depot is curable, e.g., UV-curable.

In certain embodiments, the method can further include producing a permeable working electrode, e.g., a carbon nanotube electrode, on the permeable layer.

In certain embodiments, the method can further include depositing an enzyme composition comprising one or more NAD(P)-dependent enzymes, e.g., an NAD(P)-dependent dehydrogenase, on the working electrode. In certain embodiments, the enzyme composition can include one or more additional enzymes, e.g., diaphorase, a crosslinking agent, e.g., polyethylene glycol diglycidyl ether, a polymer and/or a redox mediator. In certain embodiments, the enzyme composition can be deposited onto the surface of a working electrode as one large application which covers the desired portion of the working electrode or in the form of an array of a plurality of enzyme compositions, e.g., spaced apart from each other, to generate one or more active areas for detecting one or more analytes. In certain embodiments, the method can further include curing the enzyme composition.

In certain embodiments, NAD(P), the permeable polymer, the permeable working electrode and enzyme composition can be prepared as solutions that dry or cure to solidify after deposition. Therefore, in certain embodiments, all layers can be deposited in an automated fashion using small-volume liquid handling or similar techniques for high-throughput sensor fabrication.

In certain embodiments, the method can further include adding a membrane composition on top of the cured enzyme composition and/or around the entire sensor. In certain embodiments, the membrane composition can include a polymer, e.g., a polyvinylpyridine, and/or a crosslinking agent, e.g., polyethylene glycol diglycidyl ether. In certain embodiments, the method can include curing the membrane polymer composition.

Generally, the thickness of the membrane is controlled by the concentration of the membrane solution, by the number of droplets of the membrane solution applied, by the number of times the sensor is dipped in or sprayed with the membrane solution, by the volume of membrane solution sprayed on the sensor, and the like, and by any combination of these factors. In certain embodiments, the membrane described herein can have a thickness ranging from about 0.1 micrometers (m) to about 1,000 μm , e.g., from about 1 μm to and about 500 μm , about 10 μm to about 100 μm or about 10 μm to about 100 μm . In certain embodiments, the sensor can be dipped in the membrane solution more than once. For example, but not by way of limitation, a sensor (or working electrode) of the present disclosure can be dipped in a

membrane solution at least twice, at least three times, at least four times or at least five times to obtain the desired membrane thickness.

In certain embodiments, the membrane can overlay one or more active areas, and in certain embodiments, the active areas can have a thickness from about 0.1 μm to about 10 μm , e.g., from about 0.5 μm to about 10 μm , from about 1 μm to about 10 μm , from about 1 μm to about 5 μm or from about 0.1 μm to about 5 μm . In certain embodiments, a series of droplets can be applied atop of one another to achieve the desired thickness of the active area and/or membrane, without substantially increasing the diameter of the applied droplets (i.e., maintaining the desired diameter or range thereof). In certain embodiments, each single droplet can be applied and then allowed to cool or dry, followed by one or more additional droplets. For example, but not by way of limitation, at least one droplet, at least two droplets, at least three droplets, at least four droplets or at least five droplets are added atop of one another to achieve the desired thickness of the active area.

III. Analyte Monitoring

The present disclosure further provides methods of using the analyte sensors disclosed herein to detect an analyte in vivo. In certain embodiments, the present disclosure provides methods for detecting one or more analytes, e.g., one analyte or two analytes. For example, but not by way of limitation, the present disclosure provides methods for detecting one or more analytes including glucose, ketones, lactate, oxygen, hemoglobin A1C, albumin, alcohol, alkaline phosphatase, alanine transaminase, aspartate aminotransferase, bilirubin, blood urea nitrogen, calcium, carbon dioxide, chloride, creatinine, hematocrit, lactate, magnesium, oxygen, pH, phosphorus, potassium, sodium, total protein and/or uric acid using one or more NAD(P)-dependent enzymes. In certain embodiments, the analyte can be ketones, alcohol, glucose and/or lactate using one or more NAD(P)-dependent enzymes. For example, but not by way of limitation, the present disclosure provides methods for detecting one or more ketones. In certain embodiments, the present disclosure provides methods for detecting glucose. In certain embodiments, the present disclosure provides methods for detecting creatinine. In certain embodiments, the present disclosure provides methods for detecting lactate. In certain embodiments, the present disclosure provides methods for detecting alcohol.

In certain embodiments, the present disclosure provides methods for monitoring in vivo levels of an analyte over time with analyte sensors that include an NAD(P) depot and one or more NAD(P)-dependent enzymes, e.g., an NAD(P)-dependent dehydrogenase. Generally, monitoring the in vivo concentration of an analyte in a fluid of the body of a subject includes inserting at least partially under a skin surface an in vivo analyte sensor as disclosed herein, contacting the monitored fluid (interstitial, blood, dermal, and the like) with the inserted sensor and generating a sensor signal at the working electrode. The presence and/or concentration of the analyte detected by the analyte sensor can be displayed, stored, forwarded and/or otherwise processed. A variety of approaches can be employed to determine the concentration of analyte (e.g., glucose, an alcohol, a ketone and/or lactate) with the subject sensors. In certain embodiments, monitoring the concentration of analyte using the sensor signal can be performed by coulometric, amperometric, voltammetric, potentiometric, or any other convenient electrochemical detection technique.

In certain embodiments, a method for detecting an analyte includes: (i) providing an analyte sensor including: (a) an internal supply of NAD(P); (b) a permeable polymer that overcoats the internal supply of NAD(P); (c) at least a first working electrode that is disposed upon a surface of the permeable polymer, wherein the first working electrode is a permeable working electrode; (d) an analyte-responsive active area disposed upon a surface of the first working electrode, wherein the analyte-responsive active area comprises an NAD(P)-dependent enzyme; and (e) a mass transport limiting membrane permeable to the analyte that overcoats at least the analyte-responsive active area; (ii) applying a potential to the first working electrode; (iii) obtaining a first signal at or above an oxidation-reduction potential of the analyte-responsive active area, the first signal being proportional to a concentration of analyte in a fluid contacting the analyte-responsive active area; and (iv) correlating the first signal to the concentration of analyte in the fluid.

In certain embodiments, a method for detecting one or more ketones includes: (i) providing an analyte sensor including: (a) an internal supply of NAD(P); (b) a permeable polymer that overcoats the internal supply of NAD(P); (c) at least a first working electrode that is disposed upon a surface of the permeable polymer, wherein the first working electrode is a permeable working electrode; (d) a ketone-responsive active area disposed upon a surface of the first working electrode, wherein the analyte-responsive active area comprises β -hydroxybutyrate dehydrogenase and diaphorase; and (e) a mass transport limiting membrane permeable to ketones that overcoats at least the analyte-responsive area; (ii) applying a potential to the first working electrode; (iii) obtaining a first signal at or above an oxidation-reduction potential of the ketones-responsive active area, the first signal being proportional to a concentration of analyte in a fluid contacting the analyte-responsive active area; and (iv) correlating the first signal to the concentration of ketones in the fluid.

In certain embodiments, the method of the present disclosure can further include detecting a second analyte by providing an analyte sensor that includes a second active area and/or exposing an analyte sensor that includes a second active area to a fluid comprising the first analyte and the second analyte. In certain embodiments, the analyte sensor for use in a method for detecting a first analyte and a second analyte can further include a second working electrode; and a second active area disposed upon a surface of the second working electrode and responsive to a second analyte differing from the first analyte, where the second active area comprises a second polymer, at least one enzyme responsive to the second analyte covalently bonded to the second polymer and, optionally, a redox mediator covalently bonded to the second polymer; wherein a portion, e.g., second portion, of the mass transport limiting membrane overcoats the second active area. Alternatively, the second active site can be covered by a second mass transport limiting membrane that is separate and/or different than the mass transport limiting membrane that overcoats the ketones-responsive active area. In certain embodiments, at least one enzyme responsive to the second analyte comprises an enzyme system comprising multiple enzymes that are collectively responsive to the second analyte.

In certain embodiments, the method further includes attaching an electronics unit to the skin of the patient, coupling conductive contacts of the electronics unit to contacts of the sensor, collecting data using the electronics unit regarding a level of analyte from signals generated by the sensor, and forwarding the collected data from electron-

ics unit to a receiver unit, e.g., by RF. In certain embodiments, the receiver unit is a mobile telephone. In certain embodiments, the mobile telephone includes an application related to the monitored analyte. In certain embodiments, analyte information is forwarded by RFID protocol, such as Bluetooth, and the like.

In certain embodiments, the analyte sensor can be positioned in a user for automatic analyte sensing, e.g., continuously or periodically. In certain embodiments, the level of the analyte can be monitored over a time period ranging from seconds to minutes, hours, days, weeks or months. In certain embodiments, the methods disclosed herein can be used to predict future levels of the analyte, based on the obtained information, such as but not limited to current analyte level at time zero, as well as the rate of change of the analyte concentration or amount.

IV. Exemplary Embodiments

A. In certain non-limiting embodiments, the presently disclosed subject matter provides for analyte sensors comprising:

- (i) an internal supply of NAD(P);
- (ii) a permeable polymer that overcoats the internal supply of NAD(P);
- (iii) at least a first working electrode that is disposed upon a surface of the permeable polymer, wherein the first working electrode is a permeable working electrode;
- (iv) an analyte-responsive active area disposed upon a surface of the first working electrode, wherein the analyte-responsive active area comprises an NAD(P)-dependent enzyme; and
- (v) a mass transport limiting membrane permeable to the analyte that overcoats at least the analyte-responsive area.

A1. The analyte sensor of A, wherein the NAD(P)-dependent enzyme is an NAD(P)-dependent dehydrogenase.

A2. The analyte sensor of A or A1, wherein the permeable working electrode comprises a carbon nanotube.

A3. The analyte sensor of any one of A-A2, wherein the permeable polymer comprises poly(propylene glycol) methacrylate and 2-hydroxyethyl methacrylate.

A4. The analyte sensor of any one of A-A3, wherein the analyte is selected from the group consisting of glucose, a ketone, an alcohol, creatinine, lactate and a combination thereof.

A5. The analyte sensor of A4, wherein the analyte is glucose.

A6. The analyte sensor of A4, wherein the analyte is lactate.

A7. The analyte sensor of A4, wherein the analyte is an alcohol.

A8. The analyte sensor of A4, wherein the analyte is a ketone.

A9. The analyte sensor of A5, wherein the NAD(P)-dependent enzyme is a glucose dehydrogenase.

A10. The analyte sensor of A6, wherein the NAD(P)-dependent enzyme is a lactate dehydrogenase.

A11. The analyte sensor of A7, wherein the NAD(P)-dependent enzyme is an alcohol dehydrogenase.

A12. The analyte sensor of A8, wherein the NAD(P)-dependent enzyme is β -hydroxybutyrate dehydrogenase.

A13. The analyte sensor of any one of A-A12, wherein the analyte-responsive active area further comprises diaphorase.

A14. The analyte sensor of any one of A-A13, wherein the analyte-responsive active area further comprises a redox mediator.

A15. The analyte sensor of any one of A-A14, wherein the analyte-responsive active area further comprises a stabilizer.

A16. The analyte sensor of A15, wherein the stabilizer comprises an albumin.

A17. The analyte sensor of any one of A-A16, wherein the analyte-responsive active area further comprises a crosslinking agent.

A18. The analyte sensor of any one of A-A17, wherein the internal supply of NAD(P) comprises from about 1 μ g to about 1,000 μ g of NAD(P).

A19. The analyte sensor of any one of A-A18, wherein the mass transport limiting membrane comprises a polyvinylpyridine (e.g., poly(4-vinylpyridine) or poly(4-vinylpyridine)), a polyvinylimidazole, a polyvinylpyridine copolymer (e.g., a copolymer of vinylpyridine and styrene), a polyacrylate, a polyurethane, a polyether urethane, a silicone or a combination thereof.

A20. The analyte sensor of any one of A-A19, further comprising:

- (vi) a second working electrode; and
 - (vii) a second active area disposed upon a surface of the second working electrode and responsive to a second analyte differing from the first analyte, wherein the second active area comprising at least one enzyme responsive to the second analyte;
- wherein a second portion of the mass transport limiting membrane overcoats the second active area.

B. In certain non-limiting embodiments, the presently disclosed subject matter provides a method for detecting an analyte comprising:

- (i) providing an analyte sensor comprising:
 - (a) an internal supply of NAD(P);
 - (b) a permeable polymer that overcoats the internal supply of NAD(P);
 - (c) at least a first working electrode that is disposed upon a surface of the permeable polymer, wherein the first working electrode is a permeable working electrode;
 - (d) an analyte-responsive active area disposed upon a surface of the first working electrode, wherein the analyte-responsive active area comprises an NAD(P)-dependent enzyme; and
 - (e) a mass transport limiting membrane permeable to the analyte that overcoats at least the analyte-responsive area.
- (ii) applying a potential to the first working electrode;
- (iii) obtaining a first signal at or above an oxidation-reduction potential of the first active area, the first signal being proportional to a concentration of a first analyte in a fluid contacting the first active area; and
- (iv) correlating the first signal to the concentration of the first analyte in the fluid.

B1. The method of B, wherein the NAD(P)-dependent enzyme is an NAD(P)-dependent dehydrogenase.

B2. The method of claim B or B1, wherein the permeable working electrode comprises a carbon nanotube.

B3. The method of any one of B-B2, wherein the permeable polymer comprises poly(propylene glycol) methacrylate and 2-hydroxyethyl methacrylate.

B4. The method of any one of B-B3, wherein the analyte is selected from the group consisting of glucose, a ketone, an alcohol, lactate and a combination thereof.

B5. The method of B4, wherein the analyte is glucose.

B6. The method of B4, wherein the analyte is lactate.

B7. The method of B4, wherein the analyte is an alcohol.

B8. The method of B4, wherein the analyte is a ketone.

B9. The method of B5, wherein the NAD(P)-dependent enzyme is a glucose dehydrogenase.

B10. The method of B6, wherein the NAD(P)-dependent enzyme is a lactate dehydrogenase.

B11. The method of B7, wherein the NAD(P)-dependent enzyme is an alcohol dehydrogenase.

B12. The method of B8, wherein the NAD(P)-dependent enzyme is β -hydroxybutyrate dehydrogenase.

B13. The method of any one of B-B12, wherein the analyte-responsive active area further comprises diaphorase.

B14. The method of any one of B-B13, wherein the analyte-responsive active area further comprises a redox mediator.

B15. The method of any one of B-B14, wherein the analyte-responsive active area further comprises a stabilizer.

B16. The method of B15, wherein the stabilizer comprises an albumin.

B17. The method of any one of B-B16, wherein the analyte-responsive active area further comprises a crosslinking agent.

B18. The method of any one of B-B17, wherein the internal supply of NAD(P) comprises from about 1 μ g to about 1,000 μ g of NAD(P).

B19. The method of any one of B-B18, wherein the mass transport limiting membrane comprises a polyvinylpyridine (e.g., poly(4-vinylpyridine) or poly(4-vinylpyridine)), a polyvinylimidazole, a polyvinylpyridine copolymer (e.g., a copolymer of vinylpyridine and styrene), a polyacrylate, a polyurethane, a polyether urethane, a silicone or a combination thereof.

B20. The method of any one of B-B14, wherein the analyte sensor further comprises:

(f) a second working electrode; and

(g) a second active area disposed upon a surface of the second working electrode and responsive to a second analyte differing from the first analyte, wherein the second active area comprising at least one enzyme responsive to the second analyte;

wherein a second portion of the mass transport limiting membrane overcoats the second active area.

EXAMPLES

The presently disclosed subject matter will be better understood by reference to the following Example, which is provided as exemplary of the presently disclosed subject matter, and not by way of limitation.

Example 1: Preparation of Polymer-Controlled NAD Release System

The present Example provides a method for making a sensor with an NAD depot, as illustrated in FIG. 23A.

The analyte sensor was prepared by depositing various solutions. An NAD solution was first deposited on a thin plastic substrate (support layer) and allowed to dry, leaving solid NAD. Subsequently, a polymer solution composed of a mixture of poly(propylene glycol) methacrylate (POMA) and 2-hydroxyethyl methacrylate (HEMA) was deposited on the solid NAD and polymerized by UV exposure. A carbon nanotube solution was then deposited and allowed to dry, forming the permeable electrode. A ketone-sensing enzyme composition that includes an enzyme system comprising diaphorase and β -hydroxybutyrate dehydrogenase was then deposited onto the permeable electrode. Finally, the electrodes were singulated and dip-coated in a membrane solution comprising a polyvinylpyridine and polystyrene copolymer and a crosslinker to form the outer membrane. A control sensor was made similarly, except no NAD was deposited on the thin plastic substrate (FIG. 23B).

The response of the sensor and the control were then evaluated with 2 mM of β -hydroxybutyrate, which is used as a substitute for ketones present in vivo. As shown in FIG. 24, control sensors without an NAD depot showed reduced signal over time as NAD diffused out from the sensing layer. However, sensors with an NAD depot did not show reduced signal over time as a result of sustained release of NAD from the depot to maintain a sufficient NAD concentration in the sensing layer. Without being bound to a theory, it is believed that the outer membrane of the sensor can allow NAD to leach out from the sensing layer, resulting in decreased ketone response over time because NAD is required to facilitate the flow of electrons from the analyte to the electrode. As shown in this Example, the use of an NAD depot can overcome such limitations by allowing sustained release of NAD from the depot to maintain a sufficient NAD concentration to be used by the NAD-dependent enzyme in the sensing layer.

Although the presently disclosed subject matter and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosed subject matter. Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, and composition of matter, methods and processes described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosed subject matter of the presently disclosed subject matter, processes, machines, manufacture, compositions of matter, methods, or steps, presently existing or later to be developed that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein can be utilized according to the presently disclosed subject matter. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, methods, or steps.

Various patents, patent applications, publications, product descriptions, protocols, and sequence accession numbers are cited throughout this application, the inventions of which are incorporated herein by reference in their entireties for all purposes.

What is claimed is:

1. An analyte sensor comprising:

(i) an internal supply of NAD(P);

(ii) a permeable polymer that overcoats the internal supply of NAD(P);

(iii) a first working electrode that is disposed upon the active surface of the permeable polymer, wherein the first working electrode is a permeable working electrode;

(iv) an analyte-responsive active area disposed upon a surface of the first working electrode, wherein the analyte-responsive area comprises an NAD(P)-dependent enzyme; and

(v) a mass transport limiting membrane that overcoats the analyte-responsive area, wherein the membrane is permeable to the analyte.

2. The analyte sensor of claim 1, wherein the NAD(P)-dependent enzyme is an NAD(P)-dependent dehydrogenase.

3. The analyte sensor of claim 1, wherein the permeable working electrode comprises a carbon nanotube.

79

4. The analyte sensor of claim 1, wherein the permeable polymer comprises a polyether-based polymer.

5. The analyte sensor of claim 1, wherein the analyte is selected from the group consisting of glucose, a ketone, an alcohol, lactate and a combination thereof.

6. The analyte sensor of claim 5, wherein the NAD(P)-dependent enzyme is a glucose dehydrogenase, a lactate dehydrogenase, an alcohol dehydrogenase, or a p-hydroxybutyrate dehydrogenase.

7. The analyte sensor of claim 6, wherein the analyte-responsive active area further comprises diaphorase.

8. The analyte sensor of claim 1, wherein the analyte-responsive active area further comprises a redox mediator.

9. The analyte sensor of claim 1, further comprising:

(vi) a second working electrode; and

(vii) a second active area disposed upon a surface of the second working electrode and responsive to a second analyte differing from the first analyte, wherein the second active area comprises at least one enzyme responsive to the second analyte; wherein a second portion of the mass transport limiting membrane of (v) overcoats the second active area.

10. A method for assaying concentration of a first analyte in a fluid comprising:

(i) contacting the first active area of the analyte sensor of claim 1 with the fluid

(ii) applying a potential to the first working electrode;

(iii) obtaining a first signal at or above an oxidation-reduction potential of the first active area, the first signal being proportional to a concentration of the first analyte in the fluid contacting the first active area; and

80

(iv) correlating the first signal to the concentration of the first analyte in the fluid.

11. The method of claim 10, wherein the NAD(P)-dependent enzyme is an NAD(P)-dependent dehydrogenase.

12. The method of claim 10, wherein the permeable working electrode comprises a carbon nanotube.

13. The method of claim 10, wherein the permeable polymer comprises a polyether-based polymer.

14. The method of claim 10, wherein the first analyte is selected from the group consisting of glucose, a ketone, an alcohol, lactate and a combination thereof.

15. The method of claim 14, wherein the NAD(P)-dependent enzyme is a glucose dehydrogenase, a lactate dehydrogenase, an alcohol dehydrogenase, or a 3-hydroxybutyrate dehydrogenase.

16. The method of claim 15, wherein the analyte-responsive active area further comprises diaphorase.

17. The method of claim 10, wherein the analyte-responsive active area further comprises a redox mediator.

18. The method of claim 10, wherein the analyte sensor further comprises:

(f) a second working electrode; and

(g) a second active area disposed upon a surface of the second working electrode and responsive to a second analyte differing from the first analyte, wherein the second active area comprises at least one enzyme responsive to the second analyte; wherein a second portion of the mass transport limiting membrane of (v) overcoats the second active area.

* * * * *